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Road-Conditioned Diffusion for Traffic Trajectory Synthesis

Complete Journeys, Learned Route Preferences and GAF-Derived Generative Conditions

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Abstract

Synthetic traffic data require complete journeys that combine feasible route choices with time-dependent movement. This thesis develops a road-conditioned diffusion framework for five participant classes on the MoST Monaco network. A learned route policy selects among permitted alternatives, a conditional duration model samples travel time, and a temporal diffusion model generates progress at a physical resolution of 0.25 seconds. Road-coordinate decoding produces timestamped trajectories. The training corpus contains 29,895 continuous journey legs from native SUMO demand.

The framework integrates two trajectory-image representations: an ordered Hilbert–GASF/GADF/MTF spatial encoding and a speed encoding derived from coordinates and timestamps. Self-trained image classifiers achieve development macro F1 of 76.47% and 90.34%, respectively. Their encoders provide global motion conditions and are adapted jointly with trajectory denoising. A conditional latent diffusion prior supplies these codes for new requests, enabling direct trajectory synthesis from map, route, duration and participant information.

Evaluation covers eight model conditions, two fitted seeds, eight prospective SUMO episodes and thirty downstream classifier fits. The temporal pilot reduces native acceleration discrepancy by 81–82%, and learned route preferences reduce edge-use divergence by 13.6% relative to shortest-route selection in their respective controlled studies. In the main comparison, joint spatial–speed conditioning increases acceleration distribution distance by 2.44% relative to the matched baseline. Spatial-only synthetic augmentation yields 96.95% downstream macro F1, compared with 96.26% for baseline augmentation and 94.41% for observed-only training; its advantage over baseline augmentation is concentrated in one of the two evaluation OD cohorts.

The thesis contributes a complete road-conditioned synthesis pipeline and an experimental analysis of spatial and speed GAF conditions. The results distinguish class recognition, physical motion quality and downstream utility: image representations capture participant-related structure, while their effect on synthesized data depends on the conditioning interface and the evaluation task.

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Chapter 1

Introduction

A synthetic traffic trajectory describes where a participant travels and how the journey unfolds in time. Several routes can connect the same origin and destination. On any one route, a vehicle can move steadily, wait at a stop, accelerate after a junction or slow down for other traffic. Producing a connected polyline therefore solves only part of trajectory synthesis. A useful generated dataset must also retain the distribution and variation of movement associated with its participants.

The product of this thesis is a dataset in the familiar `time,id,x,y,type` format. Each generated identifier represents a sampled complete journey through its destination. The requested participant class influences both permitted travel and learned movement preferences. Repeating a request produces alternative realizations; it does not aim to recover one uniquely correct future trace. This makes road support, motion fidelity, diversity, coverage and downstream usefulness distinct evaluation concerns.

Two research directions motivate this study. Conditional diffusion models learn distributions of trajectories [4, 6, 7], while trajectory-image representations support traffic participant recognition [1, 2]. Their combination raises a central question: can spatial and motion representations learned for recognition improve conditional trajectory synthesis?

1.1 A complete journey with spatial and temporal choices

Let c be a participant class, R a permitted directed route and $P_R(s)$ its coordinate interpolation at normalized progress $s \in [0,1]$. A synthesized journey is

$$\mathcal{T} = \{(t_i, P_R(s_i), c)\}_{i=0}^{n}, \qquad t_0 = 0, \quad t_n = T, \quad 0 = s_0 \le s_1 \le \cdots \le s_n = 1. \tag{1.1}$$

The route, duration T and intervening progress vary across samples. Repeated progress values represent stopped intervals. Different rates of progress encode movement changes. Sequence length follows the sampled duration, so a long service is not represented solely by a short observation window.

The implemented system combines a learned route-preference policy, a training-side duration prior, temporal motion diffusion and a deterministic road decoder. A request supplies the map, endpoints, class and declared permissions; buses additionally retain public stop anchors. The policy samples among supported road alternatives. Diffusion then generates motion on a quarter-second physical clock, and the decoder returns coordinates. Each component has a specific responsibility: road adherence comes from construction and decoding, duration comes from the prior, and the learned denoiser controls temporal progression. Experiments keep those responsibilities identifiable.

Training uses observed route–motion pairs and participant labels. This supervision does not make inference a deterministic regression. Noise-prediction learning fits a conditional distribution, and random draws produce different trajectories for the same available conditions. Supplying a class label also does not certify realistic movement. A generated bus can follow a valid line while having implausible stops or acceleration, just as a bicycle can reach its destination with the wrong speed distribution. The output must be evaluated independently of its input label.

SUMO supplies learning observations and new reference realizations on the public Monaco SUMO Traffic network [3, 47]. The trained generator samples individual journeys; it does not jointly advance every surrounding vehicle and signal. This defines the intended use as conditional synthetic trajectory data. Interactive traffic safety, adaptation to unseen cities and replacement of the simulator's traffic dynamics require evidence beyond this task.

1.2 GAF representations for conditional generation

The spatial representation maps ordered coordinates through a Hilbert sequence into GASF, GADF and MTF image channels [10, 12]. A separate speed representation applies image transforms to a sequence derived from the same positions and timestamps. The spatial reader receives coordinates, while the speed reader receives the derived speed sequence. Both readers are trained to predict the same participant classes, providing complementary views of participant-related movement.

These deterministic transformations introduce no additional physical observation. Their potential value is representational: pairwise sequence relationships and learned features may expose movement regularities that a class embedding alone does not make easy to learn. The two views lose different details. Hilbert quantization and run compression can suppress positional or timing distinctions, while one-second speed summaries can suppress rapid fluctuations. Shared origin therefore

motivates comparing the views, but does not guarantee that either is informative for every movement property or that combining them is beneficial.

Recognition and conditional generation emphasize different properties. A feature can distinguish buses from motorcycles by identifying familiar road corridors. Once a route and class are fixed, that feature may offer little information about a plausible within-route speed profile. A classifier can also assign an extreme class-associated behavior a high score without it being representative. The evaluation measures recognition, motion quality and downstream utility separately.

The final method makes the GAF representations part of the generation training objective. Small spatial and speed codes condition every denoising stage, and the final CNN stages can adapt through motion reconstruction alongside the participant-classification objective. A training-only conditional latent diffusion prior supplies codes for new requests. This gives the model a possible channel for trajectory-specific information while preserving a prospective interface: it never receives an image of the future trajectory being evaluated. The general encoder–conditional-decoder construction follows established diffusion representation ideas [53]; the thesis tests its use with spatial and speed encodings, road conditions and complete physical-time motion.

1.3 Simulation data and physical-time representation

The experiments use the public MoST scenario with SUMO to construct complete journeys. Activity-based demand, vehicle parameters and public services preserve associations between participant class and travel behavior. A controlled comparison of demand configurations motivates this choice: class-permitted random routes retain motion differences but weaken the spatial patterns used for participant recognition. Chapter 3 describes the learning corpus and its construction.

Complete journeys also require a suitable temporal representation. A fixed 257-point sequence gives a long bus trip a much coarser physical clock than a short journey. The native-time model retains quarter-second observations, repeated positions and variable lengths. A matched temporal-resolution pilot assesses its consequences for generated motion and computation. Reconstruction fidelity alone is not accepted as proof that the trained generator produces good trajectories.

The learning population contains 29,895 complete vehicle legs from 24,073 parents, covering bus, motorcycle, private car, bicycle and taxi/ride-hailing. Public services, activity context and vehicle permissions are retained. New simulator realizations provide reference trajectories for evaluation, with source-parent partitions and shared service templates specified in the experimental protocol.

1.4 Research questions and contributions

The thesis addresses two questions. First, can a road-conditioned diffusion system produce complete journeys with reasonable route preferences and motion distributions, and what is gained by preserving physical time? Second, do spatial and speed GAF representations improve the generated data beyond class, route, duration and appropriate representation controls?

The first contribution is an executable complete-journey generator with a permitted road interface, learned route preferences and variable-length temporal diffusion. Its matched controls separate structural road support, sampled duration and learned motion. The second is a controlled study of GAF's role in that generator: recognition, terminal preferences, frozen-feature supervision and generation-trained conditions are tested as different mechanisms. The final conditions have frozen, single-view, dual-view, raw-series, wrong-trip and class-centroid controls, with no image inversion or specially chosen guidance timestep.

The evidence is built around actual synthetic trajectories. Five-class physical measures, fixed-order journey and GAF figures, diversity, coverage and an independent downstream augmentation task accompany the model comparisons. The final experiment uses eight fresh reference episodes organized as four OD cohorts with paired windows and two model seeds. Classifiers' confidence does not define success, and gains over a small downstream observed-data subset are separated from gains specifically attributable to GAF.

The following chapters place the approach in the literature, establish the data and representations, describe the generation and GAF modules, and define the comparisons before presenting their results. Earlier findings retain their original populations and uncertainty. The discussion relates these results to representation design and conditional motion synthesis.

Chapter 2

Background and related work

2.1 Trajectories as geometric and temporal objects

A sampled trajectory is an ordered sequence $X = ((t_i, p_i))_{i=0}^{T-1}$ with $p_i \in \mathbb{R}^2$. The polyline through its points defines geometry. The assignment of time to those points defines motion. Even on a straight road these properties can be separated. A vehicle covering one hundred metres in ten seconds can travel at constant speed or spend half the window waiting. Its endpoints and total displacement are identical in both cases, but its sequence of increments differs. A trajectory synthesizer therefore needs both a spatial support and a distribution over movement on that support.

A directed lane graph supplies information that a background map cannot convey by itself. For example, a bridge may lie directly above a second street in plan view. The distance between projected centerlines can be zero even though a vehicle cannot turn between them. At a junction, the permitted successor also depends on direction and participant class. A small distance to the nearest road is useful evidence about position. A connected assignment of road states throughout the sequence establishes a stronger property. Both depend on the accuracy and version of the map.

Microscopic simulation offers a practical test bed. The MoST scenario represents Monaco and its surroundings in SUMO, including multimodal roads and mobility demand [3]. Simulation exports can be sampled frequently and can carry participant types without manual annotation. They inherit the modelling choices of the simulator. A generated trace resembling the export can therefore reproduce a simulated mobility pattern even when its performance on traffic measured in the field is unknown.

2.2 Diffusion models for trajectories

A denoising diffusion model learns to recover structure from samples corrupted at known noise levels [4]. The representation of the clean sample is a design choice. When the representation is a sequence of positions, denoising acts on a coordinate plane. In the present work it acts on the temporal sequence of distances travelled along a sampled path. Either representation can produce coordinates as output. Their intermediate states and their routes to enforcing feasibility differ.

DiffTraj models spatial and temporal structure in GPS trajectory synthesis [6]. Its use of diffusion demonstrates the value of modelling a distribution of trajectories when multiple samples can be valid. A coordinate sample does not by itself name a permitted successor lane. Diff-RNTraj addresses road-constrained generation by representing positions through segment information and continuous moving ratios, and learning to denoise a vectorized representation [7]. This is particularly relevant prior work: the general concept of road-constrained diffusion generation predates this study.

A related design, RNTrajGen, separates road-level movement from the production of finer trajectories [8]. These studies make a useful comparator at the level of modelling assumptions. The temporal-resolution pilot here supplies a permitted planned route and compares learned motion with empirical transfer on identical paths. The later system samples a route from a learned policy over a permitted catalog. Its final GAF comparison holds that route policy common, preserving attribution of motion changes while allowing alternative routes across requests and draws. Published systems trained on different cities, sampling rates and definitions of trajectory quality provide context rather than numerical head-to-head results.

Scenario Diffusion generates agents and their trajectories jointly under map and scenario conditioning [9]. Its output represents a different unit: a traffic scene. In our experiment the same starting position can generate several independent participant traces. Simultaneous occupancy and interaction are therefore outside the output distribution. This is consequential for interpretation. A bus trace can follow connected lanes while failing to be compatible with the timing of a particular traffic signal or the trajectory of another vehicle.

2.3 Imaging a time series

Wang and Oates encode time series using Gramian angular fields and Markov transition fields [10]. A pixel represents a relation between temporal indices. In a Gramian field the relation is computed from angular transforms of the underlying scalar values. In a Markov transition field it depends on the estimated transition probabilities between discretized states. Convolutional networks can process the resulting arrays as images. The construction gives spatially local filters access to structured patches of pairwise relations.

For a traffic trajectory the input has two spatial coordinates. A Hilbert-based spatial encoding maps each point to a scalar using a Hilbert traversal of a spatial grid. The mapping changes the representation of a location while preserving the order in which observations were made. A CNN therefore sees temporal relations between locations collapsed to a scalar index. Nearby points along the Hilbert traversal occupy neighboring spatial cells, but some neighboring cells can be far apart in traversal order. The mapping is directional in this sense. It can also lose distinctions between slightly different positions in one cell.

Salvadori investigated trajectory images for traffic participant recognition, including encoding and classifier choices [1]. Ferreira et al. studied image-based classification on simulated traffic [2]. These studies motivate spatial image representations as a source of participant-related features. This thesis examines their use in conditional generation, using a published implementation to specify Hilbert indexing and sequence compression [12]. The classifiers used in the experiments are trained on the study corpus. Recognition results are evaluated under the stated data partitions and sampling horizons.

2.4 Evaluation under multiple objectives

A generative model can satisfy one objective and miss another. A route decoder can enforce topology for any input sequence of speeds. A generator that copies one speed profile can pass the geometric test while providing little new training information. A very diverse generator can produce implausible behavior. We therefore evaluate geometric consistency, one-dimensional motion distributions, candidate diversity and training utility separately. The common route and candidate budget make the comparison interpretable along each axis.

For distributional evaluation we use the first Wasserstein distance. In one dimension this is the area between two empirical cumulative distribution functions. Its units are those of the measured quantity. A difference of twenty metres in the travel-distance comparison has a direct interpretation as a reduction in distribution discrepancy; it is not the error of a predicted endpoint for a particular participant. For downstream evaluation we use macro-F1 on held-out participants. The averaging treats a rare class as a separate classification task instead of letting the most frequent class dominate.

DDIM sampling provides a deterministic sequence of reverse updates conditional on the initial noise [5]. Reusing paths, request order and random seeds reduces incidental changes in a comparison. Different temporal representations still have different noise dimensions, so a common seed is not an identical perturbation of the same vector. The temporal-resolution pilot reports two fixed model seeds on one new SUMO episode. The final GAF-conditioning comparison has four OD cohorts with paired windows and two model seeds. Its cohort intervals are descriptive; repeated windows, model seeds and candidate draws are not additional independent

transport domains. Earlier resampling studies retain their own conditional scope.

2.5 Learned representations and conditional synthesis

Representation alignment and generative conditioning provide different interfaces between an encoder and a denoiser. REPA uses pretrained representations as an auxiliary training target for diffusion models [50]. It motivates asking whether useful structure in a feature space can aid generative learning, but a class-trained traffic feature is not automatically an adequate target for within-class motion. Its performance must be established on generated trajectories, not inferred from image-domain findings.

Diffusion Autoencoders combine a learned encoder with a conditional diffusion decoder and fit a latent generative model for new sampling [53]. This supplies a relevant precedent for turning an observed representation into a sampleable condition. The present traffic adaptation uses two prescribed image transforms, partial CNN adaptation, a road/duration-conditioned code prior and a physical trajectory decoder. It does not claim the near-exact image reconstruction or inversion properties of the original architecture. A lossy image view can still be useful as a condition, but prior mismatch and an ignored condition are practical possibilities that require separate checks.

Route choice is a further component rather than an incidental coordinate error. Recursive models describe path preferences through local decisions and goal-dependent structure [54]. Here, a masked next-edge policy learned from observed branching decisions scores a finite set of permitted routes. It is a behavioural cloning model with catalog renormalization, not an estimated recursive-logit equilibrium. This distinction makes both its learned preference and its coverage limitations explicit.

2.6 What a trajectory generator learns

A route model and a motion model face different ambiguities. TrajGAIL learns sequential vehicle decisions by adversarial imitation on a road network [15]. Such a policy can choose among candidate links, while the rate at which a vehicle travels along a link remains a separate quantity. Diff-RNTraj jointly represents segment identity and movement within segments [7]. The two-stage design of RNTrajGen likewise separates road sequences from fine movement [8]. These distinctions motivate the temporal pilot comparator here. Transferring an empirical speed sequence to the very same path gives the baseline every geometric advantage of the proposed model. Because both candidates complete the prescribed route, the relevant

improvement is in motion distributions rather than in reaching a different travel distance.

Map-conditioned forecasting offers transferable representation ideas but uses different inference information. VectorNet encodes polylines, whereas LaneGCN explicitly propagates information over lane connectivity [16, 17]. Knowledge of map structure can improve predictions without guaranteeing that each returned sample admits a directed lane assignment. Trajectron++ models heterogeneous agents and feasible dynamics, and Scene Transformer jointly predicts agent futures [18, 19]. Social GAN explicitly treats pedestrian interactions as a source of multimodality [20]. Motion Transformer separates intention localization from local movement refinement [21]. These distinctions matter when interpreting a benchmark. A predictor exploiting an observed history or surrounding agents has information absent from the present synthesis request. The initial pilot supplies a planned route, while the later system samples an alternative from a learned road policy given endpoints and public stop anchors. In either case, the motion network receives neither an observed movement history nor the reference participant's future.

2.7 Guiding a trajectory while it is being generated

In sequential decision problems, Diffuser treats planning as denoising a trajectory, while Diffusion Policy generates action sequences conditioned on observations [22, 23]. MotionDiffuser is a closer comparison for map-based movement: it represents joint agent futures in a compressed space and guides samples using differentiable costs [24]. The Hilbert transform used here quantizes coordinates and changes sequence length. A small change to an increment can leave every encoded pixel unchanged until a cell boundary is crossed. A coordinate-gradient intervention therefore needs an additional surrogate or a different interface. The final learned-condition experiment preserves the discrete transform and differentiates CNN parameters during training, without requiring derivatives through Hilbert indexing.

Classifier guidance uses a classifier gradient during the reverse process, while classifier-free guidance combines conditional and unconditional model estimates [25, 26]. Universal Guidance extends the range of external guidance functions used at inference [27]. These methods make different assumptions about the information available at noisy states. In particular, a class score trained on clean trajectories has a physical interpretation only after a provisional motion sequence has been decoded. Inverse-problem samplers face a related distinction. DDRM uses structure in linear measurement models, whereas diffusion posterior sampling approximates a measurement term for more general observations [28, 29]. A participant classifier supplies neither measured future positions nor a known noise likelihood. Using its probability as a selection weight is therefore an experimental preference, rather than a statistically calibrated observation model.

Twisted Diffusion Sampling uses weighted particles to approximate conditional distributions [14]. Feynman–Kac steering makes explicit how intermediate rewards, resampling times and potentials affect reward-guided generation [30]. A terminal scorer can choose only among trajectories already produced. An intermediate scorer can change which partial solutions receive subsequent computation. That opportunity also permits premature loss of viable solutions. If an early estimate of the final class preference is noisy, resampling can discard good trajectories before their motions become recognizable. At a very late update, a surviving state may have too little stochastic evolution left to move appreciably away from a terminal candidate. These are reasons to distinguish terminal feedback from in-sampler guidance. The earlier fixed-pool controls isolate terminal preferences. The final experiment learns a condition used throughout denoising, without a classifier-gradient or resampling intervention at a selected timestep. Results from these mechanisms are reported separately.

Sampling budget itself can change conclusions. Improved DDPM studies learned reverse variances and sampling efficiency [31]; EDM separates choices of noise schedule, preconditioning and solver [32]; DPM-Solver constructs efficient numerical updates for diffusion sampling [33]. These works motivate explicit cost accounting. Equalizing network calls controls one cost, but a denoiser call on a long physical-time sequence is more expensive than one on 257 bins. The present comparison holds optimization steps and candidates fixed while reporting measured time; it does not label the two temporal models equal-compute. The earlier GAF selection studies reuse the same eight candidates. The final direct-draw comparison uses four samples per request and records latent-prior and motion-denoiser work separately.

2.8 Representation, fidelity and utility

External representations can also enter diffusion training. REPA aligns projected hidden states from noisy denoiser inputs with clean-image features from a pretrained encoder [50]. Its image-generation results motivate asking whether a trajectory representation could provide useful supervision; they do not establish that a small class-trained GAF reader is an equally effective teacher. Representation similarity is another diagnostic: centered kernel alignment compares relationships between learned feature representations [49]. Similarity, class separability and improvement of generated motion are distinct empirical questions. A high similarity score cannot replace the trajectory-quality comparison needed here.

A diffusion model over time does not uniquely specify a trajectory task. TimeGrad uses an autoregressive probabilistic model for multivariate forecasting, while CSDI conditions on observed entries for imputation [34, 35]. Those problems have observed context that a synthesis request may lack. DiffWave shows that an entire ordered waveform can itself be the generated object [36]. For road motion the temporal target used here is more structured: nonnegative increments give

progressively ordered positions on a chosen path. A plausible marginal distribution of increments can still give an implausible sequence if their temporal ordering is wrong.

Recognition offers another perspective on that loss of information. Temporal convolutional models offer an alternative to recurrent sequence encoders [37, 38]. InceptionTime applies an ensemble of deep convolutional networks to time-series classification; ROCKET and MiniRocket show that convolutional transforms with simple classifiers can also be competitive [39, 40, 41]. Unsupervised multivariate sequence representations can learn reusable embeddings without training an image classifier [42]. These alternatives show why agreement among multiple copies of an image classifier cannot uniquely establish a class-specific motion effect. Raw geometry and motion readers serve as information diagnostics in the data study. In the generation experiment, independent physical measurements are the primary outcome, and neither GAF teacher is an oracle of realism.

Classifier two-sample tests ask whether a learner can distinguish populations [43]. Distributional precision and recall instead separate sample quality from coverage [44, 45]. A selector can increase concentration in a recognizable part of the data distribution while decreasing the range of plausible behaviors available to the user. For that reason we compare variation across candidates, time near rest and steady movement in addition to speed and acceleration. Esteban and colleagues use synthetic time series to train predictors evaluated on real data [46]. Downstream augmentation asks a further practical question whose result need not track either classifier confidence or a single distribution distance. The final experiment measures this use directly with fixed downstream training budgets and separate reference cohorts. Its observed-only comparison must account for the larger labelled corpus used upstream to train the generator; a GAF-specific claim additionally needs the baseline and raw-representation controls.

Chapter 3

Data and complete-trip representation

This chapter describes the simulation scenario, the complete-journey learning population and the representations used by the generator. Native activity-based demand provides training and development trajectories. Separate SUMO runs provide reference observations for evaluation requests. The diffusion model produces the synthetic trajectories compared with those references.

3.1 Simulation scenario and learning population

The corpus is generated with SUMO on the public Monaco SUMO Traffic (MoST) scenario [3, 47]. Scenario revision `b29b2f65` provides the road network, activity-based demand, parking, background traffic, 24 bus services and 181 stops. Vehicle types retain their scenario-defined physical and behavioral parameters [51]. Participant labels and road-permission classes are recorded separately; in particular, taxi/ride-hailing labels can belong to vehicles with passenger permissions.

Each observation follows `time,id,x,y,type`, with time in seconds and coordinates in local planar metres. Speed and acceleration for evaluation are derived from positions and timestamps consistently for simulated and generated trajectories. A supplementary comparison of archived exports and demand configurations is provided in Appendix A.14.

The main corpus uses a ten-hour observation interval, simulation seconds 14,400–50,400, at a 0.25-second step. Two training realizations and one development realization retain the same public map and demand families. Each training parent is assigned a realization before its motion is inspected, so a second seed is not counted as another copy of the same independent parent. Known identities retain their earlier partition; original-test parents are excluded by stored identity metadata without reading their frozen trajectories.

For ordinary vehicles, departure, activity-stop boundaries and final arrival define complete legs. Intermediate bus service stops remain inside a bus leg. A parent can therefore contribute several completed activity legs, but a complete leg is not a full person's day of travel. The output unit of the generator is this complete vehicle journey. Pedestrian walking stages are outside the declared five-class synthesis task.

Table 3.1. Expanded native-demand corpus. Activity legs and distinct parent identities are separate units.

Class	Train legs	Train parents	Dev. legs
Bus	685	685	155
Motorcycle	6547	5226	1591
Private car	18381	13880	4436
Bicycle	2744	2744	668
Taxi/ride-hailing	1538	1538	380
Total	29,895	24,073	7,230

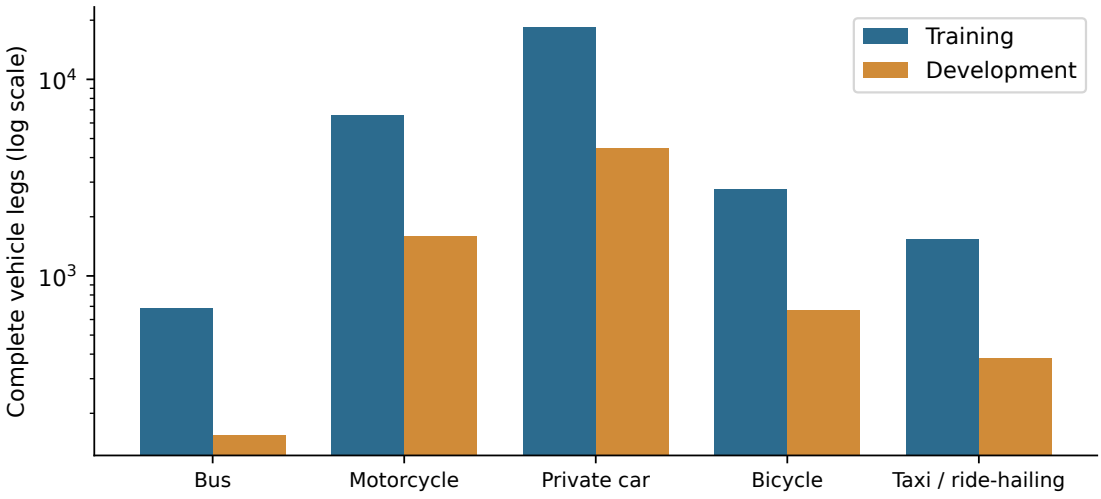


Figure 3.1. Expanded native-demand complete-leg counts. The logarithmic axis makes minority support visible while retaining the private-car majority. Legs, parent vehicles and independently simulated traffic realizations are different observation units.

There are 29,895 training legs from 24,073 parents and 7,230 development legs from 5,842 parents. The recorded coordinate counts are approximately 65.1 million and 15.8 million. Median training duration is 549 seconds, the 95th percentile is 1,152.75 seconds and the longest retained leg lasts 3,037.25 seconds. These counts describe the retained cohort, not a balanced set of independent scenes. The source spelling `bycicle` is preserved in machine-readable labels and displayed as bicycle.

Continuity and event checks establish completed observations. Their numerical extent ends at the last recorded coordinate; no position is invented between that sample and a later event timestamp. Zero-displacement intervals and long trips remain in the data. Success of the prospective map constructor is not a training admission requirement: an observed route can supply a valid supervised motion pair even if the separate map adapter cannot reconstruct every comparable request.

Very short complete legs also remain. In the expanded image cache, 2,454 training and 527 development examples use the declared subsecond speed fallback. This preserves cohort matching between the readers and generator. It does not make those examples contain long-range speed structure. Constant images, clipped values and fallback counts are reported rather than removed to raise classification accuracy.

3.2 Separate geometry, progress and clock

For observed positions $\mathbf{x}_i$, cumulative planar distance and normalized progress are

$$a_0 = 0, \qquad a_i = \sum_{j=1}^{i} \|\mathbf{x}_j - \mathbf{x}_{j-1}\|_2, \qquad s_i = a_i / a_n. \tag{3.1}$$

Repeated positions contribute zero progress while retaining elapsed time. All nonduplicate spatial vertices are stored in double precision for piecewise-linear coordinate reconstruction. A separate 129-point equal-arc summary conditions the neural network. Compressing that condition does not compress the route decoder itself.

The native clock retains quarter-second samples and an exact fractional final interval when required at synthesis. The temporal-resolution pilot compares it with 257 equally spaced relative-time points. The same number of relative points gives a long trip a coarser physical resolution, which can remove movement details before learning begins. Reconstruction checks identify this representational limit; matched trained models test whether retaining finer detail actually improves generated motion.

The temporal pilot uses 925 training and 126 development vehicle legs, with two models for each clock. The expanded-corpus comparison is a separate study in which coverage and optimization exposure both increase.

3.3 Two image views with a common source

The spatial view starts only from ordered XY observations. Frozen coordinate bounds, the recursive Hilbert mapping and squeeze/mirror procedure produce the scalar sequence used for GASF, GADF and MTF. Observation order is preserved;

points are not sorted by Hilbert value. No speed, acceleration or duration column is appended to the spatial reader. Position repetition and traversal can still affect its image through the sampling pattern.

The speed view derives one-second arc increments from the same XY/time observations and summarizes them at 128 relative sequence positions, with a fixed 35 m/s normalization scale. It predicts the same participant labels through a separate reader. The two views are deterministic transformations of one observation, with different quantization and temporal losses. Their effects on synthesis are evaluated through the generative comparison. Chapter 5 specifies their exact construction and generative use.

3.4 Prospective reference requests

The final request set contains eight fresh SUMO episodes organized into four non-bus OD-parent cohorts and paired departure windows. Each manifest fixes 24 public bus services and 19 requests for each other class. Non-bus plans are complete single-leg requests without internal activity parking. Their source parents are disjoint from training and earlier evaluated non-bus cohorts. The public bus templates and map remain shared.

The generator sees endpoints, declared permissions and ordered public bus stop anchors. It does not receive the reference's interior realised route. The reference simulator uses its frozen public plans and background demand, while the generator samples from a network-only catalog. A different permitted route can be a valid outcome. Catalog membership of the reference route, geometric support and accepted physical references are separate coverage quantities.

All requested identities remain in the denominator even when simulation extraction, map construction or a particular metric lacks support. No easier request is substituted after reading results. New simulator realizations provide prospective same-map evidence, not physical-road measurements or independent city-level replication. The detailed fit, sampling and uncertainty rules follow in Chapter 6.

Chapter 4

Road-conditioned synthesis of complete journeys

4.1 The conditional generation task

The output is a complete timed journey in the form `time,id,x,y,type`. A request supplies a road network M, an origin and destination, a participant class c, and the vehicle's declared road permissions. A scheduled bus additionally supplies its ordered public stop anchors. Intermediate reference coordinates, realised arrival time and future speeds are unavailable. The system samples a permitted route R, a duration T, and a motion realization along that route. A new random draw can change the route, the motion, or both.

Let $R(s)$ interpolate a constructed route at arc distance $s \in [0, L]$, and let $p(t)$ denote normalized progress. The complete system factors its sampling distribution as

$$\begin{aligned} R &\sim p_\rho(\,\cdot \mid M, \mathrm{OD}, c, \mathrm{stops}),\\ T &\sim p_{\mathrm{dur}}(\,\cdot \mid L, c),\\ h &\sim p_\eta(\,\cdot \mid R, T, c),\\ p &\sim p_\theta(\,\cdot \mid R, T, c, h), \qquad X_i = R\big(Lp(t_i)\big). \end{aligned} \tag{4.1}$$

The optional condition h represents the learned spatial and speed views described in Chapter 5; the baseline omits it. Its distribution is fitted using training trajectories and sampled for new requests. No image of the unknown future is required at synthesis. The main learned trajectory component remains a temporal diffusion model, rather than an image generator followed by inversion.

The factorization assigns responsibilities explicitly. The road policy learns preferences among constructed alternatives. The duration prior supplies a travel-time distribution. Motion diffusion learns how progress unfolds within that duration.

The deterministic decoder supplies route adherence and completion. Neither static road support nor a shared duration result can therefore be credited to GAF. The system does not jointly evolve traffic signals, surrounding agents or collision avoidance.

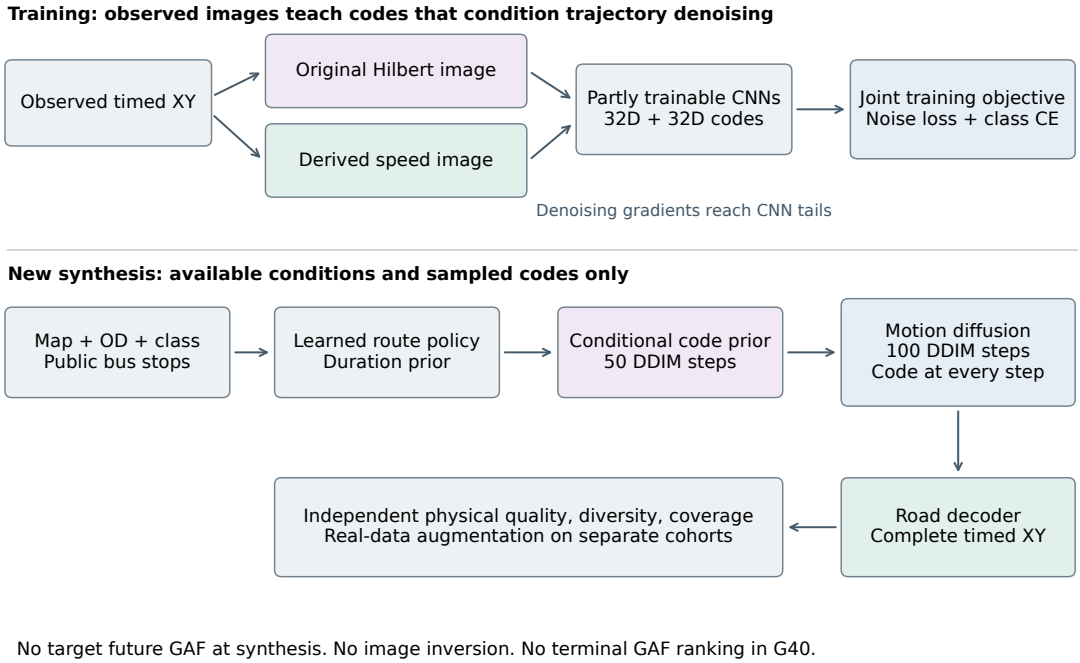


Figure 4.1. Implemented training and prospective synthesis. Observed training trajectories produce the two image views and generative conditions; new synthesis samples conditions from a training-only prior. Route choice, duration, motion diffusion and coordinate decoding have distinct responsibilities. G40 adds a learned condition throughout denoising; it does not apply a classifier gradient at a selected sampling step or rank the completed samples by GAF score.

Training observations legitimately supply paired routes and motion targets. Prospective synthesis instead constructs routes from the public request and map. An observed validation route can support an explicitly labelled reconstruction diagnostic, but cannot repair an unsupported prospective request. The generator and evaluator therefore have different information interfaces even when their data come from the same simulator domain [6, 7].

4.2 Learning route preferences within a permitted catalog

Road legality alone leaves a choice between several valid ways of travelling from an origin to a destination. The route module first constructs a finite catalog, then assigns learned probabilities to its members. Construction uses the directed road graph and the declared SUMO vehicle permissions. A category label such as taxi/ride-hailing does not override those permissions. For buses, successive public stop anchors divide the route into ordered subproblems.

The constructor makes at most twelve deterministic proposals using shortest-path search with penalties that encourage alternatives. It retains at most four geometrically supported paths and discards candidates longer than 1.5 times the first valid route. Actual lane connections, internal junction geometry and endpoint positions are checked during construction. This is a bounded candidate set, not an exhaustive enumeration of the road graph. A route absent from the catalog cannot be recovered by the policy, and the true reference path need not be present for another candidate to be valid.

The policy is fitted by behavioural cloning of permitted branching decisions in training trips. At an edge e with next public goal g, it considers directed successors from which g is reachable. Each successor a has twenty static features: current and successor coordinates, goal coordinates and relative goal vectors, edge length, speed limit, lane count, turn and goal alignment, remaining shortest-path distances, excess path cost and a goal-arrival indicator. Their physical scales are fixed. Neither a realised future speed nor a reference travel time enters these features.

A multilayer perceptron with widths $20 \to 128 \to 64 \to 5$ produces a utility for each class. Selecting the requested class output and masking unsupported successors gives

$$\pi_\rho(a \mid e, g, c, M) = \frac{\exp u_{\rho,c}(f(e, a, g, M))}{\sum_{b \in \mathcal{A}(e,g,M)} \exp u_{\rho,c}(f(e, b, g, M))}. \tag{4.2}$$

Cross-entropy fitting samples a class, parent, leg and branching decision uniformly at each level. The supported training subset contains 23,696 legs and 546,363 branching decisions; the corresponding development subset contains 5,748 legs and 132,341 decisions. Transitions with only one reachable successor do not supply a classification decision. Two fixed neural fits are retained, alongside linear and shortest-cost controls in the route study.

For a catalog route r, the score is the sum of its branching-choice log probabilities, with the goal updated at ordered stop anchors. Normalizing these scores over

the available catalog yields

$$p_\rho(r \mid M, \mathrm{OD}, c, \mathrm{stops}) = \frac{\exp\Big\{\sum_{j\in\mathcal{B}(r)} \log \pi_\rho(e_{j+1} \mid e_j, g_j, c, M)\Big\}}{\sum_{r'\in\mathcal{C}} \exp\Big\{\sum_{j\in\mathcal{B}(r')} \log \pi_\rho(e_{j+1} \mid e_j, g_j, c, M)\Big\}}. \tag{4.3}$$

There is no additional length normalization. This is a catalog-normalized behavioural policy, not a fitted recursive-logit equilibrium model; recursive route-choice methods provide relevant background [54]. An unsupported policy evaluation uses a recorded uniform fallback over the valid catalog. It does not replace the catalog with the reference route.

Route quality is consequently assessed through route-use distributions, coverage and alternative-path diversity, together with the generated motion. Pointwise disagreement with one reference route is not treated as an error whenever several alternatives are permissible. In the G40 comparison, the route policy and its random keys are fixed across all arms. That comparison isolates the motion contribution of GAF conditioning; it does not test a new GAF-dependent route policy.

4.3 Representing motion on physical time

For a training trajectory, let $\ell_i = \|X_i - X_{i-1}\|_2$, $s_i = \sum_{j\le i} \ell_j$ and $p_i = s_i/L$, where $L = s_n > 0$. Repeated positions give repeated progress values and therefore retain stopped intervals. All nonduplicate spatial vertices are stored for exact piecewise-linear reconstruction; a separate set of 129 equally spaced arc positions provides the network's route description. Compressing the conditioning input therefore does not compress the coordinate decoder.

Two temporal representations are compared. The relative-clock version uses 257 points uniformly distributed over $[0, T]$. The physical-clock version uses $0{,}0.25{,}0.50, \ldots, T$, appending the exact final time when it is not a multiple of 0.25 seconds. Its number of intervals varies with duration. The distinction matters because a fixed number of controls gives a long trip much coarser physical resolution than a short one. Preserving fine resolution is a representation choice to be tested in generation, not proof of better generated motion.

With $\Delta t_i = t_i - t_{i-1}$, define the dimensionless relative speed and its transformed target as

$$r_i = \frac{p_i - p_{i-1}}{\Delta t_i / T}, \qquad u_i = \log(1 + r_i), \qquad z_{0,i} = \frac{u_i - \mu}{\sigma}. \tag{4.4}$$

Thus $r_i = 1$ is constant progress at the journey's average rate, and $r_i = 0$ represents a stop. The logarithm moderates the upper tail without removing zero motion. Training-only μ and σ assign equal mass to each trajectory and weight its intervals by $\Delta t_i/T$. Both clock versions use this target definition, with their own fitted normalization. They form a matched comparison on the current network, rather than a byte-for-byte reproduction of an earlier model.

4.4 Generating duration without future information

Within each class, a ridge regression relates log travel time to centred log route length. For training journey j of class c,

$$\log T_j = a_c + b_c(\log L_j - \bar{\ell}_c) + e_j, \qquad T^* = \exp\{a_c + b_c(\log L - \bar{\ell}_c) + e^*\}, \quad (4.5)$$

where e^* is sampled uniformly from that class's training residuals. The slope has ridge penalty 10^{-3}; the intercept is unpenalized. Residual sampling retains variation beyond the conditional mean without imposing a Gaussian residual law. A numerical lower bound gives $T = \max(T^*, 0.25, L/55)$, with metres and seconds as the physical units. This is a broad duration guard, not a class-specific speed limit or a guarantee about instantaneous speed.

In the temporal-resolution pilot, both diffusion versions and the empirical comparator share the same prior and request-keyed draws. G40 instead holds the expanded training-side prior fixed across all eight arms, with matched route and duration draws. Consequently their unselected duration distributions are paired by construction. Differences in generated speed variation can be attributed to the motion mechanisms under this shared prior, whereas a common duration result cannot establish a diffusion advantage. Terminal feedback may select different durations from the shared pool; that is a selection effect and is reported as such.

4.5 A masked temporal denoiser

The base denoiser is a one-dimensional U-Net with channel widths 64, 96 and 128 and 891,969 parameters. G40 registers an additional 24,832-parameter condition adapter; the B arm bypasses it. The separately trained image encoders are accounted for in Chapter 5. Two downsampling stages, skip connections and nearest-neighbour upsampling connect the resolutions. Residual convolutional blocks use dilations 1 and 2, with dilations 1, 2, 4 and 8 in the bottleneck. Each block also receives a masked global feature summary, allowing information to pass beyond its local receptive field.

The conditioning vector adds a sinusoidal diffusion-step embedding, a learned class embedding, a convolutional route embedding and a five-component context embedding. The context contains normalized starting coordinates, log route length, log sampled duration and log mean travel speed. Route coordinates are expressed relative to the start and scaled by 1,500 metres. Three channels describe each interval's physical midpoint, relative midpoint and actual width. These allow the network to distinguish physical time from relative trip position. Bus stop times and background traffic states are not separate network inputs; geometric passage through a stop does not enforce a dwell there.

Sequences are padded to a multiple of four. The mask is reapplied through the network, pooling divides by the number of valid entries, and normalization acts across channels at each position rather than across padded time. Padding therefore has no semantic role as a stationary suffix. Batches are sampled before length sorting; sorting and microbatching manage memory without removing long journeys.

Following noise-prediction diffusion [4], a uniformly sampled noise index k corrupts a target by

$$z_k = \sqrt{\bar{\alpha}_k} z_0 + \sqrt{1 - \bar{\alpha}_k}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \tag{4.6}$$

on valid intervals. The 1,000-step schedule increases β_k linearly from 10^{-4} to 0.02, with $\bar{\alpha}_k = \prod_{j \le k}(1 - \beta_j)$. For an effective batch of B trajectories, the loss is

$$\mathcal{L} = \frac{1}{B} \sum_{b=1}^{B} \sum_{i} \frac{\Delta t_{b,i}}{T_b} \left[\epsilon_{b,i} - \epsilon_\theta(z_{b,k}, k, R_b, c_b, T_b, h_b)_i\right]^2 . \tag{4.7}$$

Padding has zero weight. Physical-time weighting within a trajectory prevents the fractional final interval from receiving a full interval's influence; averaging trajectories prevents long trips from dominating merely through their larger point counts.

4.6 Sampling and coordinate decoding

Generation uses 100 deterministic DDIM updates [5] between equally spaced indices of the training schedule. At a current index k, the predicted clean target and next state at $j < k$ are

$$\hat{z}_0 = (z_k - \sqrt{1 - \bar{\alpha}_k}\, \hat{\epsilon})/\sqrt{\bar{\alpha}_k}, \tag{4.8}$$

$$z_j = \sqrt{\bar{\alpha}_j}\, \hat{z}_0 + \sqrt{1 - \bar{\alpha}_j}\, \hat{\epsilon}. \tag{4.9}$$

The final update uses $\bar{\alpha}_j = 1$. Different initial Gaussian draws produce candidates; paired request and candidate identifiers reproduce duration draws across methods. Different clock dimensions do not imply identical noise arrays.

After undoing standardization, the decoder clips u_i to [0,8] and constructs nonnegative progress mass:

$$m_i = [\exp\{\mathrm{clip}(\mu + \sigma \hat{z}_{0,i}, 0, 8)\} - 1]\, \frac{\Delta t_i}{T}, \qquad \hat{p}_i = \frac{\sum_{j \le i} m_j}{\sum_j m_j}. \tag{4.10}$$

The initial progress is zero and the final progress is one. If total mass is below 10^{-12}, the decoder falls back to constant progress in physical time. Negative-value truncations, upper clipping and this fallback are counted rather than hidden. The

normalization couples the whole journey: a local mass change can alter progress elsewhere. It guarantees endpoint completion for a decoded path, but does not guarantee realistic acceleration or stop timing.

Coordinates are interpolated along all retained vertices of the constructed map path. After route selection, the motion model has no separate lateral control or lane-change process. For the relative-clock model, progress is first interpolated onto the same quarter-second output clock; the exact fractional endpoint remains. Thus both methods deliver comparable timestamped coordinates. Permitted lanes, directed connections and source stop anchors are handled by the route constructor. This geometric constraint is shared with the empirical baseline and all GAF selectors. It cannot be credited to diffusion or feedback, and it does not imply vehicle-body clearance, signal compliance or interactive traffic safety.

The baseline generates complete journeys using class, route and duration alone. The GAF-conditioned variants retain the same target, physical clock, sampler and decoder, while adding a sampled condition at every denoising step. Chapter 5 describes how that condition is learned and generated without future trajectory access. Earlier terminal-selection experiments remain separate controls on this synthesis system.

Chapter 5

GAF representations as generative conditions

5.1 Two representations of the same journey

Two image representations summarize each training journey. The spatial view uses ordered Hilbert coordinates and GAF channels following trajectory-recognition methods [1, 2, 12]. The speed view applies image transforms to the journey's derived speed sequence. Separate CNNs are trained on the same five participant classes; their encoders subsequently provide conditions for trajectory diffusion.

For a scalar sequence $a_1, \ldots, a_q$, let $u_i \in [0,1]$ be its normalized value and $\phi_i = \arccos u_i$. The two angular fields used here are

$$G_{ij}^{\text{sum}} = \cos(\phi_i + \phi_j), \qquad G_{ij}^{\text{diff}} = \sin(\phi_i - \phi_j). \tag{5.1}$$

A third channel is a Markov transition field. If b_i denotes a sequence value's bin and P_{ab} is the row-normalized count of adjacent transitions from bin a to bin b, then $H_{ij} = P_{b_i b_j}$. The angular construction follows time-series imaging [10]. For brevity, "GAF feedback" refers to the implemented three-channel GASF/GADF/MTF representation, including the transition channel.

5.2 Spatial Hilbert–GAF encoding

The spatial input consists only of the ordered coordinate samples. Frozen coordinate bounds map them into the unit square, with out-of-range values clipped and counted. The recursive Hilbert function assigns a scalar index to each point. Its precision parameter is 0.0009, corresponding to a 64×64 spatial grid. The temporal order is preserved: points are not sorted by Hilbert index.

The squeeze procedure compresses repeated-index runs, followed by mirror duplication when necessary, to obtain 224 values. This is not uniform time resampling

or a learned encoder. If the procedure cannot reach the required length, the case is reported as an encoding failure. The scalar sequence is mapped to GASF, GADF and an eight-bin uniform MTF. The angular channels use per-sequence scaling to [0,1], while the frozen channel bounds convert the three fields to an unsigned eight-bit RGB image. Coordinate and channel parameters are estimated from training data and held fixed for evaluation and generated samples.

This branch receives no additional velocity, acceleration, duration or participant feature. Nevertheless it is not invariant to motion. Different speeds change the number and order of sampled coordinates within quantized cells, and the run compression preserves some of that structure. Conversely, quantization and compression can map distinct motion profiles to the same image. On a fixed route, spatial feedback may therefore have little ability to rank the candidate pool even when it classifies trajectories from different routes well. Image hashes and the number of distinct spatial images in each pool make this limitation observable.

5.3 Speed encoding and its temporal limits

Speed is computed from the same coordinates and timestamps. Let $d(t)$ be the piecewise-linear interpolation of cumulative planar distance. At integer seconds, the speed summary is

$$v_j = d(j+1) - d(j), \qquad j = 0, \ldots, \lfloor T \rfloor - 1. \tag{5.2}$$

These one-second increments are expressed in metres per second and resampled to 128 equally spaced relative sequence positions. The complete-cohort encoder interpolates whatever complete one-second increments are available. For a valid leg shorter than one second, it repeats the mean arc speed over all 128 positions. This declared fallback retains 2,454 training and 527 development legs in the expanded cohort.

The angular normalization is $u_i = \mathrm{clip}(v_i/35,0,1)$, with a common absolute scale across classes. Values above 35 metres per second are counted. The MTF uses eight fixed bins on this normalized interval. The two angular channels are mapped from $[-1,1]$ to $[0,1]$; the resulting image is resized from 128×128 to 224×224. Both representations then use the same ImageNet channel normalization before entering a ResNet18 [11].

The speed image retains absolute speed scale but compresses temporal detail. One-second averaging and the 128-point summary do not preserve all quarter-second acceleration. Explicit duration and location are not supplied to this classifier. The representation summarizes whole-trip speed structure.

5.4 From a generated journey to the two image views

Figure 5.1 makes the implemented representation concrete using an actual completed diffusion sample. The top row starts from its ordered Hilbert sequence; the lower row starts from speed computed from the same timestamped coordinates. Each row ends in the three channels that the corresponding class reader receives. The image is computed from the generated journey.

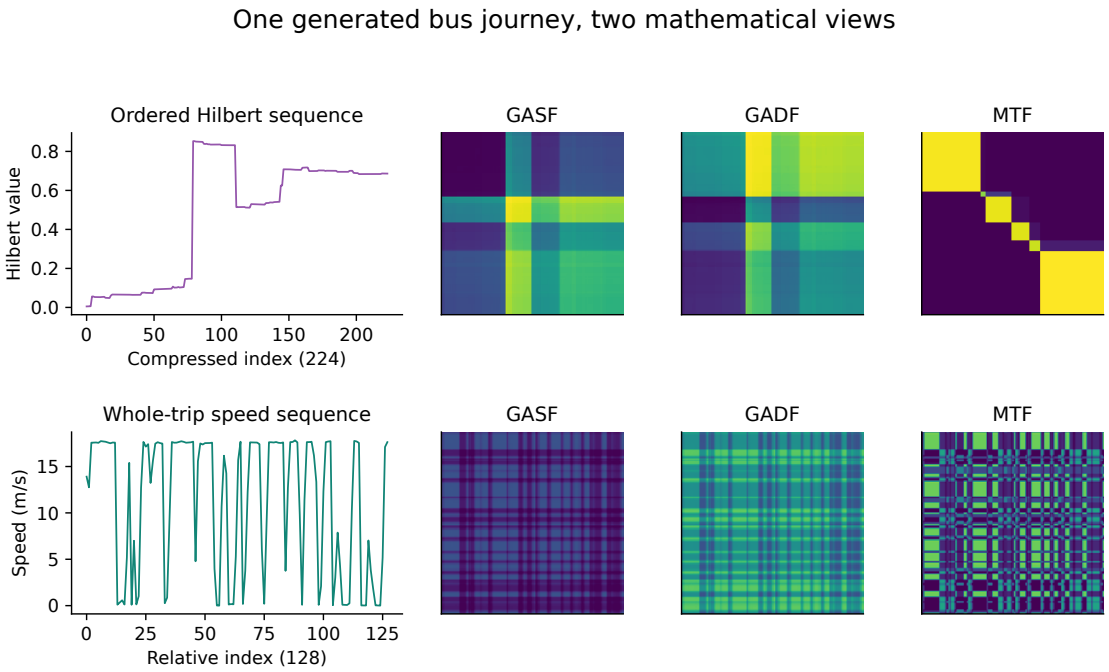


Figure 5.1. Exact representation channels of one earlier G36 generated bus journey, used to illustrate the unchanged encoding. The spatial row uses Hilbert run compression and frozen channel conversion; the speed row uses the 128-value absolute-speed summary and the implemented resize. Each field is shown with a common display range of [0,1] after its implemented conversion. These are channel intensities, not a common physical scale across the two encoders. No per-image contrast enhancement is applied.

The two sequences have different coordinates on their horizontal axes. The spatial sequence index follows run compression, whereas the speed sequence index follows relative-time resampling. Their off-diagonal image patterns relate pairs of sequence positions. Consequently, a visible image difference need not imply a better trajectory: it first establishes that the representation distinguishes the inputs. The empirical question is whether the resulting representations provide useful constraints for generating motion. Chapter 7 links complete generated journeys and

their output image views to independent population-level quality statistics. Output images shown there are visualizations, not future observations supplied to synthesis.

5.5 Recognition and the limits of score-based feedback

The spatial and speed readers are separately trained on the same participant labels. Their different inputs remain separate: a speed sequence is never appended to the spatial classifier. Recognition measures participant-related information in recorded journeys. Its use for synthesis is evaluated through the conditioning experiments and independent motion metrics.

The earlier terminal experiments therefore compare matched candidates. Their scores are the floored log target-class probabilities,

$$s_v(X, c) = \log \max\{f_v(c \mid E_v(X)), 10^{-8}\}, \qquad v \in \{S, V\}. \tag{5.3}$$

Single-view policies use one score and the dual-view policy uses their equal-weight mean. The fixed candidate pool and common random keys prevent a larger sampling budget from being mistaken for a feedback advantage. The later route-and-duration control distinguishes global reweighting from reweighting that preserves those marginals. For that diagnostic, weighted distributions are the mechanism endpoint: an individual coupled random selection can differ even when the marginals are preserved exactly.

A second precursor fits the motion model against frozen reader features. This tests whether a class representation can provide a useful training target, but the reader itself cannot adapt its representation to trajectory reconstruction. The final experiment changes that interface: GAF-derived codes become sampled conditions of the generator, and the encoder's final stage is trained with the denoiser. Terminal ranking, frozen-feature alignment and learned generative conditioning are distinct interventions; their different populations are not pooled into a single effect estimate.

5.6 Generation-trained spatial and speed conditions

Let $E_S(X)$ be the spatial image and $E_V(X)$ the speed image. For each active view, a CNN produces a pooled 512-dimensional feature $f_{\psi,v}(E_v(X))$. A separate bottleneck projects it through widths $512 \to 128 \to 32$, using SiLU and a final LayerNorm:

$$h_v = P_v\big(f_{\psi,v}(E_v(X))\big), \qquad h = [h_S; h_V] \in \mathbb{R}^{64}. \tag{5.4}$$

An inactive view contributes 32 zeros. The dimensionality of the denoiser interface consequently stays fixed across single- and dual-view conditions. Each view supplies a learned global condition.

In a joint arm, the final ResNet residual stage, class head and bottleneck are trainable. The earlier convolutional stages are frozen, as are all BatchNorm running statistics. Exact cached prefix activations reduce repeated image processing: their pooled teacher features and cached spatial image bytes were checked against the existing training artifacts. The input encoding remains fixed.

The 64-dimensional code passes through a $64 \to 128 \to 128$ adapter and is added to the existing diffusion-step, class, route and context embedding. Every residual denoising block receives the resulting condition at every sampling step. The adapter's final layer is initialized to zero, reproducing the common starting denoiser exactly before adaptation. The added condition therefore starts without an arbitrary perturbation of the pretrained generator.

For active views $\mathcal{V}$, the joint fitting objective is

$$\mathcal{L}_{\text{joint}} = \mathcal{L}_{\text{noise}}(\theta, \psi, P) + \frac{0.1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \text{CE}(c, g_{\psi,v}(f_{\psi,v}(E_v(X)))) \,. \tag{5.5}$$

The first term is the physical-time-weighted noise-prediction loss in Equation 4.7. The class heads $g_{\psi,v}$ act on the pooled CNN features, not on the 32-dimensional bottlenecks. The auxiliary term retains participant recognition as an explicit training task while denoising can adapt features that help reconstruct motion. The fixed coefficient is not chosen from the new generation results.

Gradients update the CNN tail and projection weights. They do not differentiate the discrete Hilbert indexing, squeeze procedure or byte conversion with respect to coordinates. This distinction retains the discrete image construction without requiring a smooth mathematical inverse of its images. It also distinguishes the method from applying a classifier gradient to an intermediate noisy trajectory [25, 27]. The experiment contains no specially chosen guidance timestep.

Joint adaptation updates the encoder through both denoising and classification objectives. The frozen-reader comparison measures their combined effect. Recognition scores reported for the original readers refer to their pre-adaptation checkpoints.

5.7 Sampling conditions without an observed future

Observed training trajectories supply the codes used in joint fitting. A conditional prior learns their distribution and supplies codes for prospective synthesis.

After motion fitting, the final exponential-moving-average encoder is frozen and

all training codes are extracted. Each active dimension is standardized by its training mean and standard deviation, with a standard-deviation floor of 10^{-3}. A conditional latent diffusion model then fits the distribution of these codes given class, the route's 129 two-dimensional points and the existing five-component route/duration context. It uses a 256-dimensional hidden state, four residual MLP blocks, the same 1,000-index linear noise schedule, and a fixed 5,000-update budget. Its active-dimension noise loss does not train on inactive zero dimensions.

At synthesis, 50 DDIM updates sample a code from this prior; 100 separate DDIM updates generate motion conditional on that code. Route, duration and motion-noise keys are matched across the experimental arms. The code prior has its own request-keyed random stream and cannot advance the motion-noise stream. The sampled code is held fixed through the motion denoising process. The image encoders and class heads need not run during prospective synthesis.

The encoder, conditional diffusion decoder and learned code prior follow the general construction of Diffusion Autoencoders [53]. Here the construction operates on road-conditioned variable-length motion and prescribed spatial and speed image views. The latent prior samples the condition, and the temporal decoder generates progress.

The code prior is checked through training-side interventions and prospective code-range statistics. Pilot settings and numerical diagnostics are reported in Appendix A.

5.8 Representation controls

Table 5.1 separates the main method from its controls. Every arm continues from the corresponding same-seed G39 baseline and uses the same motion target and sampling budget. A zero-initialized adapter and identical training streams make the starting point explicit.

The shuffled arm preserves its wrong-trip associations when extracting codes and fitting the prior; it is not silently repaired at deployment. Its auxiliary labels remain unchanged. The shuffle breaks exact example pairing while deliberately retaining class and approximate trip scale; it need not remove every route or service correlation. The class-centroid arm learns projections of fixed training class means and uses a deterministic class-code lookup at sampling. The baseline and centroid arms have no learned code prior. RAW uses different pretraining and encoder architecture, so it is an alternative representation control, not a perfectly isolated test of angular transforms alone. Parameter counts and computation accompany its comparison.

Evaluation compares direct trajectory outputs under matched sampling budgets. Single-view and joint conditions measure the contribution of each representation; frozen, raw-series, shuffled and centroid conditions provide complementary controls.

Table 5.1. G40 conditions and their roles. B bypasses the code adapter. All other arms use the same 64-dimensional interface; an inactive view is zero.

Arm	Condition source	Question isolated
B	No additional code	Class/route/duration diffusion baseline
F-SV	Frozen spatial and speed CNNs; learned projections	Is fixed recognition sufficient as a condition?
J-S	Joint spatial CNN tail and projection	What does the spatial view contribute?
J-V	Joint speed CNN tail and projection	What does the derived motion view contribute?
J-SV	Both jointly adapted views	Does combining views help beyond either alone?
J-RAW	Separate raw position and speed encoders	Is an image representation necessary for the observed gain?
J-SH	Same-class, approximately length/duration-matched wrong-trip images	Does trip-specific pairing matter?
C	Training class-mean CNN features	Is extra class information sufficient?

Physical distributions, diversity, coverage and downstream augmentation characterize the resulting synthetic data.

Chapter 6

Experimental protocol and independent evaluation

The final experiment tests complete trajectory synthesis with learned GAF-derived conditions. Earlier studies establish the source domain, temporal representation, route module and feedback controls. Their observations remain available, but are not pooled with the new comparison. Methods, budgets, request selection and endpoints for G40 were fixed before examining its prospective generation outcomes.

6.1 Learning observations and information boundaries

All G40 motion arms use the same expanded native-demand corpus: 29,895 training legs from 24,073 parents, with the original physical clock and in-leg stops retained. The 7,230 development legs are not used to choose G40 checkpoints or coefficients. The older exposed test and previous evaluation cohorts are not reopened for method selection. Train-only quantities include coordinate and channel parameters, motion normalization, duration regression, class feature means and code-prior standardization.

The original G35 spatial and speed readers are self-trained ResNet18 models with ImageNet initialization, 50 epochs, batch 64, AdamW learning rate 2×10^{-4} and weight decay 10^{-4}. Their fixed final epoch, inverse-square-root class weights and unaugmented images match the recorded protocol. The RAW controls use the same labelled cohort, 50 epochs and final-checkpoint rule, but a separate one-dimensional architecture without ImageNet pretraining. Epoch and label matching therefore do not imply architectural or computational equivalence.

Training uses an observed route and duration paired with its motion target. Prospective generation receives only a map-derived catalog, class, declared road

permissions and available public stop anchors. Duration and latent condition are sampled. The public generation entry point rejects undeclared trajectory or code inputs. Reference coordinates, accepted-reference status and realised travel times cannot repair a generation request. The training-side reconstruction and condition-intervention diagnostics are kept outside prospective synthesis.

6.2 Fixed model fits and matched sampling

The eight arms in Table 5.1 each have seeds 42 and 43. For a seed, all arms start from the corresponding completed G39 B final EMA model, which was trained from initialization for 10,000 updates on the same motion corpus. G40 adds 5,000 motion updates to every arm. The 1,000-update pilots do not supply warm starts or selectable checkpoints.

Motion fitting uses effective batch 64, uniform class sampling followed by a training leg within class, FP32 arithmetic and physical-time-weighted noise loss. Selected examples are length-sorted into microbatches capped at 32 rows and 32,768 padded temporal values. AdamW uses motion learning rate 2×10^{-4}, encoder learning rate 5×10^{-5}, weight decay 10^{-4}, global gradient clipping at one and EMA update weight 0.001. Frozen prefixes and all BatchNorm running statistics remain unchanged. Every comparison uses final EMA weights. No new reference result selects an earlier training step.

Common initialization, valid-token totals, normalizers and final training random states are checked across the arms for each seed. Code sampling uses a separate random stream. This verifies declared pairing without asserting that arbitrary GPU reruns are bitwise reproducible. Training and sampling cost, encoder parameters and peak allocated memory are reported separately from quality.

Each active code prior is fitted for 5,000 updates, with balanced batch 256, AdamW learning rate 2×10^{-4}, weight decay 10^{-4}, clipping at one and EMA weight 0.001. B uses no code; C uses a deterministic class lookup. The remaining six arms require twelve learned priors across the two seeds. No code prior uses development or prospective reference trajectories.

Each supported request receives four direct trajectory draws per arm and seed. Route, duration and initial motion-noise keys are common across arms. Each learned-prior draw uses 50 latent DDIM updates and every motion draw uses 100 temporal DDIM updates. The first draw, K1, is the primary delivered-data comparison. Equal-weight distributions over all four draws and within-request diversity are secondary. There is no terminal GAF selection in G40, and no reference-dependent replacement of a failed slot. Equal motion-forward counts do not imply equal total computation because the learned-prior arms add latent sampling.

6.3 Eight prospective SUMO references

The final reference set comprises episodes 400201–400208 on the unchanged public MoST network. Four non-bus source-parent cohorts use selection seeds 40001–40004. Each cohort is simulated in two departure windows beginning at 18,000 and 32,400 seconds, with 600-second target departure spans and the frozen background-demand rules. A manifest contains 100 requests: the 24 public bus services and 19 source plans for each of the four other classes. All eight manifests are frozen before the first new outcome.

Non-bus source parents are disjoint from training and from the G36/G38/G39 evaluation cohorts. The two windows within a cohort deliberately reuse its underlying non-bus plans. Public bus service templates are shared across cohorts and with the learning domain. Four OD cohorts define the grouping for uncertainty descriptions; each has two departure windows.

SUMO exports are sampled at 0.25 seconds. Reference extraction applies its fixed continuity, arrival and service-event rules; catalog construction independently applies geometric support rules. Every scene retains the original 100-request denominator, and physical measures report the intersection appropriate to their support. Unsupported requests and rejected references are retained as coverage outcomes without substitute trajectories. The generator does not see the intersection before sampling. Exact reference-route membership in the catalog is reported separately; absence of the reference route does not by itself make a different permitted path incorrect.

6.4 Physical quality beyond travel distance

The primary endpoints are native-resolution speed and acceleration distribution distances. An interval has speed $v_i = \|X_i - X_{i-1}\|/\Delta t_i$; successive speed differences are divided by $(\Delta t_i + \Delta t_{i-1})/2$ for acceleration. Coordinates and timestamps are processed identically for SUMO references and model output. Simulator-internal speed fields and either GAF reader's probabilities are not physical ground truth for this comparison.

Per-class distributions are compared using the first Wasserstein distance,

$$W_1(F, G) = \int_{-\infty}^{\infty} |F(x) - G(x)| \, \mathrm{d}x. \tag{6.1}$$

Requests have equal mass inside a class; candidates divide their request's mass; native observations are weighted by represented physical time. All five classes then receive equal weight. A longer trace or a frequent private-car class therefore cannot dominate merely through its number of coordinates. Missing support is reported rather than silently averaging a smaller set of classes. Distances retain their physical units and are not added into a cross-unit composite score.

The source FCD coordinates have centimetre-level reporting precision. Differentiation can amplify quantization on a quarter-second clock. Coarser-clock movement summaries therefore remain visible alongside the original native endpoints; they do not replace them after seeing outcomes. Cumulative planar distance $d(t)$ is interpolated at integer seconds, with $v_j = d(j+1) - d(j)$ and $a_j = v_{j+1} - v_j$. Only complete one-second intervals contribute to these summaries; duration and planar distance retain the fractional endpoint. Speed and stopped-state summaries need one complete interval, acceleration and steady movement need two, and the four-second lag summary needs five. Each endpoint retains its own support count; an unsupported short journey is not replaced by a zero.

Table 6.1. Fixed per-journey motion summaries. Thresholds describe states rather than class-specific safety limits.

Summary	Definition
Duration; planar distance	Full timestamp span; sum of planar displacements
Mean; 95th-percentile speed	Mean and 0.95 quantile of one-second speed
Mean absolute acceleration	Mean of $\lvert a_j \rvert$ on the one-second clock
Stopped fraction; longest stop	Fraction and longest run with $v_j < 0.3$ m/s
Stop episodes per minute	Number of stopped runs per observed minute
Longest steady movement	Longest run with $v_{j+1} \geq 0.3$ m/s and $\lvert a_j \rvert < 0.5$ m/s^2
Four-second speed change	Mean of $\lvert v_{j+4} - v_j \rvert$

The shared route/duration draws make travel distance and duration controls rather than evidence of a new GAF duration model. Different movement distributions on those conditions remain possible. Distributional disagreement is not a framewise prediction error against a unique future: several journeys can be valid for the same request.

6.5 Coverage, diversity and statistical perspective

All classes, scenes, seeds and declared motion endpoints are retained. The primary grid has 128 model/scene cells and 640 class rows. Aggregate differences average classes, model seeds, windows and cohorts equally. A descriptive bootstrap resamples the four cohort means 2,000 times using a fixed seed. The intervals describe between-cohort variability within the shared simulator domain. The sixteen cell-level directions remain visible alongside the aggregates.

Request-level mean absolute discrepancies in physical summaries complement class-marginal distribution distances. The former preserve request correspondence, whereas the latter compare delivered populations. One stochastic reference is not the only acceptable movement for that request. The two perspectives can consequently disagree without one being a coding error.

Within-request diversity compares 129 relative-time coordinate positions and 257 relative-time progress values, as well as duration variation and effective route counts. Candidate-pair quantities are averaged within a request before averaging requests. Larger differences are not automatically better: they can reflect plausible alternatives or implausible variability. A nearest same-class training-progress comparison describes proximity to stored motion shapes, without claiming privacy or proving absence of memorization. Decoder clipping and constant-progress fallbacks are counted.

6.6 Independent synthetic-data augmentation

In the plots, "real" denotes observed SUMO journeys, not field-collected road traffic. The downstream task predicts the five participant classes from raw time series, using a network independent of the GAF readers and their weights. Its input has 257 relative-time positions with three channels: x, y and speed derived from coordinates. A 64-channel temporal convolution followed by four residual blocks with dilations 1, 2, 4 and 8 feeds concatenated mean/max pooling and a five-class head. This task tests one practical use of synthetic data; it does not assess every motion dependency or substitute for native-resolution quality.

The observed-data training subset contains 1,000 distinct parents, balanced at 200 per class by a fixed identity hash. Training-only normalization comes from these rows. Synthetic sets contain 1,000 matched identities, also balanced by class, selected without quality scores from the first two OD cohorts and both generator seeds. B, F-SV, J-S, J-V, J-SV and J-RAW share the same selected request/draw identities. A resampling control bootstraps the same 1,000 observed rows; it is distinct from the earlier empirical motion-transfer generator.

There are fifteen regimes: observed-only; synthetic-only for six model arms plus the resampling control; and observed-plus-source for those seven sources. Each has downstream seeds 4001 and 4002, giving thirty fits. All receive 2,000 AdamW updates, balanced batch 64, learning rate 2×10^{-4}, weight decay 10^{-4} and gradient clipping at one. A seed shares its initialization across regimes. The final checkpoint is fixed before loading evaluation trajectories.

The last two OD cohorts, each with both windows, supply the observed SUMO evaluation journeys. Non-bus parents are disjoint from the synthetic-condition cohorts, and no evaluation source parent is in the 1,000 observed training parents. Shared bus templates remain a limitation. Pooled macro F1, class F1, episode results and equal-parent non-bus results are retained. Both downstream seeds are shown without treating two fits as a precise population confidence interval.

The generator and its encoders used all 29,895 labelled training legs. The observed-only downstream learner uses 1,000 parents. Results therefore reflect a larger upstream label budget for the generative models. B, RAW and resampling provide the corresponding augmentation controls.

6.7 Mechanism checks and earlier evidence

Before formal generation, a fixed 500-training-parent audit checks original encoding reproduction and sensitivity to time warping, within-second perturbations and smoothing. It compares centered feature changes with physical changes without labelling the perturbations as new transport classes. Four 1,000-update motion pilots check implementation, code use and prior behaviour. Their failures are retained; the single prior recovery uses the already declared 5,000-update formal prior budget and does not refit motion or inspect new reference quality.

Before any formal generation outcomes, a final-model condition audit is also fixed. It uses 25 training parents, five per class, at noise indices 50, 250, 500, 750 and 950. Assigned code, a same-class other-parent code, zero code and sampled-prior code are compared under identical route, duration and noisy motion input. All arms and seeds are included. Zeroing a code retains the learned adapter and its biases. These noise indices are diagnostic locations, not selected guidance times. The audit measures condition use and prior mismatch on training parents.

A further read-only audit uses the recorded code magnitudes and decoder counters from every completed prospective draw. It reports fixed quantiles and exceedance of the observed training-code maximum. That maximum is a descriptive reference, not the support boundary of a learned distribution or a criterion for excluding generated data. This audit was specified before inspecting any prospective quality outcomes and adds no new fitting or selection.

The temporal-resolution pilot, source-demand study, route-choice study and earlier GAF controls retain their original populations and budgets. They motivate the final mechanism, but a favorable cell from an earlier round is not combined with a favorable G40 cell to create a new method. Fixed example selection uses episode 400201, model seed 42 and the first lexicographic request per class supported by the reference and every arm. Candidate zero supplies the displayed sample. Both image views are recomputed from those outputs, with no score-based example search or contrast enhancement.

Chapter 7

Results: synthesis, representation and data utility

This chapter evaluates complete-journey synthesis through route preferences, physical-time motion and downstream data utility. The temporal and routing controls establish the generation framework. The main comparison then measures the effect of spatial, speed and joint GAF conditions under matched route, duration and sampling budgets. Five-class numerical results are paired with complete generated trajectories and their image representations.

7.1 Evaluation population

The main comparison comprises eight model conditions and two fitted seeds, evaluated over eight prospective SUMO episodes. It produces 50,176 candidates, or 6,272 per condition. The primary endpoint uses the first direct draw. Thirty downstream fits evaluate the utility of the generated data. Artifact verification and scalar recomputation are documented in the accompanying reproducibility records.

The 800 declared episode requests yield 784 supported generated requests and 755 physical/reference-route pairs. The realised reference route occurs in the finite catalog for 476 of those pairs. Two windows reuse each OD cohort and public bus services recur across cohorts. Each episode has 98 generated requests, while accepted physical pairs range from 90 to 97. The downstream task has 384 observed references under its own extraction support. Complete coverage is retained in Appendix A.

The temporal and routing controls use their original matched populations and budgets. Their results are reported separately from the main GAF-conditioning comparison.

7.2 The physical clock establishes the motion baseline

The matched G33 pilot compares a 257-point relative-time representation, a native 0.25-second representation and empirical transfer of a training motion profile onto a generated path. It uses 925 training legs, two fitted seeds and one new SUMO reference episode. Of 100 declared requests, map construction and reference extraction each support 97, with 94 in their intersection. Every full pool contains eight draws per supported request. The pilot isolates the temporal representation at a common update budget; runtime differs between clocks.

Native acceleration W_1 falls from 0.8397/0.8319 to 0.1492/0.1575 m/s^2 across seeds 42/43, reductions of 82.2/81.1%. Empirical transfer gives 0.3245/0.3588 m/s^2. The native model improves that endpoint in all five classes for both seeds. Native-speed distance also falls in the macro, from 0.6794/0.8238 to 0.6093/0.7437 m/s, but not in every class. One-second mean-absolute-acceleration distance decreases from approximately 0.113 to 0.0661/0.0560 m/s^2, providing a related observation at a coarser clock.

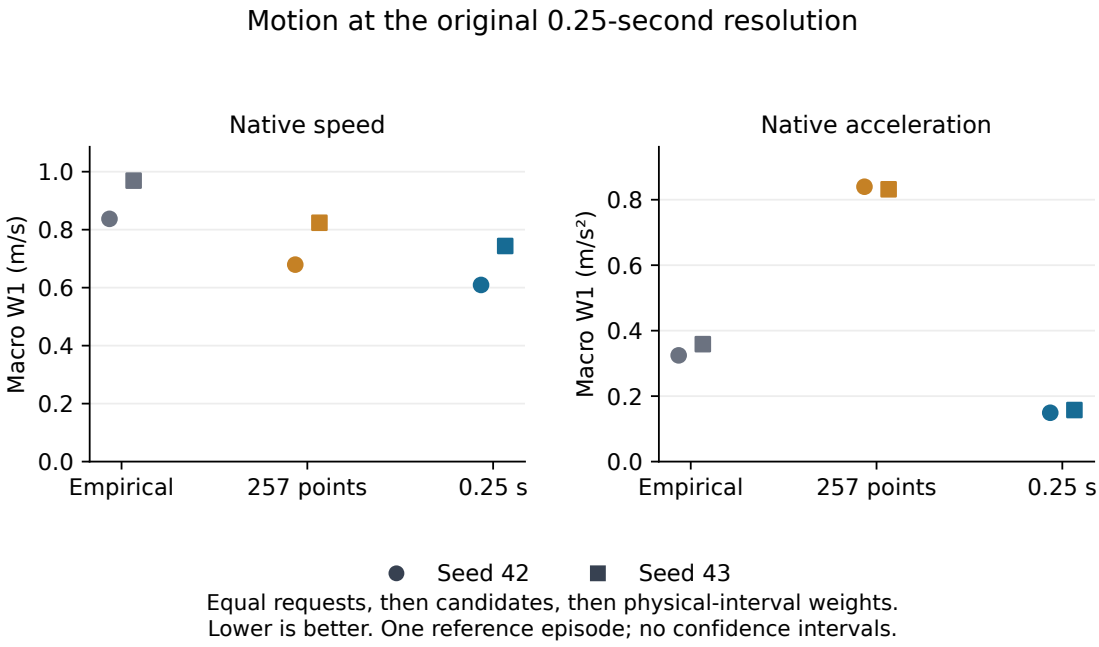


Figure 7.1. Matched temporal-resolution pilot: native-interval distributions for two diffusion clocks and empirical motion transfer. Both model seeds use the same request population and corresponding shared duration draws. The pilot uses one reference episode and motivates the clock used subsequently.

Longer-timescale structure remains difficult. Longest-stop distance worsens from

4.69/4.89 to 7.75/7.66 seconds; for buses it rises from 12.03/11.76 to 26.37/25.66 seconds. Lower acceleration discrepancy therefore does not establish correct stop organization. Native fitting takes 1,520/1,529 seconds compared with 301/319 seconds for the relative-time fits, at the same 10,000 updates. The complete physical and class tables, including adverse outcomes, remain in Appendix A.

7.3 Learning route preference changes the delivered route distribution

The G38 route experiment uses 23,696 accepted training legs and 5,748 development legs for next-edge learning. The class-equal development next-choice accuracy is approximately 96.8% for the two neural policies, compared with 94.9% for the shortest-cost rule. This is an observed branching-choice task; it is not an exact whole-route recovery score.

Two prospective SUMO episodes and two policy seeds compare distributions over a finite, geometrically constructed route catalog. Both episodes have 100 declared requests and 98 supported map conditions, with 97 and 92 physical/reference-route pairs. The exact realised SUMO route is in the catalog for 64 and 61 of those pairs. This coverage limitation is retained. A different permitted route between the endpoints is not automatically an erroneous prediction.

Averaged over the four cells, learned preference R reduces road-edge usage Jensen–Shannon divergence from 0.04000 for shortest-route choice to 0.03458, a 13.6% reduction, with the same direction in all four cells. Effective route count increases from 1.000 to 1.176. Weighted native-speed and acceleration distances decrease from 1.20327/0.17435 to 1.16728/0.17192 in their respective units. These measurements characterize route choice on the shared public network.

Uniform route sampling retains a relevant tradeoff: its speed distance is lower at 1.05113 m/s and its effective route count is 3.454, but edge-use divergence is higher at 0.10177. The neural route policy balances empirical route preference against alternative-path diversity. Subsequent G40 arms use common R draws so that GAF effects can be attributed to conditional motion changes rather than different route selection.

7.4 Both GAF views recognize classes, with different weaknesses

The complete-cohort G35 readers use the same 29,895 training legs and 7,230 development legs as the expanded motion corpus. Spatial macro F1 is 76.47% and speed macro F1 is 90.34%. Accuracies are 86.17% and 89.13%, respectively. Table 7.1 exposes class imbalance and the spatial reader's taxi/ride-hailing weakness; overall

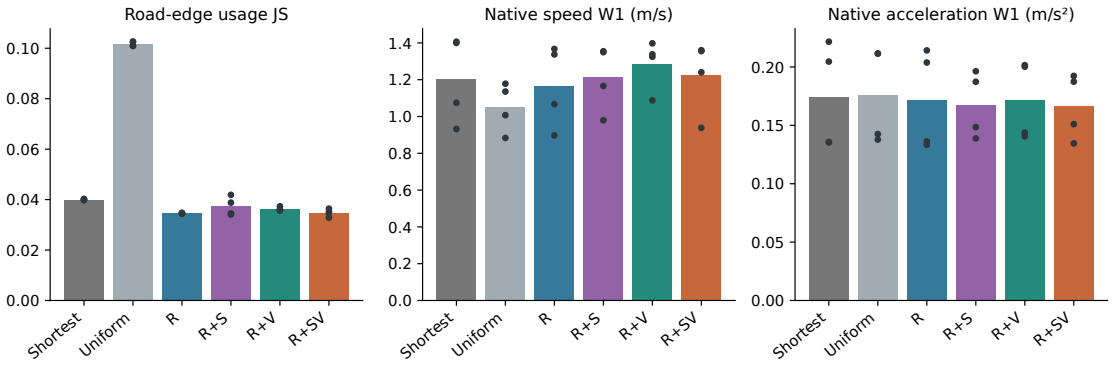


Figure 7.2. Learned route preference and route/motion distribution distances in the closed G38 study. Values summarize two episodes and two seeds; the underlying class and cell results remain archived. Lower edge-use divergence is closer to the observed distribution. The plotted points show each scene/seed cell.

accuracy alone would obscure that result. These figures describe the self-trained readers before G40 joint adaptation, not the final adapted encoders.

Table 7.1. Pre-adaptation G35 GAF readers on the same development population. F1 values are percentages. Support counts are legs, not independent simulator scenes.

Class	Development legs	Spatial F1	Velocity F1
Bus	155	95.65	100.00
Motorcycle	1591	73.57	78.90
Private car	4436	95.20	92.23
Bicycle	668	72.10	99.48
Taxi / ride-hailing	380	45.85	81.07
Macro F1	7,230	76.47	90.34

Spatial images classify private cars well but do not reliably separate the service-related taxi label. Speed images retain useful information about the same labels, without being added to the spatial classifier's input. The generative comparison evaluates how these class-related features affect motion synthesis. The source-demand comparison in Appendix A.14 and the matched earlier/current-reader comparisons in the appendix provide the data context; they are not a scaling-law claim.

7.5 Final generative conditions: native motion quality

Table 7.2 gives absolute primary distances and their paired contrasts against B. The dual-view J-SV condition changes native speed distance from 1.437941 to 1.440104 m/s and acceleration distance from 0.176001 to 0.180289 m/s^2: increases of 0.150% and 2.437%, respectively. The acceleration increase occurs in all four cohort means. Only four of the sixteen scene/seed cells improve that endpoint. Equal weighting over all four generated draws also leaves both dual-view distances above baseline.

Table 7.2. Prospective direct-draw quality on eight episodes, two model seeds and all five classes. Wins count scene/seed means; lower distances are better.

Arm	Speed W_1	Acceleration W_1	Speed wins	Accel. wins
B	1.43794	0.17600	–	–
F-SV	1.43543	0.17611	8/16	10/16
J-S	1.43909	0.17694	6/16	7/16
J-V	1.43740	0.17921	6/16	7/16
J-SV	1.44010	0.18029	8/16	4/16
J-RAW	1.44368	0.18068	7/16	4/16
J-SH	1.43449	0.17450	10/16	11/16
C	1.43850	0.17578	10/16	8/16

Spatial J-S alone changes speed by +0.080% and acceleration by +0.536%. Speed-only J-V changes them by −0.037% and +1.821%. Frozen F-SV has a small speed reduction and essentially unchanged acceleration. RAW also increases both primary distances. Across these interfaces, the differences are small and mostly adverse.

Same-class wrong-trip J-SH reduces speed and acceleration distances by 0.240% and 0.851%, respectively; its descriptive acceleration interval spans zero. Class-centroid C remains close to B. Correctly paired joint conditions therefore offer no consistent advantage over these simpler controls. J-SH preserves class and approximate trip scale, including associated route and service structure.

7.5.1 All five classes and longer-timescale motion

Figure 7.4 shows the class-specific responses. J-SV reduces private-car acceleration distance by 3.28%, while increasing it for buses, motorcycles, bicycles and taxi/ride-hailing by 1.60%, 1.97%, 5.68% and 5.26%. Spatial-only J-S has its largest acceleration increase for private cars; speed-only J-V increases bicycle and taxi distances. The effects depend on both representation and participant class.

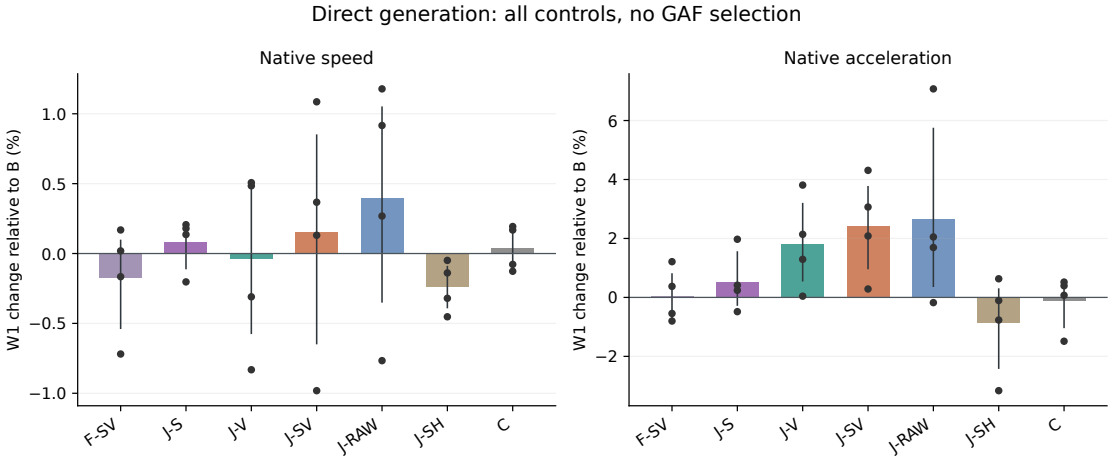


Figure 7.3. G40 first-draw native motion distances relative to B. Bars are class/seed/window/cohort-equal mean differences divided by the overall B mean; points are the four cohort differences with the same denominator. Whiskers are descriptive cohort-bootstrap intervals. Lower is better. Intervals summarize variability across the four same-network cohorts.

The longer-timescale summaries show a similar tradeoff. J-SV reduces longest-steady-motion distance by 2.26% and stop-episode-frequency distance by 1.56%, while increasing longest-stop distance by 1.52%, one-second mean-absolute-acceleration distance by 3.32%, and four-second speed-change distance by 1.67%. The differences extend across native and coarser physical clocks.

Duration is identical across arms by the paired sampling design. Its shared distance of 178.63 seconds is therefore a limitation of the common duration component, not a GAF effect. Generated planar-distance discrepancies are also almost unchanged; small differences arise because finite-time planar chords need not sum to the exact constructed route arc length. The appendix provides all ten physical summaries, request-paired discrepancies and four-draw results.

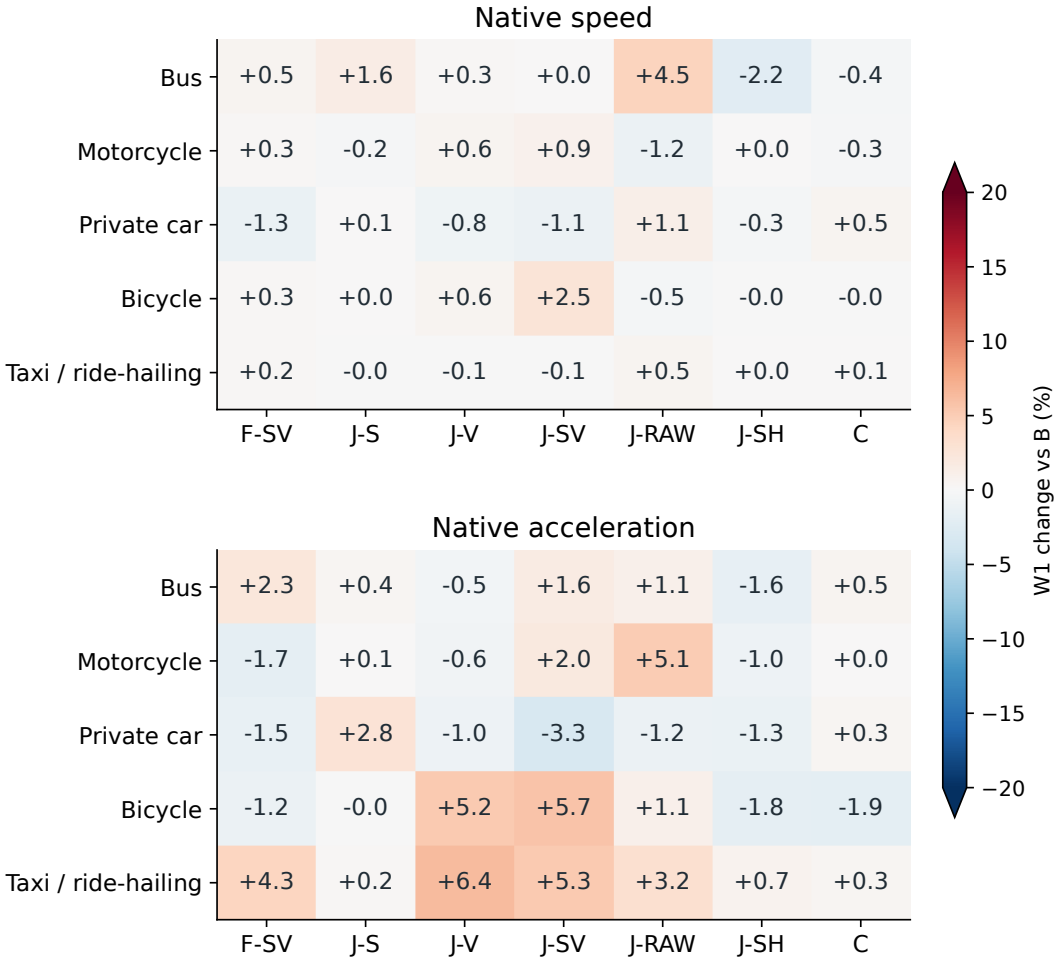


Figure 7.4. All five classes and seven condition contrasts against B: native speed and acceleration. Values are percentage changes in each class's distance, averaged over both model seeds and all eight episodes. Blue is lower distance. The fixed color range is shared; printed values are not clipped. Absolute values and both seeds are in the appendix.

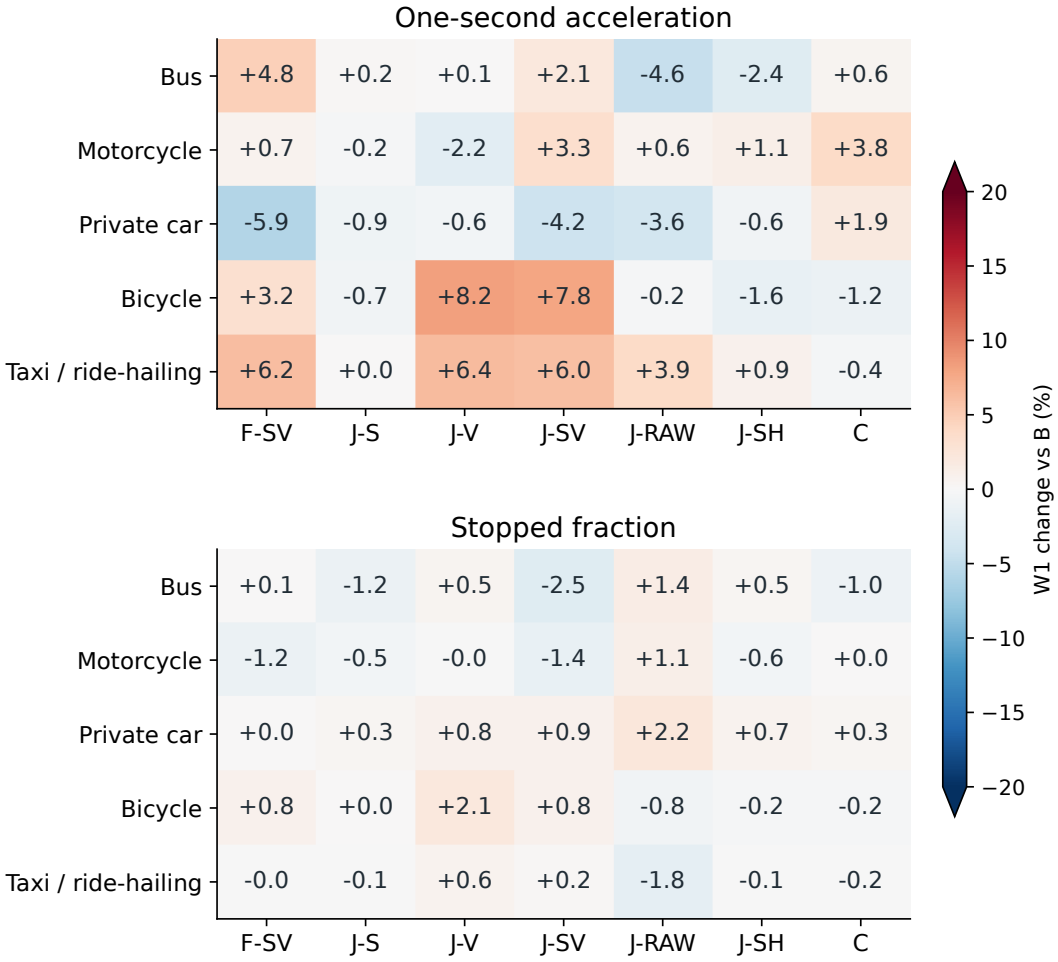


Figure 7.5. All-class one-second mean absolute acceleration and stopped-fraction distribution distances. These coarser-clock summaries complement the native endpoints. Per-endpoint support is retained in the evidence files.

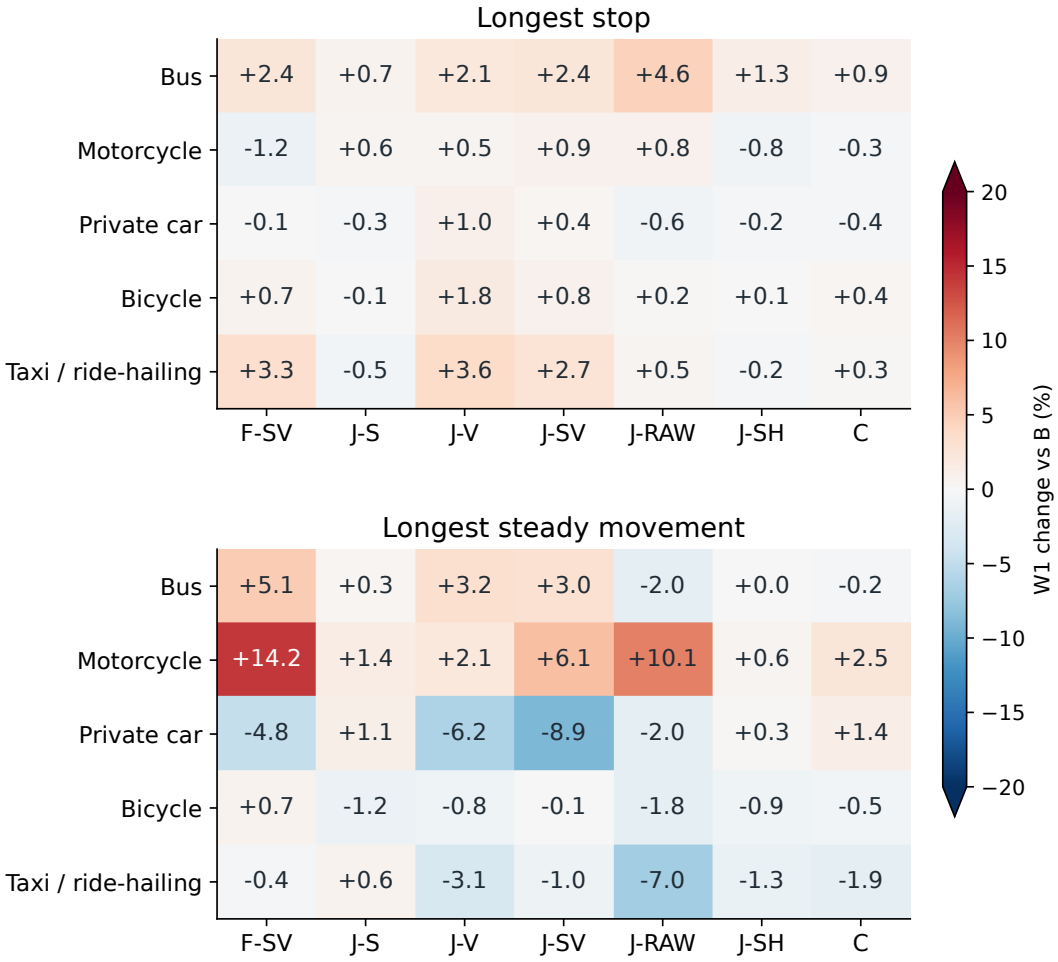


Figure 7.6. All-class longest-stop and longest-steady-movement distances. Different aspects of stop organization can change in opposite directions. These panels retain the same seven contrasts and five classes as the primary comparison.

7.6 Complete generated trajectories and their GAF views

The next ten figures show one fixed-order case for every class. The rule is episode 400201, model seed 42, first lexicographic jointly supported request per class, candidate zero. It was fixed before rendering. The displayed subset comprises the reference and B, J-S, J-V, J-SV and J-RAW; all eight arms remain in the numerical comparison. Maps show the reference, B and J-SV, while the time curves show all displayed model arms.

Map points progress from purple to yellow with relative elapsed time; a circle marks the start and a cross the end. Curve comparisons use the whole elapsed-time axis. Their acceleration distributions use complete one-second increments for readability and do not replace the quarter-second primary metric. The examples compare timing and speed structure alongside the population statistics; individual stop sequences vary across stochastic realizations.

Each companion image figure uses GASF/GADF/MTF channels computed from those same output coordinates and timestamps. No per-image contrast enhancement is applied. The rows are observed SUMO, B, J-S, J-V, J-SV and J-RAW. These output representations accompany the physical-time comparisons.

7.6.1 Bus: shared service geometry and variable dwell organization

The bus reference lasts 2,328 seconds and the paired generated journeys 2,560.27 seconds. Similar public service geometry coexists with differences in the time spent moving and stopped. The generated arms have closely overlapping progress curves, and the condition changes do not repair the common duration discrepancy. This case illustrates why a recognizable spatial bus pattern, especially on a shared service template, is insufficient evidence of accurate dwell organization.

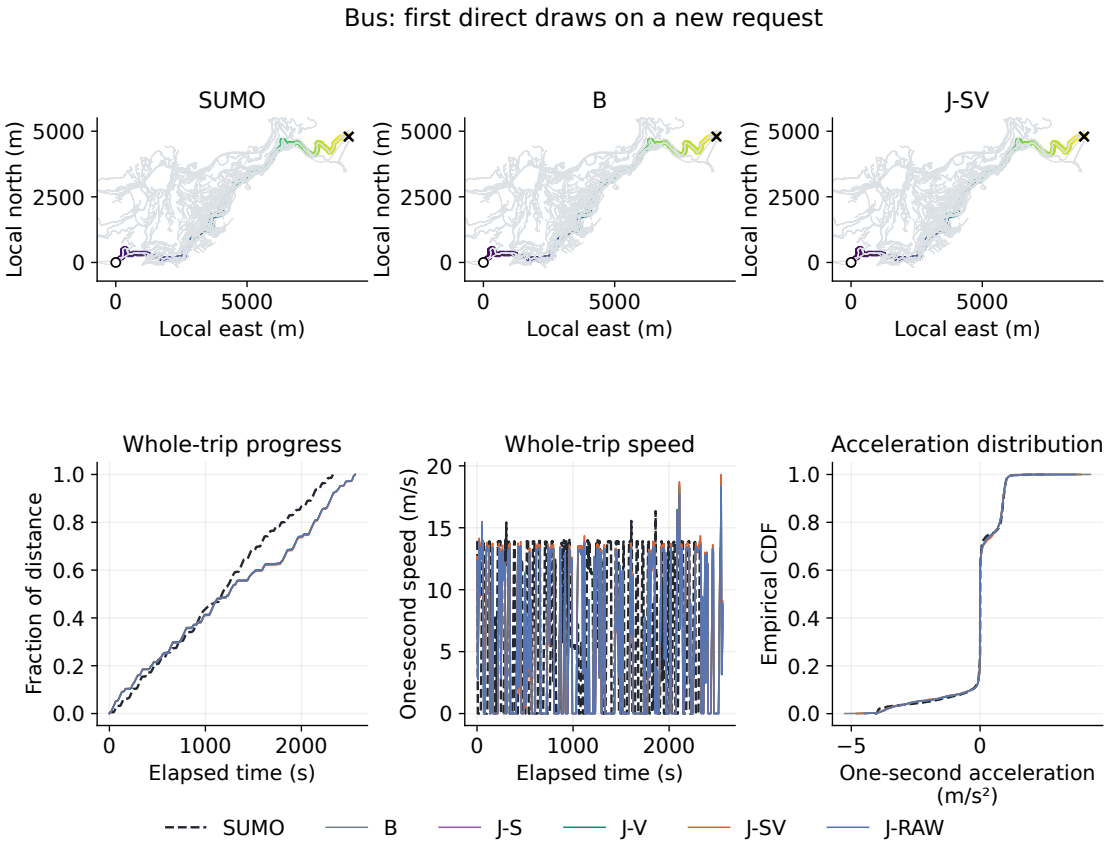


Figure 7.7. Fixed bus case: complete road geometry, cumulative progress, one-second speed and acceleration distribution. Route adherence belongs to the shared decoder; the condition variants are compared under common sampled route and duration.

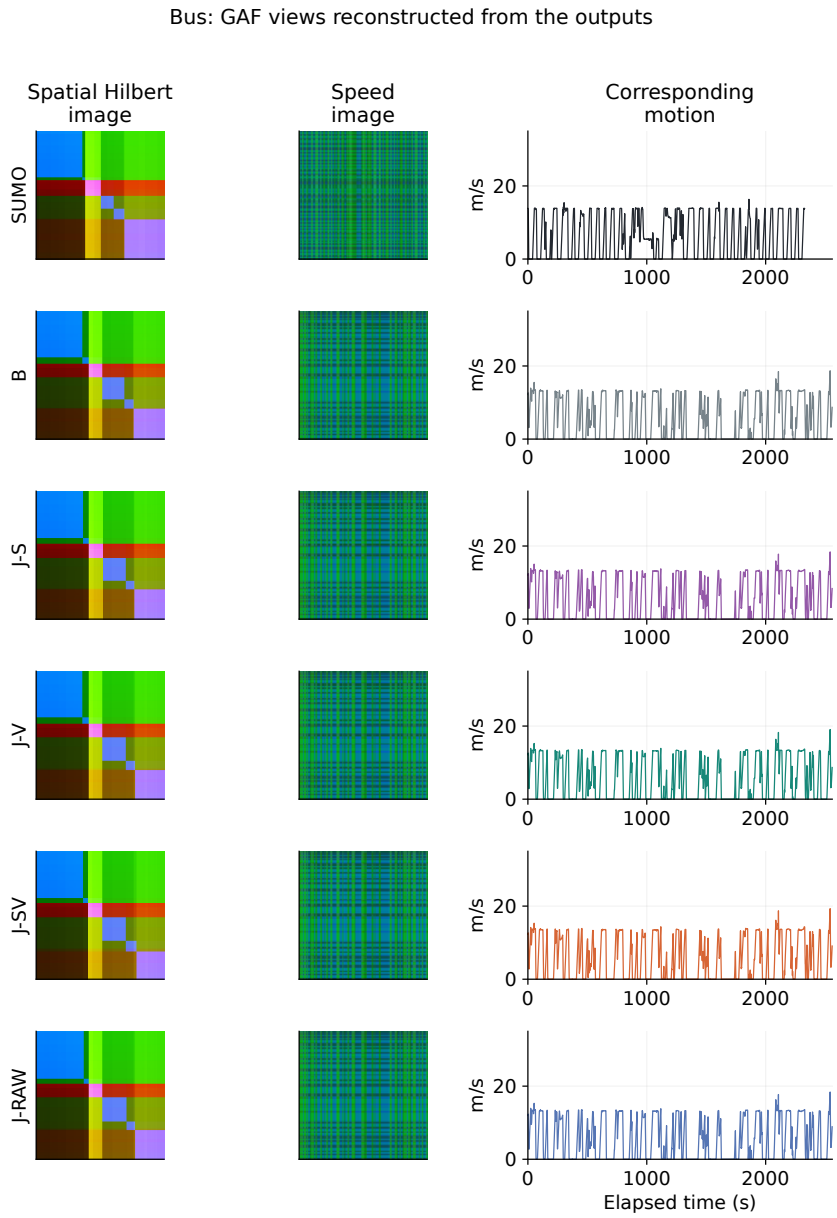


Figure 7.8. Spatial and speed image views reconstructed from the fixed bus reference and generated outputs. The corresponding full-trip speed traces expose timing differences that can be difficult to infer from the spatial image alone.

7.6.2 Motorcycle: route alternatives and the time axis

The reference lasts 266.75 seconds, compared with 355.31 seconds for the generated draws. The map exposes a difference between the observed route and the generated alternative, while the generated conditions share their own route. Several valid ways of reaching the endpoint can exist; the experiment therefore evaluates route distributions separately from motion. The similar generated curves show that adding code does not by itself produce a large change in this request's movement.

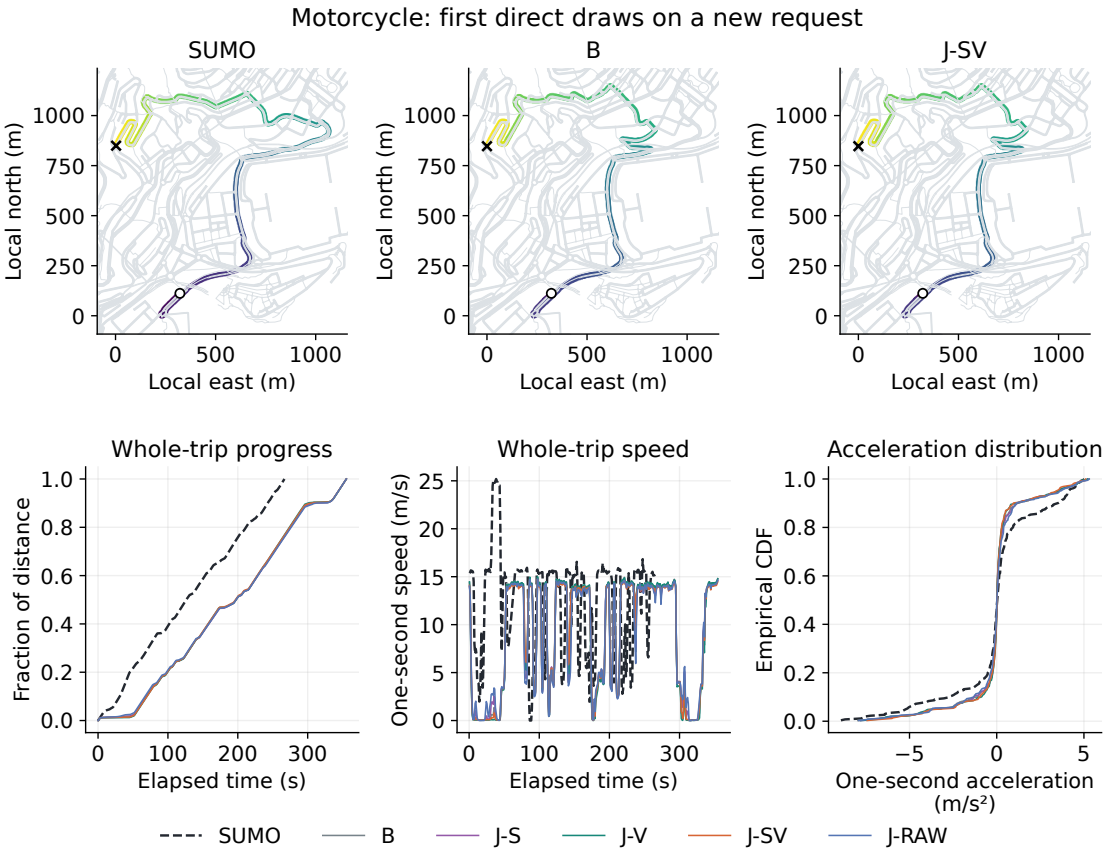


Figure 7.9. Fixed motorcycle case. Full time axes retain the difference between reference and generated duration; no time alignment is used to make the curves agree.

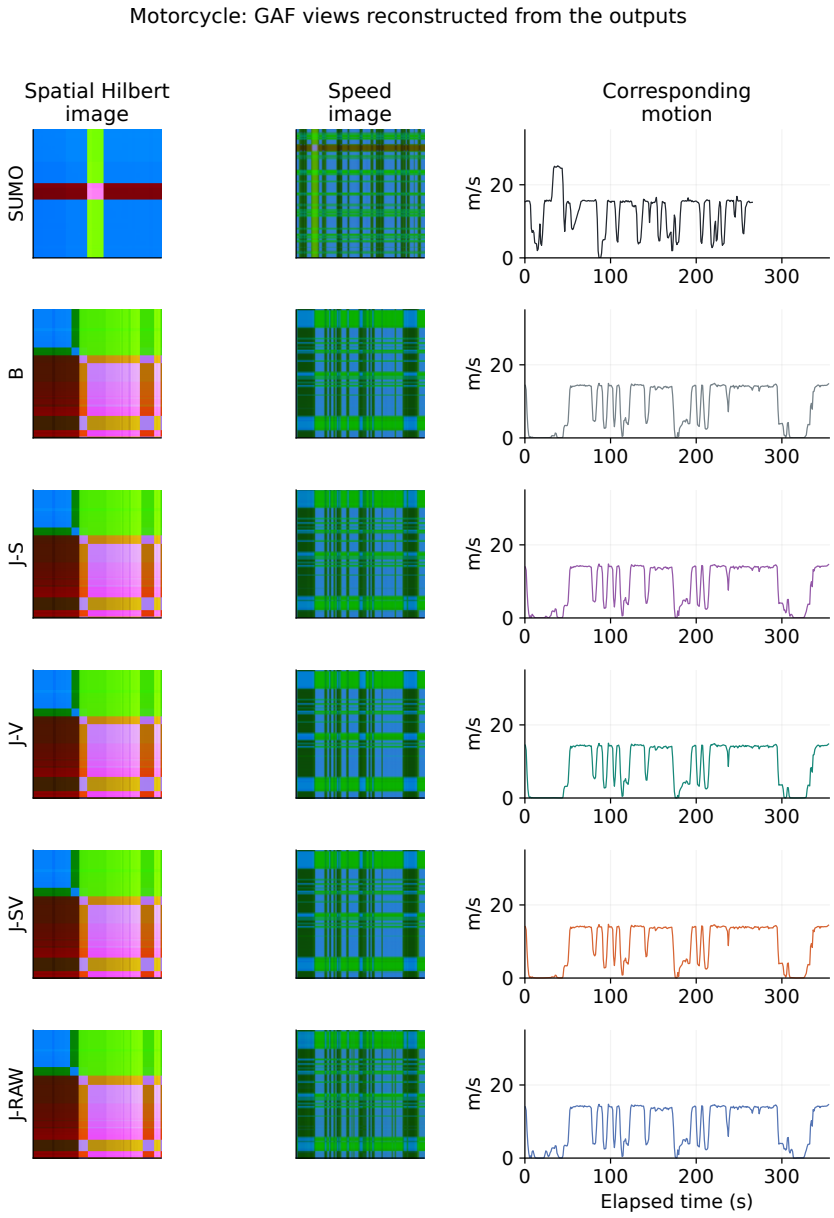


Figure 7.10. Fixed motorcycle output representations. Spatial and speed images encode different aspects of the same complete journey; neither is an additional observed sensor channel.

7.6.3 Private car: marginal acceleration and stop timing

The private-car reference lasts 631.75 seconds and the generated draws 763.90 seconds. Their acceleration distribution curves can be close even while the positions of speed peaks and stops differ in time. This is precisely why distribution distances and request-level summaries are both reported. The class-marginal and individual temporal comparisons describe complementary properties.

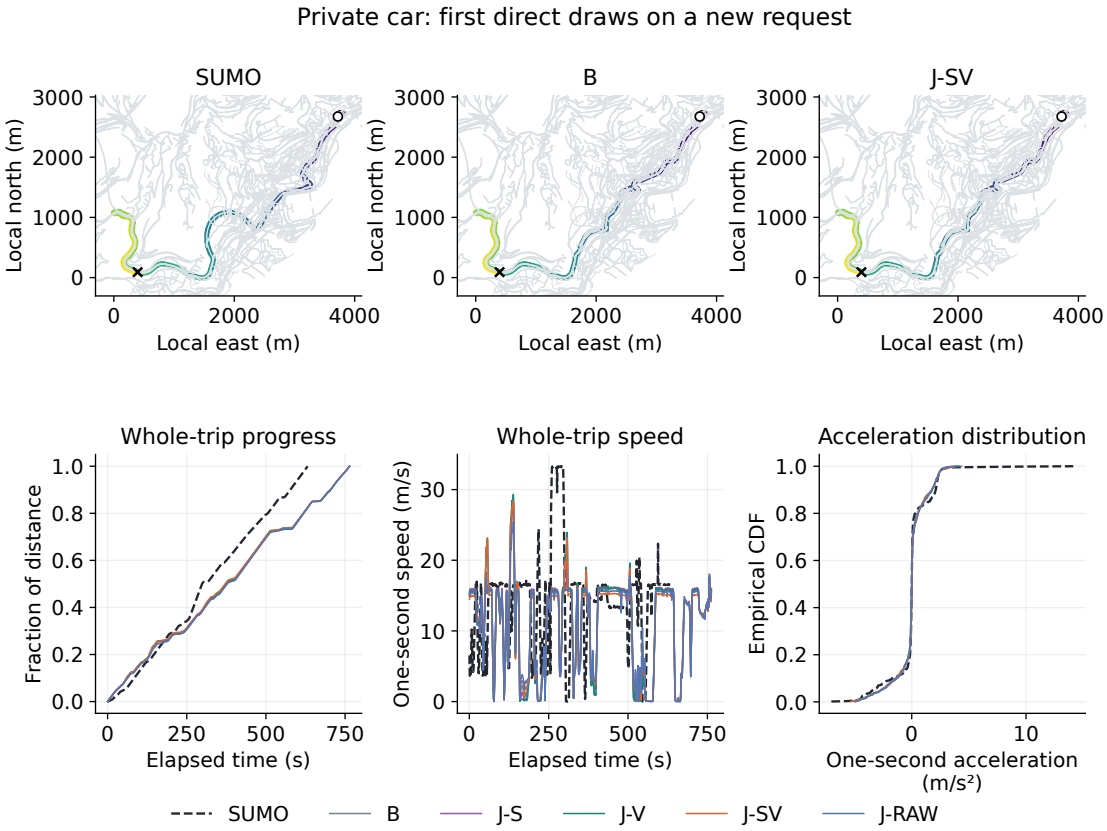


Figure 7.11. Fixed private-car case. The empirical acceleration CDF and full speed trace address different properties; their visual agreement need not imply matching temporal organization.

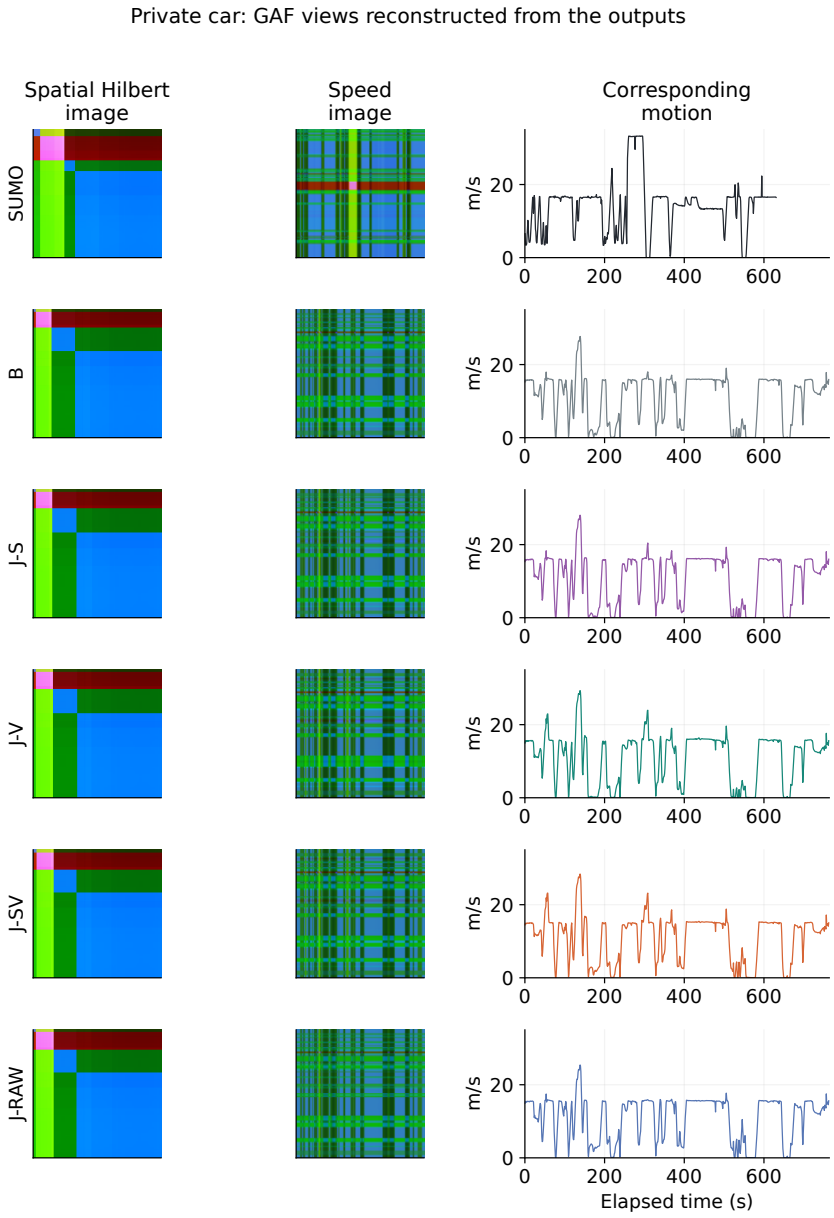


Figure 7.12. Fixed private-car GAF views and their source speed curves. The same frozen image conversion is applied to the reference and every generated row.

7.6.4 Bicycle: a visible shared travel-time limitation

The bicycle reference lasts 778.50 seconds, whereas all displayed generated draws last 499.15 seconds. The generated speed level is visibly higher in this example. Neither a more distinctive speed image nor a small change between conditional arms fixes the shared sampled duration. This case is retained because it exposes a limitation of the factorization that a selected favorable image could conceal.

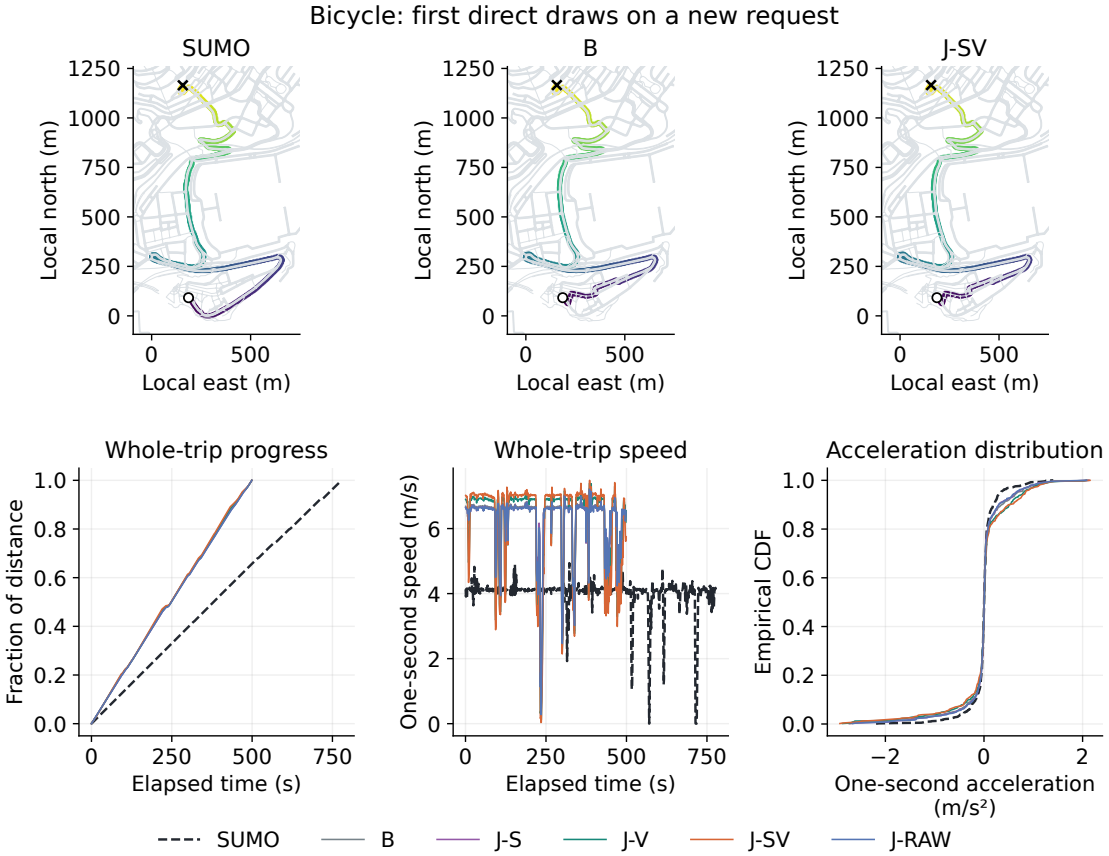


Figure 7.13. Fixed bicycle case. The shorter common generated duration and higher speed level remain visible. The case illustrates the effect of the common sampled duration.

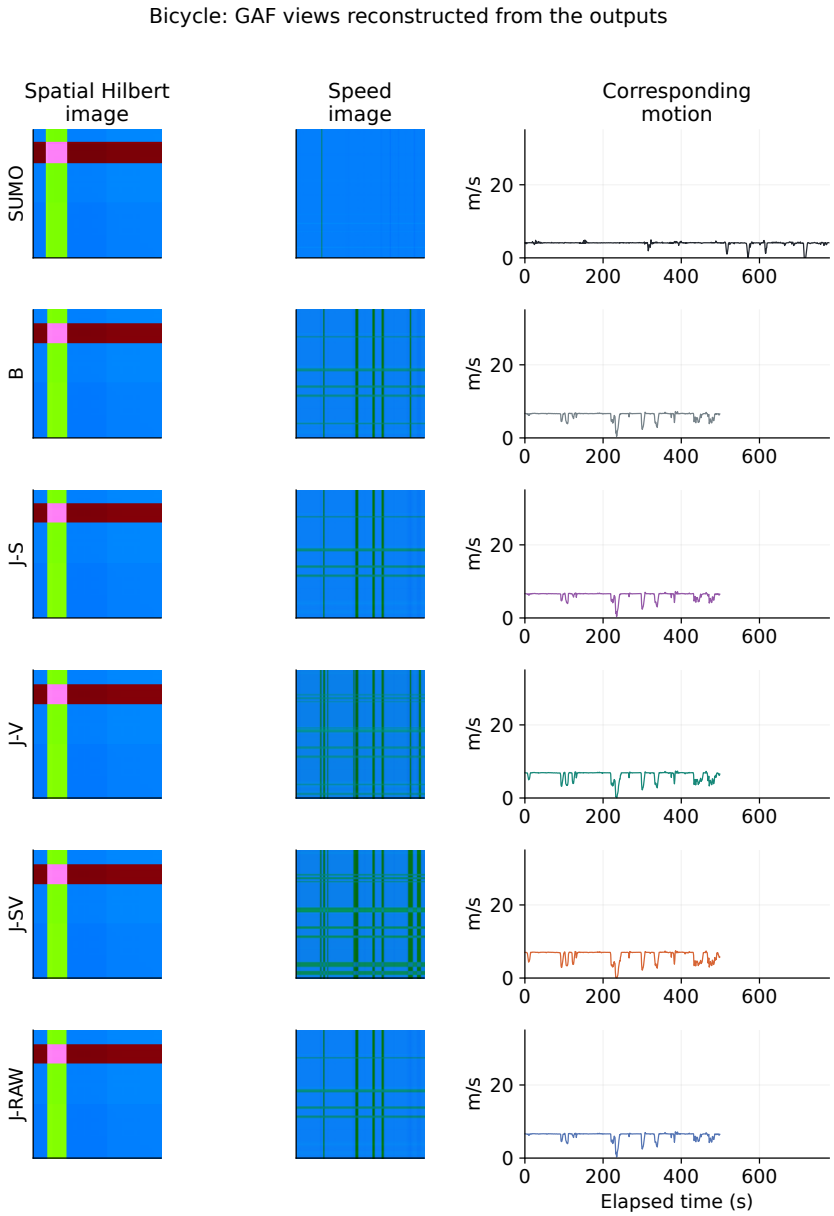


Figure 7.14. Fixed bicycle spatial and speed images. An image can preserve class-related structure while the associated physical-time profile remains mismatched.

7.6.5 Taxi and ride-hailing: service labels and geometry

The taxi/ride-hailing reference lasts 293 seconds and the generated draws 345.18 seconds. The reference and generated route choices differ visibly, and the generated speed plateaus change slightly across conditions. The label partly describes a transport service rather than a uniquely identifiable geometry or instantaneous dynamical law. Its weak spatial recognition and mixed generative results therefore require class-specific interpretation rather than an assumed universal image signature.

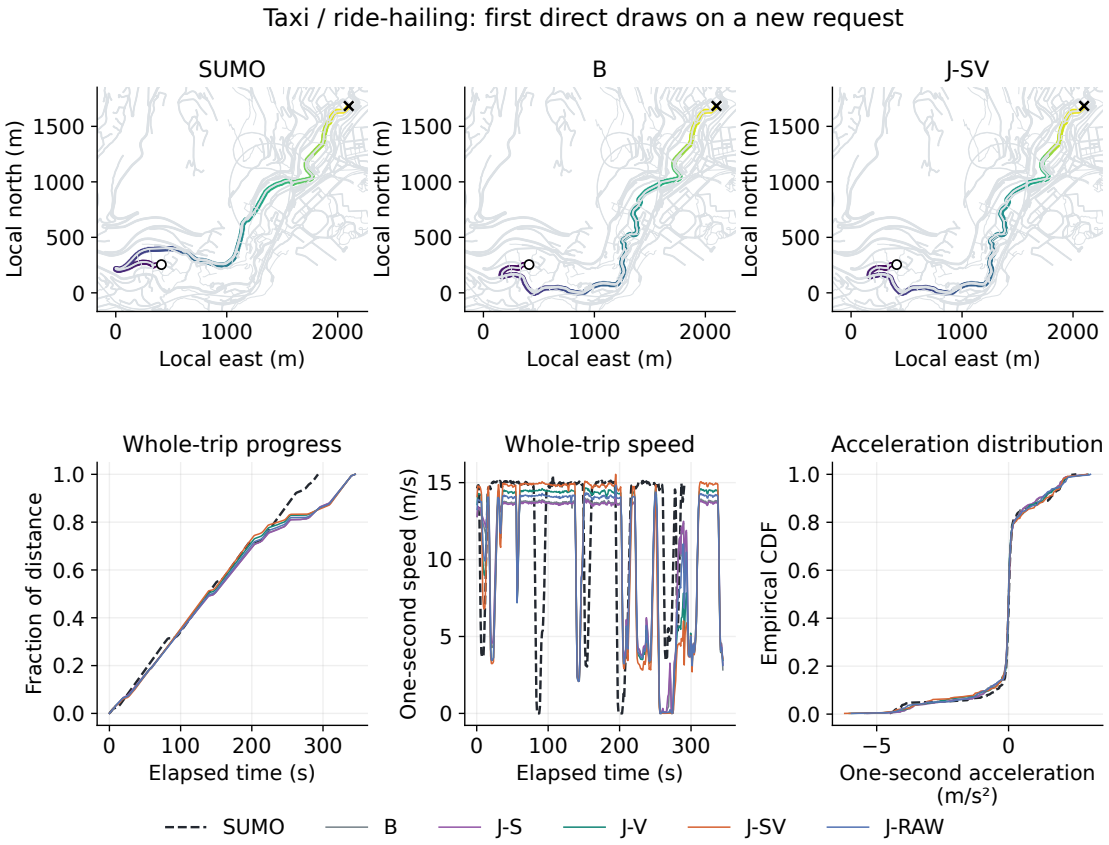


Figure 7.15. Fixed taxi/ride-hailing case: a permitted route alternative and its generated motion. One reference realization does not define the sole valid route; conditional route-use and motion distributions are separate evaluation objects.

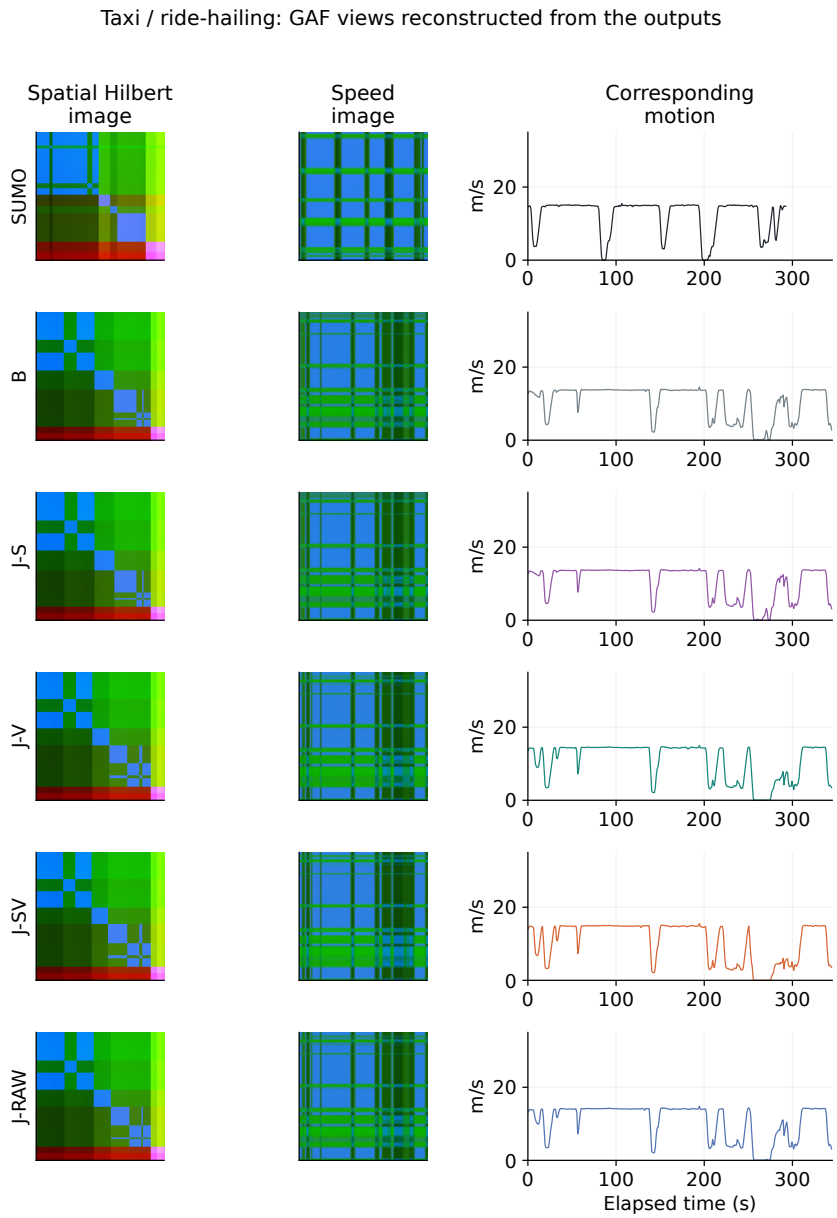


Figure 7.16. Fixed taxi/ride-hailing output images. The five class-specific pairs are retained in their predetermined order, including cases with clear timing discrepancies.

7.7 Synthetic-data augmentation

Table 7.3 retains every downstream regime, including synthetic-only training. Observed-only training achieves 94.41% mean macro F1. Adding unscored B samples increases it to 96.26%, whereas resampling the same observed set gives 94.17%. Thus the baseline synthesized data have a practical supplementation benefit for this particular raw-series recognition task and upstream information budget.

Spatial J-S augmentation reaches 96.95%, 0.699 percentage points above B. The direction is positive for both downstream training seeds. J-RAW reaches 96.66%, leaving a much smaller spatial-versus-RAW contrast of 0.294 points. The two-seed non-bus parent-equal F1 is 96.54% for J-S and 95.41% for B, so the pooled signal is not solely a bus-class effect. The physical and downstream measurements therefore yield different model rankings.

Table 7.3. All fifteen downstream regimes. Percent macro F1 on real simulated references from the last two OD cohorts. Each fit has the same optimization budget.

Training data	F1, seed 4001	F1, seed 4002	Mean F1
real	94.39	94.42	94.41
only-empirical	93.06	93.63	93.34
only-B	93.68	92.34	93.01
only-F-SV	93.44	91.83	92.63
only-J-S	92.90	92.88	92.89
only-J-V	92.35	92.87	92.61
only-J-SV	93.96	92.87	93.41
only-J-RAW	93.12	92.62	92.87
plus-empirical	95.18	93.15	94.17
plus-B	96.01	96.50	96.26
plus-F-SV	96.26	94.66	95.46
plus-J-S	96.83	97.08	96.95
plus-J-V	95.76	94.95	95.35
plus-J-SV	95.23	95.74	95.48
plus-J-RAW	96.80	96.52	96.66

The augmentation effect varies by evaluation cohort. Averaging episode macro F1 across the two windows and learner seeds, J-S minus B is -0.755 percentage points in cohort 3 and $+2.091$ points in cohort 4. Table A.12 gives these episode-equal summaries for all fifteen regimes. The pooled gain is concentrated in cohort 4.

Dual-view J-SV yields 95.48% macro F1 and speed-only J-V yields 95.35%, both below B augmentation. Synthetic-only regimes yield 92.61–93.41%, compared with

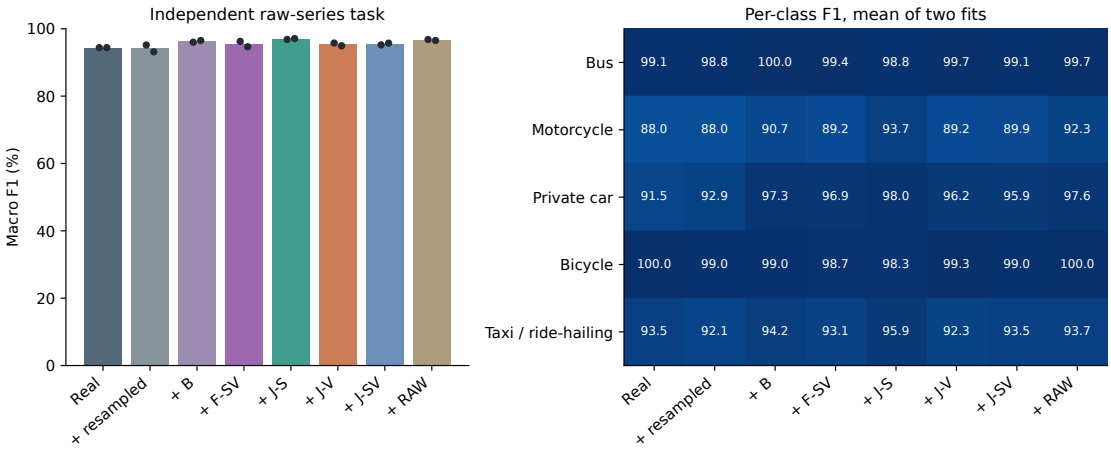


Figure 7.17. Independent augmentation on observed SUMO references. Left: pooled macro F1, with both learner seeds shown. Right: mean per-class F1. "Real" denotes simulator observations. All methods share downstream training budgets; the upstream generator uses the full labelled training corpus.

94.41% for observed-only training. Synthetic trajectories are most useful as supplementary training data in this experiment. Spatial conditioning provides the highest pooled augmentation score, whereas B retains the better aggregate physical endpoints.

7.8 Representation sensitivity and condition use

The pre-generation 500-parent sensitivity audit shows that whole-trip time warping changes both feature views. Within-second perturbations can substantially alter native acceleration while leaving the speed features almost unchanged, as expected from one-second averaging and relative-profile compression. Smoothing affects both views differently. The perturbations characterize the temporal detail retained by each view.

The final-model intervention audit confirms that the code path is active. B is exactly invariant to the four interventions; C is invariant to same-class exchange and its deterministic prior, as expected. Changing the code changes predictions in J-S, J-V and J-SV. At noise index 500, the J-SV mean prediction RMS changes by 0.012684 for a same-class exchange and 0.013908 for a sampled prior code relative to its assigned training code. The corresponding MSE values are shown in Table 7.4.

For J-SV, assigned training codes yield lower denoising MSE than all three interventions at each diagnostic level. The absolute differences are small. This training-parent experiment identifies an active conditioning path and a gap between target-derived and sampled codes. J-S has smaller prediction changes, with mixed MSE

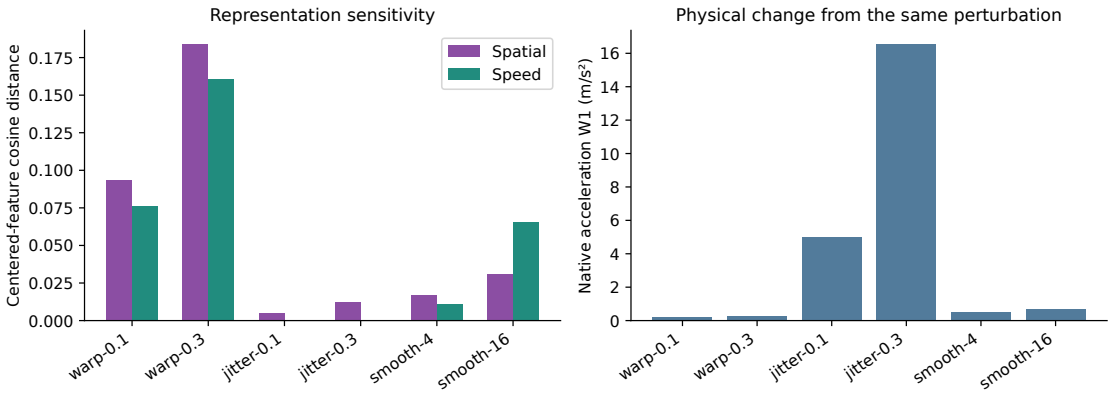


Figure 7.18. Training-side representation audit, fixed before prospective outcomes. Feature changes and native acceleration changes are measured for the same perturbations. The panels report their respective units. The views respond differently across physical time scales.

Table 7.4. Condition use on training parents: J-SV noise-prediction MSE under four fixed interventions. Means over 25 parents and two model seeds. This is an in-sample mechanism check, not prospective quality.

Noise index	Assigned	Same-class swap	Zero	Sampled prior
50	0.142174	0.142512	0.143795	0.142529
250	0.032396	0.032623	0.032764	0.032622
500	0.016933	0.017027	0.017050	0.017065
750	0.003504	0.003521	0.003573	0.003567
950	0.000201	0.000202	0.000208	0.000207

responses to same-class exchange. The complete intervention tables are given in the appendix.

7.9 Diversity, correction and computation

Mean within-request relative-progress diversity ranges from 0.05575 to 0.05638, compared with 0.05626 for B. Thus the condition variants retain similar diversity under this measure. The near-shape rate is zero against the fixed library of at most 512 training progress profiles per class.

The decoder truncates negative log-motion values. The mean affected-bin fraction is 11.64% for B and 12.45% for J-SV, with all arms between 11.52% and 12.45%. Upper clipping and constant-progress fallback occur zero times. These counters characterize the correction applied before coordinate decoding.

Table 7.5. Formal 5,000-update motion continuation on the available L20 setup. Times show seeds 42/43; peak allocated memory includes resident frozen-prefix caches. The common 10,000-update motion initialization, upstream reader fitting, code-prior fitting and prospective sampling are additional costs.

Arm	Trainable encoder	Minutes, 42 / 43	Peak allocated GiB, 42 / 43
B	0	22.90 / 24.02	0.61 / 0.61
F-SV	139,712	26.39 / 25.70	11.79 / 11.79
J-S	8,466,149	24.66 / 24.55	6.32 / 6.32
J-V	8,466,149	26.26 / 25.91	6.32 / 6.32
J-SV	16,932,298	27.46 / 26.60	12.02 / 12.02
J-RAW	5,922,250	26.18 / 25.74	1.23 / 1.23
J-SH	16,932,298	27.42 / 25.99	12.02 / 12.02
C	139,712	24.52 / 23.12	0.61 / 0.61

Each 5,000-update motion continuation takes 22.90–27.46 minutes on the assigned L20. The dependent formal queue closes in approximately 4.06 hours, including its reference, generation and utility stages. Summed generation-cell time divided by output count is approximately 0.451 seconds per B candidate and 0.486–0.499 seconds for the learned-prior arms, including per-cell model loading and export. Each arm uses 627,200 motion-network forwards; a learned-prior arm additionally uses 313,600 prior forwards.

Chapter 8

Discussion and conclusions

8.1 Complete journeys as the unit of synthesis

The synthesis system models a journey through route, duration and progress. Given a road network, an OD request, a participant class and public service constraints, it returns timestamped coordinates. Learned route preferences describe which feasible alternatives participants use; the duration model determines the physical time available; temporal diffusion generates movement along the selected path. This factorization connects route choice and motion within one usable synthetic-data pipeline.

Multiple routes can satisfy the same request. Evaluation therefore compares route-use distributions and movement statistics alongside individual examples. Learned route preferences reduce edge-use divergence relative to shortest-route selection within the available catalog. Uniform route sampling produces greater route diversity and lower speed discrepancy in that experiment, but substantially higher edge-use divergence. These results expose a practical tradeoff between route variety and agreement with observed preferences.

Physical timing is equally consequential. The temporal pilot reduces native acceleration discrepancy by 81–82% when progress is modeled on the quarter-second clock, while longest-stop discrepancy increases. The native representation is subsequently used for complete-journey generation. The five fixed examples illustrate the remaining effect of duration: different motion models can produce similar progress patterns because they share the sampled route and travel time. In the bicycle example, the shorter sampled duration shifts all generated speed profiles upward. This common error originates upstream of the GAF motion condition.

8.2 From trajectory recognition to generative conditioning

The GAF study builds on spatial trajectory classification [1, 2, 12] through an explicit generative interface. Ordered Hilbert images retain spatial organization, while speed images summarize movement over the journey. Both readers are trained in this project. Native-demand data preserve the associations between participant class, route choice and service structure that were weakened by randomized class–route assignment. On the expanded development population, the spatial and speed readers reach 76.47% and 90.34% macro F1.

The generation model uses image-derived codes throughout denoising. Joint adaptation couples the encoders to the trajectory objective, and a learned prior supplies codes for prospective requests. This follows the encoder–conditional decoder–prior construction of diffusion autoencoders [53], adapted here to road-supported variable-length motion. Code interventions change denoising predictions, confirming an active connection between the representation and the generator.

The physical comparison measures the effect of that connection. Joint spatial–speed conditioning increases aggregate native acceleration distance by 2.44%, with increases in all four OD-cohort means. Single-view conditions and controls remain close to the baseline, and the private-car improvement under the joint condition accompanies adverse changes in the other four classes. Accordingly, the baseline provides the stronger aggregate physical result in this comparison. Similar diversity across conditions indicates that the differences arise without a large change in the measured within-request spread.

Downstream utility shows a different ordering. Spatial-only augmentation yields the highest pooled macro F1, 96.95%, compared with 96.26% for baseline augmentation and 96.66% for raw-series conditioning. Its episode-equal advantage over baseline augmentation is -0.755 percentage points in one evaluation OD cohort and $+2.091$ in the other. All synthetic-only regimes score below observed-only training. The generated data are therefore most useful as a supplement in the studied task, with a cohort-dependent benefit from spatial conditioning.

8.3 Interpreting the representation effect

The two GAF views are deterministic transformations of trajectory measurements. They organize existing information into a form accessible to image encoders. Class recognition can exploit broad speed scale, route identity and service regularity. The motion generator already receives class, route geometry, length and duration, so part of this class-related information is shared with its existing inputs. The near-baseline class-centroid result and the small improvements from same-class wrong-trip codes are consistent with such overlap.

Temporal resolution provides a second explanation. Each view is compressed into a 32-dimensional global code. The speed image further summarizes one-second movement over 128 relative journey positions. The sensitivity experiment shows that within-second perturbations can change native acceleration substantially while barely changing speed-image features. Thus the representation emphasizes whole-trip structure more strongly than some local dynamics assessed by the physical metrics.

A third factor is the source of the code. During fitting, the encoder observes a training target; during prospective generation, the prior samples a code from the request conditions. Assigned J-SV codes produce slightly lower training-side denoising error than sampled, exchanged or zero codes. This gap identifies a difference between target-derived and sampled information. Together, information overlap, temporal compression and conditional sampling explain plausible constraints on this interface; the experiments measure their signatures rather than isolate a single cause.

8.4 Evaluation and scope

Trajectory synthesis requires several complementary measurements. Route statistics assess alternatives between endpoints; native and coarse-clock movement summaries assess dynamics; diversity characterizes variation; downstream classification measures data utility. Classifier confidence describes a separate property. Calibration research and proper scoring rules clarify the distinction between predictive probabilities and task-specific rewards [48, 52]. Even accurate class recognition can favor a typical class pattern while overlooking dynamics within that class. This motivates the physical and downstream evaluations used here.

The evidence concerns five participant classes on one SUMO road network with shared bus services. Four OD cohorts provide two departure windows each; the downstream experiment uses two of those cohorts. The temporal pilot has a smaller population and a common update budget rather than equal runtime. The downstream learner uses 1,000 observed parents, whereas the generators use the larger labeled corpus. These design choices define the populations and information budgets of the reported comparisons.

Road adherence is enforced by the path decoder. The system models individual journeys without joint traffic interactions, lateral behavior or evolving signal-state conditioning. Generalization to field GPS data, new cities and unseen services remains outside the evaluated scope. The representation controls also have distinct meanings: joint adaptation combines denoising and class losses without a separate class-loss-only arm; raw-series encoders differ in architecture and pretraining; same-class shuffling retains approximate trip scale. These conditions qualify the attribution of differences among interfaces.

8.5 Conclusion

This thesis delivers a road-conditioned diffusion framework for complete multi-class traffic trajectories. Native physical timing and learned route preferences connect movement generation to the road network and observed travel behavior. The resulting synthetic journeys support both physical evaluation and downstream data augmentation.

The GAF contribution is a systematic transfer of spatial and speed image representations from participant recognition into generative conditioning. Both views carry class information, their codes affect denoising, and spatial conditioning produces the highest pooled augmentation score in the studied task. Joint conditioning offers no aggregate physical advantage over the baseline. The resulting account connects representation, generation and data utility through measured outcomes, and identifies the role of temporal detail and already available conditions in that transfer.

Appendix A

Complete numerical results and fixed cases

A.1 Final G40 physical endpoints, classes and coverage

The following tables retain all eight conditions, both fitted seeds and all five participant classes from the completed prospective comparison. The K1 tables concern the first direct draw; the all-four comparison weights the four draws equally inside each request. The ten physical summaries appear in their original units, without a cross-unit composite or a choice of favourable endpoints. Class-marginal distributions and request-paired discrepancies address different questions. Neither assumes a unique valid future trajectory for each OD request.

Table A.1. Final coverage by reference episode. Counts are requests, not independent vehicle populations: paired windows reuse source plans and bus templates are shared. Exact means the realised reference route is in the catalog.

Episode	Requested	Generated	Physical pairs	Route pairs	Exact route
400201	100	98	96	96	61
400202	100	98	96	96	61
400203	100	98	95	95	61
400204	100	98	92	92	61
400205	100	98	97	97	63
400206	100	98	90	90	62
400207	100	98	96	96	54
400208	100	98	93	93	53

Table A.2. Native distribution distances for every fitted seed. K1 is the first direct draw; all four gives equal mass to four draws per request. Speed is in m/s and acceleration in m/s^2. Means cover eight episodes and five classes.

Arm / seed	K1 speed	K1 accel.	All four speed	All four accel.
B / 42	1.4726	0.1739	1.3227	0.1699
B / 43	1.4033	0.1781	1.2665	0.1721
F-SV / 42	1.4643	0.1747	1.3149	0.1691
F-SV / 43	1.4065	0.1775	1.2694	0.1729
J-S / 42	1.4713	0.1776	1.3215	0.1731
J-S / 43	1.4069	0.1763	1.2718	0.1706
J-V / 42	1.4663	0.1799	1.3191	0.1735
J-V / 43	1.4085	0.1785	1.2692	0.1737
J-SV / 42	1.4669	0.1796	1.3235	0.1708
J-SV / 43	1.4133	0.1810	1.2725	0.1750
J-RAW / 42	1.4691	0.1780	1.3297	0.1707
J-RAW / 43	1.4183	0.1833	1.2725	0.1753
J-SH / 42	1.4703	0.1729	1.3211	0.1679
J-SH / 43	1.3987	0.1761	1.2647	0.1717
C / 42	1.4716	0.1740	1.3239	0.1699
C / 43	1.4054	0.1775	1.2710	0.1708

Table A.3. Native speed Wasserstein distances (m/s) by class. Means of eight episodes and two fitted seeds, K1 direct draws. The five classes are displayed individually, without ranking or omission.

Arm	Bus	Motorcycle	Private car	Bicycle	Taxi
B	0.5230	1.9020	1.9939	0.6824	2.0884
F-SV	0.5257	1.9068	1.9684	0.6844	2.0918
J-S	0.5311	1.8983	1.9956	0.6825	2.0878
J-V	0.5243	1.9128	1.9773	0.6864	2.0863
J-SV	0.5231	1.9197	1.9720	0.6996	2.0862
J-RAW	0.5464	1.8783	2.0158	0.6792	2.0987
J-SH	0.5114	1.9024	1.9877	0.6822	2.0887
C	0.5210	1.8962	2.0034	0.6822	2.0897

Table A.4. Native acceleration Wasserstein distances (m/s^2) by class. Means of eight episodes and two fitted seeds, K1 direct draws. The five classes are displayed individually, without ranking or omission.

Arm	Bus	Motorcycle	Private car	Bicycle	Taxi
B	0.0762	0.3249	0.1403	0.1466	0.1920
F-SV	0.0780	0.3193	0.1382	0.1449	0.2003
J-S	0.0765	0.3252	0.1442	0.1465	0.1924
J-V	0.0758	0.3229	0.1388	0.1542	0.2042
J-SV	0.0775	0.3313	0.1357	0.1549	0.2021
J-RAW	0.0771	0.3416	0.1385	0.1481	0.1981
J-SH	0.0750	0.3218	0.1385	0.1440	0.1933
C	0.0766	0.3251	0.1408	0.1438	0.1926

Table A.5. Class-marginal Wasserstein distances, part 1. K1 direct draws, averaged equally over classes, model seeds, windows and OD cohorts. All arms remain; lower is closer to the reference. Units belong to each column.

Arm	Duration (s)	Distance (m)	Mean speed (m/s)	Speed p95 (m/s)	Stopped fraction
B	178.6336	61.0779	1.4848	1.3994	0.0895
F-SV	178.6336	61.0655	1.4848	1.3609	0.0893
J-S	178.6336	61.0930	1.4848	1.3794	0.0892
J-V	178.6336	61.1096	1.4847	1.3803	0.0902
J-SV	178.6336	61.0818	1.4847	1.4118	0.0892
J-RAW	178.6336	61.0768	1.4848	1.4223	0.0894
J-SH	178.6336	61.1025	1.4848	1.3816	0.0894
C	178.6336	61.0932	1.4848	1.3829	0.0893

Table A.6. Class-marginal Wasserstein distances, part 2. K1 direct draws, averaged equally over classes, model seeds, windows and OD cohorts. All arms remain; lower is closer to the reference. Units belong to each column.

Arm	Mean $\lvert a \rvert$ (m/s^2)	Longest stop (s)	Stops per min	Longest steady (s)	Four-second change (m/s)
B	0.0793	28.9814	0.2488	20.3933	0.2792
F-SV	0.0811	29.2932	0.2514	20.4734	0.2840
J-S	0.0792	29.0212	0.2490	20.4436	0.2792
J-V	0.0809	29.5223	0.2430	20.0124	0.2815
J-SV	0.0820	29.4228	0.2449	19.9324	0.2838
J-RAW	0.0796	29.3519	0.2563	20.0474	0.2768
J-SH	0.0795	28.9978	0.2487	20.3505	0.2792
C	0.0804	29.0463	0.2463	20.4462	0.2821

Table A.7. Request-paired mean absolute discrepancies, part 1. K1 direct draws, averaged equally over classes, model seeds, windows and OD cohorts. All arms remain; lower is closer to the reference. Units belong to each column.

Arm	Duration (s)	Distance (m)	Mean speed (m/s)	Speed p95 (m/s)	Stopped fraction
B	284.7414	65.7098	2.7417	3.2119	0.1504
F-SV	284.7414	65.7063	2.7416	3.1937	0.1504
J-S	284.7414	65.7248	2.7417	3.2380	0.1501
J-V	284.7414	65.7426	2.7416	3.1704	0.1524
J-SV	284.7414	65.7058	2.7415	3.1551	0.1516
J-RAW	284.7414	65.7200	2.7416	3.1859	0.1512
J-SH	284.7414	65.7248	2.7417	3.2207	0.1509
C	284.7414	65.7217	2.7417	3.2293	0.1506

Table A.8. Request-paired mean absolute discrepancies, part 2. K1 direct draws, averaged equally over classes, model seeds, windows and OD cohorts. All arms remain; lower is closer to the reference. Units belong to each column.

Arm	Mean $\lvert a \rvert$ (m/s^2)	Longest stop (s)	Stops per min	Longest steady (s)	Four-second change (m/s)
B	0.1614	41.2845	0.4349	36.2157	0.5419
F-SV	0.1629	41.6194	0.4364	36.1815	0.5464
J-S	0.1612	41.3469	0.4347	36.1413	0.5412
J-V	0.1635	42.0044	0.4328	36.3024	0.5491
J-SV	0.1661	41.9202	0.4340	36.0281	0.5577
J-RAW	0.1605	41.7909	0.4328	35.5502	0.5404
J-SH	0.1625	41.4936	0.4362	36.2181	0.5462
C	0.1625	41.4257	0.4331	36.1700	0.5447

Table A.9. Progress-shape diversity and decoder accounting. Means weight classes and scene/seed cells equally. Near-shape is the fraction within 10^{-4} of a fixed, at-most-512-per-class training library; it is not a privacy guarantee. Negative and upper are pre-correction decoder-bin fractions.

Arm	Pairwise shape	Nearest shape	Near-shape rate	Negative	Upper
B	0.0563	0.0163	0.0000	0.1164	0.0000
F-SV	0.0561	0.0163	0.0000	0.1179	0.0000
J-S	0.0563	0.0164	0.0000	0.1177	0.0000
J-V	0.0563	0.0163	0.0000	0.1217	0.0000
J-SV	0.0561	0.0163	0.0000	0.1245	0.0000
J-RAW	0.0557	0.0162	0.0000	0.1244	0.0000
J-SH	0.0563	0.0164	0.0000	0.1152	0.0000
C	0.0564	0.0164	0.0000	0.1172	0.0000

Per-cell support, all individual class results, coverage and provenance are available in `docs/results/g40-generative-conditioning/`. Missing endpoints retain their support counts. The nearest-shape diagnostic uses a fixed hash-ranked library of at most 512 training examples per class and normalized relative-time progress, not a search over every training coordinate sequence. A low reported near-shape rate cannot prove absence of memorization or establish privacy. Decoder clipping is a correction applied to the generated representation, not an observed traffic event.

A.2 Condition interventions and utility cohorts

Table A.10. All-arm condition interventions, part 1. Noise-prediction MSE on the fixed training-parent audit; means over parents and seeds.

Arm	Index	Assigned	Swap	Zero	Prior
B	50	0.144968	0.144968	0.144968	0.144968
B	250	0.032536	0.032536	0.032536	0.032536
B	500	0.016942	0.016942	0.016942	0.016942
B	750	0.003552	0.003552	0.003552	0.003552
B	950	0.000204	0.000204	0.000204	0.000204
F-SV	50	0.143556	0.142638	0.143871	0.143524
F-SV	250	0.032525	0.032563	0.032634	0.032574
F-SV	500	0.017010	0.016893	0.016955	0.016949
F-SV	750	0.003534	0.003544	0.003547	0.003555
F-SV	950	0.000203	0.000201	0.000206	0.000205
J-S	50	0.143936	0.143285	0.144029	0.143928
J-S	250	0.032478	0.032490	0.032512	0.032471
J-S	500	0.016920	0.016913	0.016927	0.016909
J-S	750	0.003556	0.003553	0.003562	0.003551
J-S	950	0.000202	0.000200	0.000203	0.000202
J-V	50	0.142568	0.142921	0.144263	0.142860
J-V	250	0.032438	0.032623	0.032790	0.032582
J-V	500	0.016950	0.016998	0.017088	0.016956
J-V	750	0.003505	0.003520	0.003579	0.003556
J-V	950	0.000202	0.000203	0.000210	0.000202

Table A.11. All-arm condition interventions, part 2. Noise-prediction MSE on the fixed training-parent audit; means over parents and seeds.

Arm	Index	Assigned	Swap	Zero	Prior
J-SV	50	0.142174	0.142512	0.143795	0.142529
J-SV	250	0.032396	0.032623	0.032764	0.032622
J-SV	500	0.016933	0.017027	0.017050	0.017065
J-SV	750	0.003504	0.003521	0.003573	0.003567
J-SV	950	0.000201	0.000202	0.000208	0.000207
J-RAW	50	0.144829	0.145033	0.146072	0.144945
J-RAW	250	0.032524	0.032630	0.032750	0.032557
J-RAW	500	0.016868	0.017033	0.017061	0.016828
J-RAW	750	0.003544	0.003556	0.003565	0.003558
J-RAW	950	0.000203	0.000205	0.000208	0.000204
J-SH	50	0.143075	0.142956	0.143539	0.143130
J-SH	250	0.032535	0.032546	0.032579	0.032538
J-SH	500	0.016922	0.016915	0.016946	0.016917
J-SH	750	0.003560	0.003559	0.003566	0.003559
J-SH	950	0.000201	0.000202	0.000206	0.000201
C	50	0.145555	0.145555	0.145880	0.145555
C	250	0.032570	0.032570	0.032575	0.032570
C	500	0.016921	0.016921	0.016929	0.016921
C	750	0.003557	0.003557	0.003561	0.003557
C	950	0.000202	0.000202	0.000208	0.000202

Table A.12. Utility by evaluation OD cohort. Entries average episode macro F1 over both windows and both learner seeds (percent); they are not the pooled main score. All fifteen regimes are retained.

Training data	Cohort 3	Cohort 4
only-B	93.35	92.68
only-F-SV	93.63	91.69
only-J-RAW	92.81	92.95
only-J-S	92.85	92.93
only-J-SV	93.38	93.47
only-J-V	92.54	92.70
only-empirical	92.37	94.24
plus-B	97.50	95.06
plus-F-SV	97.31	93.73
plus-J-RAW	97.78	95.56
plus-J-S	96.75	97.15
plus-J-SV	96.70	94.30
plus-J-V	97.25	93.53
plus-empirical	95.17	93.23
real	95.15	93.72

A.3 G33: all physical summaries

Tables A.13 and A.14 retain all ten predefined physical summaries for all fourteen completed evaluations. They supplement the representative measures in the main text without selecting columns by the direction of an effect. Every entry is a five-class macro Wasserstein distance, with lower values indicating closer distributions. Each quantity retains its own unit; the columns are not combined into an overall score.

Every row begins with the same 100 prospective requests and uses the same 94 paired requests for these quality measures: 19 buses, 18 motorcycles, and 19 participants in each remaining class. Map construction supports 97 requests and reference extraction accepts 97, but their supported sets differ. Each full-pool row contains 776 generated trajectories, of which 752 belong to the paired requests (94×8). Each terminal-selection row contains 97 delivered trajectories, of which 94 have accepted references. These denominators distinguish generation coverage from quality support; none of the six unsupported requests is replaced. The eight selection rows reuse candidates, so their outputs are not additional independent generations.

In the tables, "257" and "Native" denote the relative and quarter-second diffusion models, and "Empirical" denotes whole-motion transfer from training journeys. N42 and N43 identify the native models; U, S, V and SV denote Uniform, spatial, speed and joint terminal selection. Seed numbers refer to fitted diffusion models and their matched empirical draws, not independent reference episodes. All columns have full five-class support in this completed cohort. The figures and tables remain descriptive for one new simulator episode.

Table A.13. Complete macro physical distances, part I: trip scale and one-second motion summaries. Every row has 94 paired requests out of the original 100.

Model / seed	Duration (s)	Distance (m)	Longest steady (s)	Four-second change (m/s)	Mean speed (m/s)
257 / 42	48.2841	4.6610	20.4234	0.2330	0.4601
257 / 43	46.3904	4.6817	19.2598	0.2199	0.6682
Native / 42	48.2841	4.6980	21.5398	0.2379	0.4602
Native / 43	46.3904	4.7070	20.4322	0.1935	0.6682
Empirical / 42	48.2841	4.6858	22.5400	0.5810	0.4599
Empirical / 43	46.3904	4.8039	20.7270	0.5953	0.6685
N42 / U	57.7782	4.5722	20.5561	0.2155	0.6082
N42 / S	58.4039	4.5503	25.4848	0.3205	0.3985
N42 / V	74.4163	4.5190	25.3082	0.2722	0.9320
N42 / SV	59.0127	4.6159	24.1906	0.2950	0.6988
N43 / U	51.5908	4.5515	21.6164	0.2416	0.3590
N43 / S	51.4898	4.6036	19.1246	0.1354	0.7402
N43 / V	77.6837	4.5177	30.7719	0.2088	0.9148
N43 / SV	68.0578	4.4888	23.0170	0.2141	0.8416

Table A.14. Complete macro physical distances, part II: acceleration, stopped motion and upper speed. Stopped-fraction distances use fractions, not percentage points. Every row has 94 paired requests out of the original 100.

Model / seed	Mean $\lvert a \rvert$ (m/s^2)	Stopped fraction	Speed p95 (m/s)	Longest stop (s)	Stop episodes per minute
257 / 42	0.1129	0.0278	1.8650	4.6934	0.1111
257 / 43	0.1131	0.0268	1.8252	4.8932	0.1184
Native / 42	0.0661	0.0179	1.9802	7.7539	0.1000
Native / 43	0.0560	0.0171	1.9495	7.6598	0.0939
Empirical / 42	0.1970	0.0156	2.7377	4.7270	0.1265
Empirical / 43	0.2059	0.0128	2.4870	4.1470	0.1236
N42 / U	0.0547	0.0221	1.7914	9.2801	0.1185
N42 / S	0.0870	0.0167	1.7201	7.6515	0.1248
N42 / V	0.0700	0.0203	1.8115	7.7146	0.1312
N42 / SV	0.0759	0.0169	1.6402	8.9170	0.1100
N43 / U	0.0693	0.0163	1.5543	8.1813	0.0880
N43 / S	0.0432	0.0159	1.4687	7.9047	0.1039
N43 / V	0.0563	0.0234	1.9131	9.6322	0.1208
N43 / SV	0.0561	0.0198	1.7139	10.7994	0.1169

Duration and distance retain the complete observed span. The remaining summaries use complete one-second intervals; for example, "Four-second change" denotes the mean absolute four-second lag difference in speed, not acceleration over a four-second sampling interval. Precise state thresholds and support rules are defined in Chapter 6. All unselected generators share the duration prior, explaining the identical duration distances within a seed. Selection can change the distribution of durations retained from that pool.

A.4 G33: native-resolution distances by participant class

Table A.15 reports interval-distribution distances for the six full-pool generators. These are computed at the native output resolution, including fractional terminal intervals, and therefore address a different question from the one-second journey summaries. Requests have equal mass within a class; candidate mass is divided within a request, and speed or acceleration samples are then weighted by their physical interval length. A macro improvement does not imply improvement in every class or every other motion property.

Table A.15. Class-wise native speed and acceleration W_1 for every full-pool generator. Each cell lists speed (m/s) above acceleration (m/s^2). The 94 paired requests comprise 19 buses, 18 motorcycles and 19 members of each other class; each row has 776 generated and 752 paired trajectories.

Model / seed	Bus	Motorcycle	Private car	Bicycle	Taxi / ride-hailing
	1.0800	0.5157	0.9632	0.2570	0.5812
257 / 42	0.6174	1.4987	0.6751	0.8113	0.5961
	0.8305	0.5742	1.8238	0.2460	0.6445
257 / 43	0.5955	1.5061	0.6588	0.7934	0.6055
	0.7540	0.6876	0.8473	0.2243	0.5334
Native / 42	0.0837	0.1833	0.1280	0.1632	0.1879
	0.6947	0.5444	1.6986	0.2193	0.5615
Native / 43	0.0746	0.2199	0.1250	0.1563	0.2117
	1.4309	0.7069	1.2088	0.2879	0.5521
Empirical / 42	0.4435	0.4065	0.1947	0.2379	0.3401
	1.2757	0.6626	2.0861	0.2634	0.5578
Empirical / 43	0.4483	0.4308	0.3134	0.2502	0.3513

A.5 Fixed-duration control: complete G34 tables

Tables A.16 and A.17 report all ten predefined physical summaries for G34's two full pools and eight terminal selections. Table A.18 gives the corresponding native speed and acceleration distances for every class. N42 and N43 denote the same native-clock models used in G33; "all" denotes a full pool, and U, S, V and SV retain their earlier definitions. Every row has 94 paired requests out of the original 100. A full pool contains 776 generated trajectories and 752 paired trajectories; a selected set contains 97 and 94, respectively. The class counts remain 19 buses, 18 motorcycles and 19 members of each other class. All reported metrics have that support.

G34 fixes one training-prior duration per request and model seed before scoring the shared candidate pool. Its reference episode is the already analyzed G33 episode, not a new blind sample. The following tables use the completed attempt2 summary and retain each metric's units; all distances are lower-is-better. Values are rounded to four decimal places for presentation.

Table A.16. Complete G34 macro physical distances, part I: trip scale and one-second motion summaries. Each row covers 94 paired requests out of 100.

Model / policy	Duration (s)	Distance (m)	Longest steady (s)	Four-second change (m/s)	Mean speed (m/s)
N42 / all	57.7782	4.5295	20.7243	0.2212	0.6080
N42 / U	57.7782	4.5722	20.5561	0.2155	0.6082
N42 / S	57.7782	4.4864	22.4883	0.3001	0.6079
N42 / V	57.7782	4.6340	23.5357	0.2675	0.6069
N42 / SV	57.7782	4.5187	22.7556	0.3144	0.6076
N43 / all	51.5908	4.4876	21.1753	0.1956	0.3586
N43 / U	51.5908	4.5515	21.6164	0.2416	0.3590
N43 / S	51.5908	4.4406	19.5263	0.2063	0.3586
N43 / V	51.5908	4.4136	23.1175	0.1998	0.3577
N43 / SV	51.5908	4.4905	22.0246	0.2114	0.3587

Table A.17. Complete G34 macro physical distances, part II: acceleration, stopped motion and upper speed. Stopped-fraction distances are fractions, not percentage points. Each row covers 94 paired requests out of 100.

Model / policy	Mean $\lvert a \rvert$ (m/s^2)	Stopped fraction	Speed p95 (m/s)	Longest stop (s)	Stop episodes per minute
N42 / all	0.0591	0.0180	1.9190	8.3425	0.1059
N42 / U	0.0547	0.0221	1.7914	9.2801	0.1185
N42 / S	0.0868	0.0201	2.0436	9.2865	0.1142
N42 / V	0.0710	0.0204	2.0278	7.1930	0.1125
N42 / SV	0.0896	0.0185	1.9874	8.7971	0.1079
N43 / all	0.0577	0.0157	1.7418	8.0993	0.0901
N43 / U	0.0693	0.0163	1.5543	8.1813	0.0880
N43 / S	0.0553	0.0172	1.7112	6.4591	0.0953
N43 / V	0.0562	0.0158	1.8483	7.6474	0.0861
N43 / SV	0.0604	0.0165	2.1182	7.2901	0.0883

Table A.18. Complete class-wise G34 native speed and acceleration W_1. Each cell lists speed (m/s) above acceleration (m/s^2), using the same 94-request paired population as the macro tables. Both full pools and all terminal policies are included.

Model / policy	Bus	Motorcycle	Private car	Bicycle	Taxi / ride-hailing
	0.7412	0.7595	0.7589	0.3107	0.9118
N42 / all	0.0778	0.1877	0.1208	0.1434	0.1945
	0.7129	0.9390	0.7597	0.3934	0.8594
N42 / U	0.0745	0.2073	0.1217	0.0872	0.1316
	0.7579	1.1450	0.8358	0.4140	0.9080
N42 / S	0.0890	0.1356	0.1257	0.1852	0.1502
	0.8729	0.7472	0.7720	0.4152	0.9175
N42 / V	0.0797	0.1405	0.1270	0.1019	0.1128
	0.9038	1.1317	0.6814	0.4000	0.8677
N42 / SV	0.0927	0.1348	0.1235	0.2062	0.1578
	0.6245	0.7224	0.6906	0.2492	0.4653
N43 / all	0.0800	0.2065	0.1103	0.1778	0.1939
	0.5770	0.6701	0.6441	0.2998	0.7232
N43 / U	0.1049	0.1968	0.1223	0.1678	0.2067
	0.6618	0.7591	0.6005	0.2941	0.3924
N43 / S	0.0742	0.2525	0.1146	0.1688	0.2150
	0.5973	0.7557	0.5623	0.2950	0.4191
N43 / V	0.0878	0.2119	0.1125	0.1432	0.1057
	0.6200	0.9857	0.7289	0.2790	0.3258
N43 / SV	0.0828	0.1880	0.1112	0.1660	0.1578

A.6 Five G33 cases selected before visual inspection

For G33 model seed 42, the case for each class is the lexicographically first request with an accepted reference and outputs from all four terminal policies. This rule is based on request order and common support, not physical quality, teacher confidence or visual appearance. The five plots illustrate the same declared conditions and the consequences of selecting different candidates; they are not five independent tests of a method advantage.

Each figure shows coordinate geometry, relative-time progress and a physical-time speed curve. Speed is derived from cumulative planar distance over complete one-second intervals, with the final partial second omitted from that panel. It is not the original quarter-second speed signal and cannot display all fluctuations measured by the native-interval metrics. Different selected durations are allowed in G33. Near-coincident map traces are expected under a common route condition and do not establish that the selected timing is equally realistic.

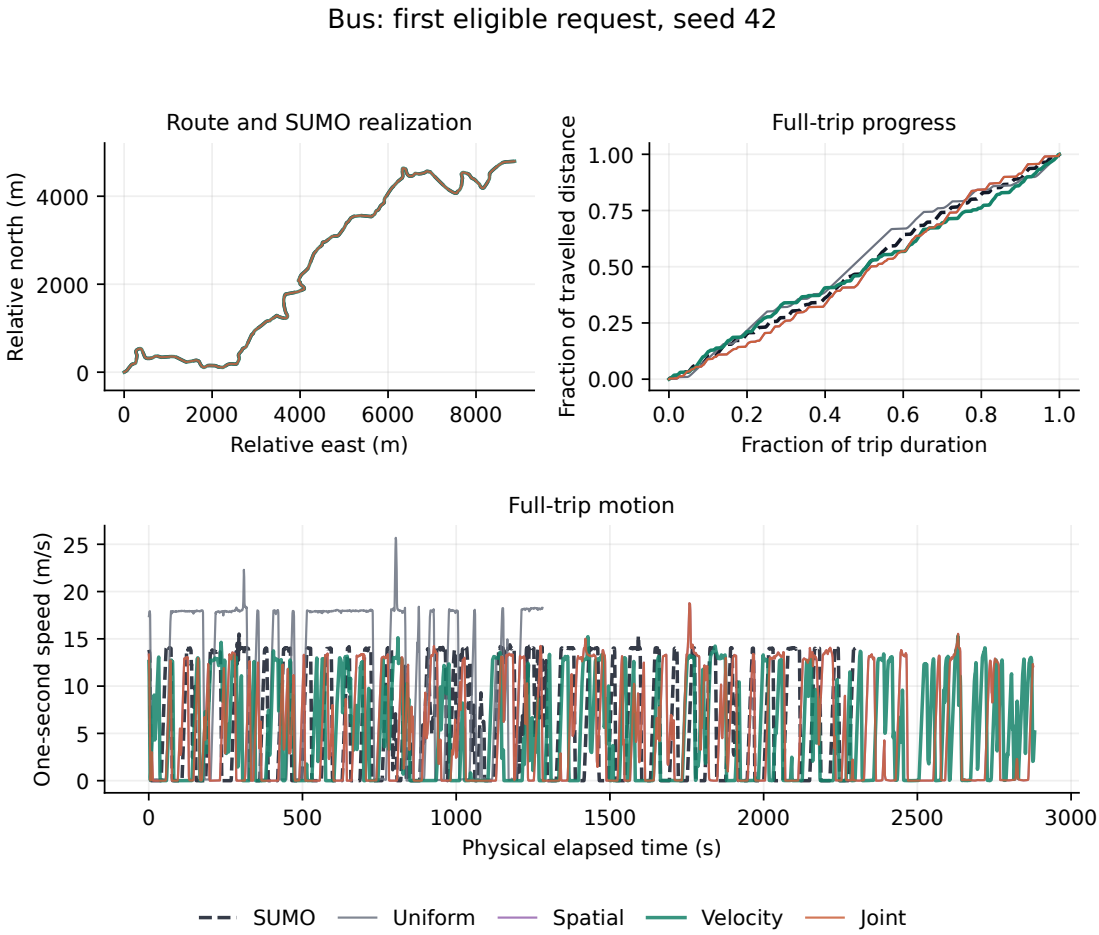


Figure A.1. Bus. Fixed seed-42 case: the first request by identifier order with a reference and all four policy outputs, not a visually selected success. Speed summarizes complete one-second intervals and omits the final partial second; it is not the native quarter-second signal.

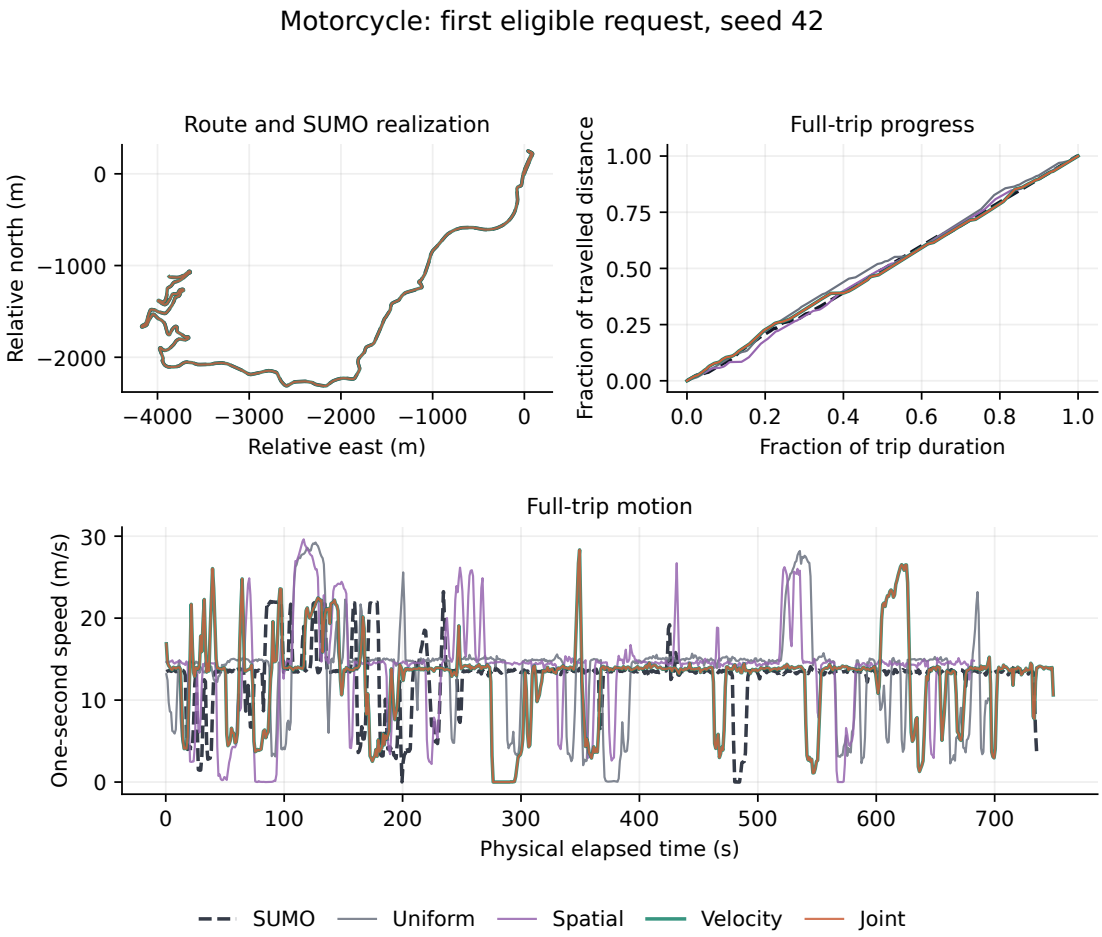


Figure A.2. Motorcycle. Fixed seed-42 case: the first request by identifier order with a reference and all four policy outputs, not a visually selected success. Speed summarizes complete one-second intervals and omits the final partial second; it is not the native quarter-second signal.

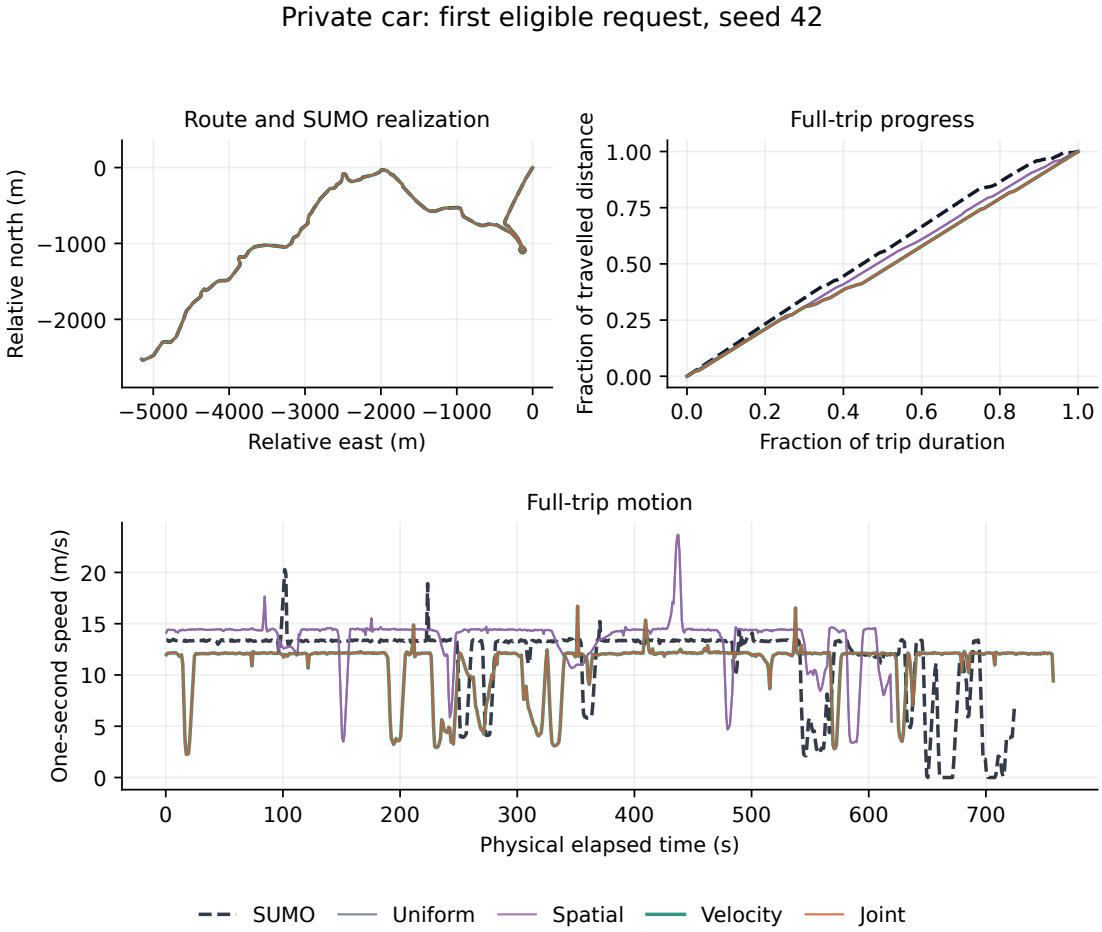


Figure A.3. Private car. Fixed seed-42 case: the first request by identifier order with a reference and all four policy outputs, not a visually selected success. Speed summarizes complete one-second intervals and omits the final partial second; it is not the native quarter-second signal.

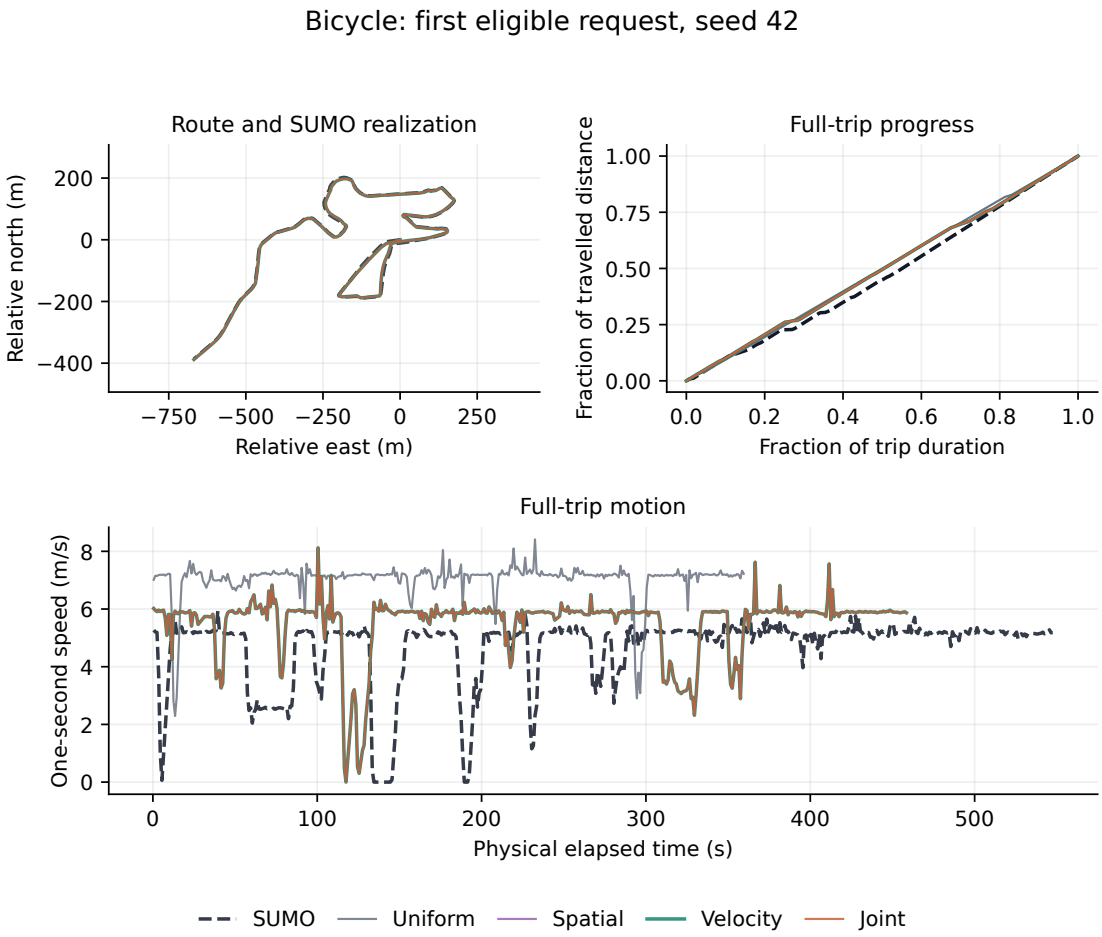


Figure A.4. Bicycle. Fixed seed-42 case: the first request by identifier order with a reference and all four policy outputs, not a visually selected success. Speed summarizes complete one-second intervals and omits the final partial second; it is not the native quarter-second signal.

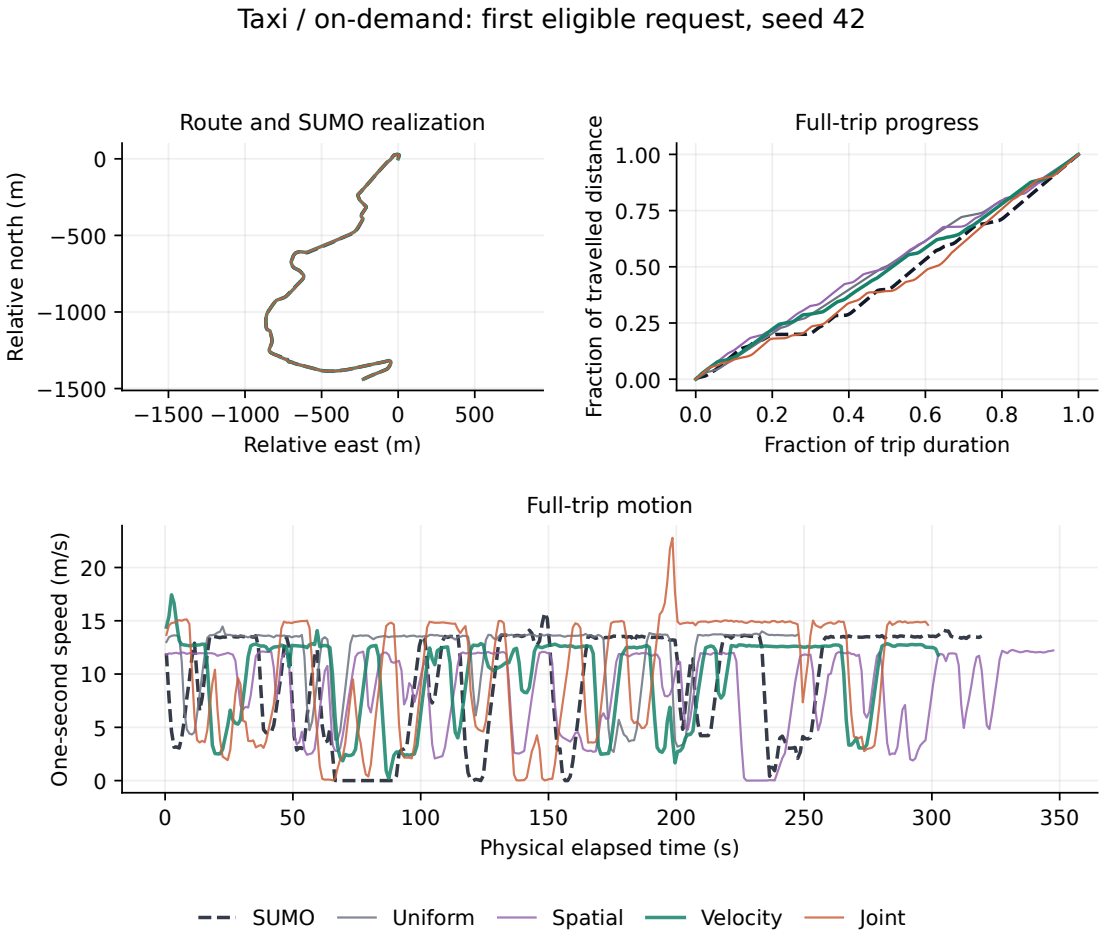


Figure A.5. Taxi or ride-hailing vehicle. Fixed seed-42 case: the first request by identifier order with a reference and all four policy outputs, not a visually selected success. Speed summarizes complete one-second intervals and omits the final partial second; it is not the native quarter-second signal.

A.7 Historical evidence and reproduction scope

The earlier locked E7 evaluation is preserved in the repository, including its unfavorable feedback and augmentation findings. It began with 270 held-out participant IDs, of which 53 satisfied the strict moving-window generation eligibility rules. Its 512-point task, reference population and evaluation protocol differ from the present complete-journey experiment. The 53 eligible requests must not be presented as coverage of all 270, merged into either the G33 pilot or the final G40 quality population, or reused to select the current method after their results were exposed. The original report remains at `docs/results/final-test-e7/REPORT.md`.

The paired-representation study G28 is also retained, at `docs/results/g28-paired-latents/REPORT.md`. It trained spatial and speed readers for matching participant labels and tested latent supervision in an earlier eight-class simulator corpus. Its planned comparisons did not establish stable independent physical gains from paired spatial–speed supervision. That result remains evidence about that corpus and intervention; it is neither erased by the current temporal-resolution result nor merged with terminal selection on native-demand trajectories. The present thesis distinguishes classification information, the mechanism used to transmit it, and physical generation quality across these stages.

For the temporal-resolution pilot, `docs/G33_PROTOCOL.md` fixes the conditions and budgets, and `docs/results/g33-generation/report/summary.json` contains the numerical source of the G33 tables, including per-class support and provenance hashes. The G34 tables use `docs/results/g34-fixed-duration/attempt2/report/summary.json` under the separate `docs/G34_PROTOCOL.md`. `docs/results/g33-generation/paper-figures/complete.json` records the fixed case rule and figure hashes. The entry scripts `scripts/g33_train.py`, `scripts/g33_evaluate.py`, `scripts/g33_feedback.py` and `scripts/g33_plots.py` separate fitting, generation, feedback, evaluation and presentation. Reproduction requires the corresponding versioned source observations, public scenario and saved model artifacts; the manuscript does not embed raw participant records or access credentials. These relative repository paths identify the required evidence without relying on a particular server layout.

The original locked-result snapshot remains separately identified in the experiment repository [13]; its test population is not reused to select the present models.

A.8 Expanded recognition on a common development population

The full-day expansion completed all four 10,000-step diffusion fits and both fixed 50-epoch GAF readers. The native models took 2,771.9 and 2,750.8 seconds, approximately 46 minutes each; the 257-point controls took approximately 319 seconds each. Model capacity remains 891,969 parameters. The larger corpus and longer native sequences change data exposure and computational cost, so equal update counts alone do not imply equal computation. No new 257-point generation-quality claim follows merely from completing those fits.

The paired reader comparison is stronger evidence than comparing the pilot's small development set with the expanded set. Both old and new checkpoints see the same 7,230 complete development legs, and the original training populations remain separate from that cohort. Table A.19 and Figure A.6 show a substantial recognition improvement in both representations.

Table A.19. Fixed old and expanded readers on the same 7,230 development legs. Percentages; no checkpoint selection.

Reader	Accuracy	Macro F1	Taxi F1
Old Spatial	60.65	50.10	19.07
New Spatial	86.17	76.47	45.85
Old Speed	63.61	61.82	30.61
New Speed	89.13	90.34	81.07

Spatial macro F1 increases by 26.37 percentage points, with a paired parent-bootstrap 95% interval of 24.54–28.16 points. Velocity macro F1 increases by 28.52 points, with an interval of 27.27–29.73 points. These intervals condition on the available public-demand cohort and fixed model fits; they do not quantify uncertainty over cities or training recipes. The improved recognition supports the decision to preserve native demand and expand complete-trip coverage. It does not identify a pure causal effect of sample count because the time coverage and number of CNN optimizer updates also changed.

The confusion matrices explain why overall accuracy alone would be incomplete. Spatial private-car F1 is 95.20%, but taxi/ride-hailing F1 is 45.85%. The velocity values are 92.23% and 81.07%, respectively. Spatial training accuracy is 97.22%, compared with 86.17% on development; the corresponding velocity figures are 92.39% and 89.13%. Generalization gaps, calibration error and shared route templates remain relevant when transferring these readers from simulator observations to generated candidates.

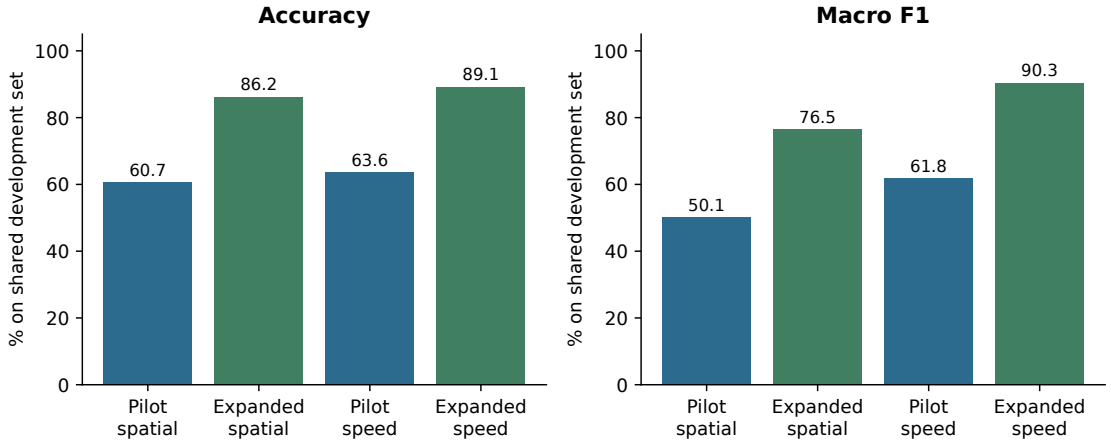


Figure A.6. Pilot and expanded readers evaluated on identical development inputs. Old/new refers to training corpus and optimization exposure, not different evaluation sets. Higher is better.

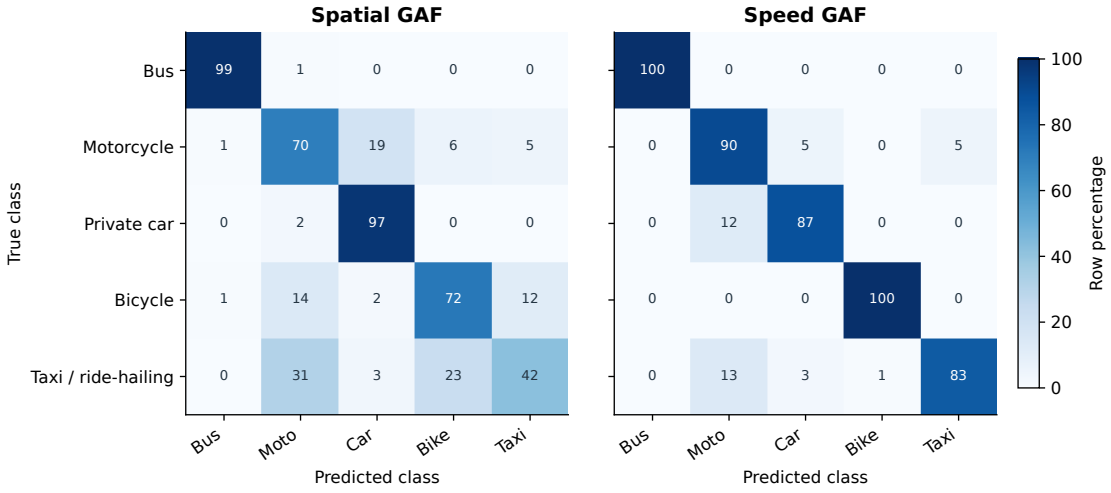


Figure A.7. Pre-adaptation spatial and speed readers on the same expanded development cohort. Each row is normalized within its true class. These are the original G35 checkpoints, not new evaluations of jointly adapted G40 encoders.

A.9 Supporting old/new generator comparison

Table A.20. Expanded versus pilot native generators, full candidate pools: class–macro native W1 changes. Each cell uses its own training-side duration prior.

Scene	Seed	Speed change	Acceleration change
Morning	42	+4.7%	+22.0%
Morning	43	+12.2%	+34.4%
Later	42	-57.2%	-15.7%
Later	43	-65.5%	-5.6%

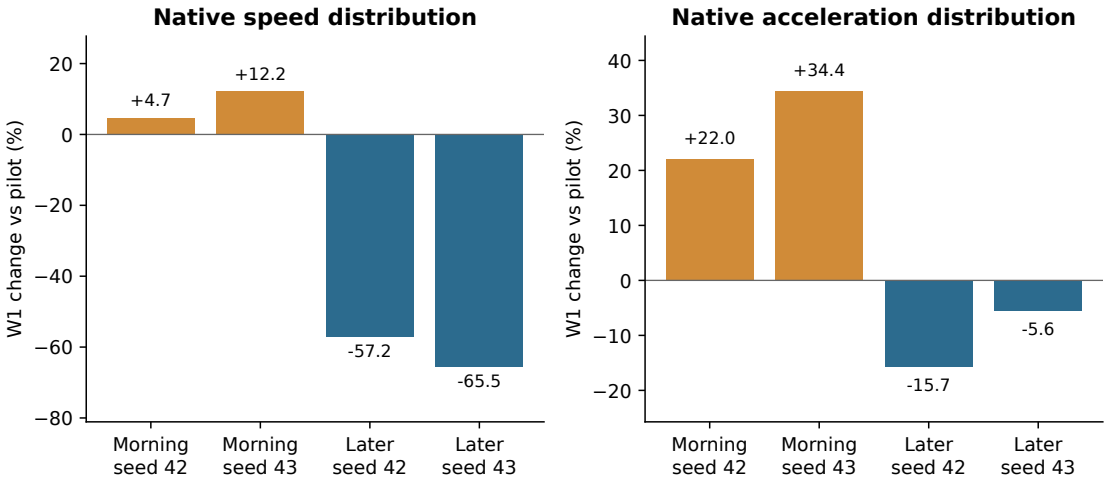


Figure A.8. Expanded versus pilot native-time generation on two new SUMO scenes. Bars show changes in full-pool five-class macro distances; negative is better. Each model uses its own training-side duration prior. The reversal across scenes prevents a universal expansion claim.

Expansion changes the generator's applicability across the observed contexts. In the later scene, native-speed distribution error falls by 57.2% and 65.5% for the two model seeds, while acceleration error falls by 15.7% and 5.6%. In the morning scene, speed error instead rises by 4.7% and 12.2%, and acceleration error rises by 22.0% and 34.4%. The signs agree across seeds within each scene, but there are only two simulator realizations and the underlying public plans are shared.

This contrast is consistent with sensitivity to demand and temporal coverage, but it does not isolate that cause. Expanded and pilot models differ in training data and duration prior, and neither receives the instantaneous background traffic state. Departure context and simulator random seed change together between scenes. The

result establishes a scene-dependent change in the delivered system, not that the diffusion network has learned an explicit time-of-day controller.

A.10 Does a more accurate GAF reader improve selection?

The central comparison applies old and expanded readers to the same expanded-generator candidates. Duration, geometry, candidate noise and the requested class are unchanged by the teacher contrast. Uniform controls are verified identical array by array. The experiment therefore distinguishes improving the generator from improving the criterion used to select its outputs.

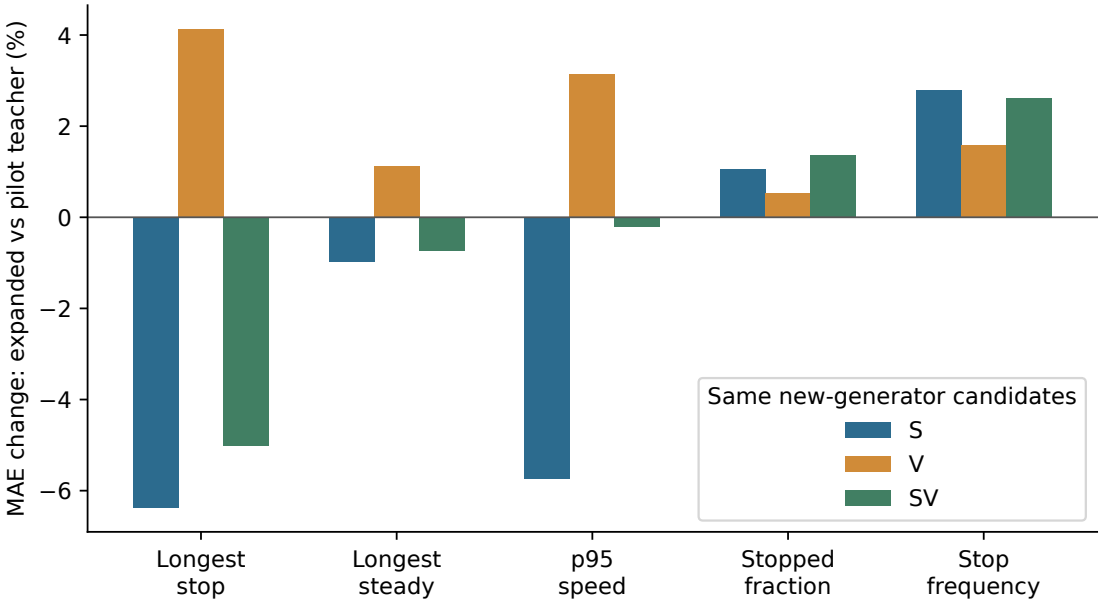


Figure A.9. Expanded versus pilot teachers on identical expanded-generator pools: changes in request-level error for five motion summaries. Negative values favour the expanded reader. These post-hoc equal-cell summaries expose gains and costs; they are not significance tests or a replacement for the full distributional matrix.

A higher recognition score does not induce a uniformly better physical ordering. For example, the expanded spatial reader reduces longest-stop request error by 6.4% relative to the pilot spatial reader on these same pools, but raises stopped-fraction error by 1.1% and stop-frequency error by 2.8%. The expanded velocity reader reduces mean-absolute-acceleration request error by 2.8% but increases longest-stop error by 4.1% relative to the pilot velocity reader. These comparisons concern

the selector with a fixed generator; they do not reuse the classifier's own target probability as a quality label.

A.11 Exploratory evidence for class-specific stopping feedback

The post-hoc request-level analysis suggests a more precise mechanism to test. Spatial feedback reduces longest-stop MAE in all four scene/seed cells, with an aggregate reduction of 7.6%. This differs from its class-marginal longest-stop W1, which improves in three cells and decreases by 4.6% in the aggregate. The distinction matters: the first statement concerns correspondence to the same request's reference, while the second concerns the distribution of that scalar over requests.

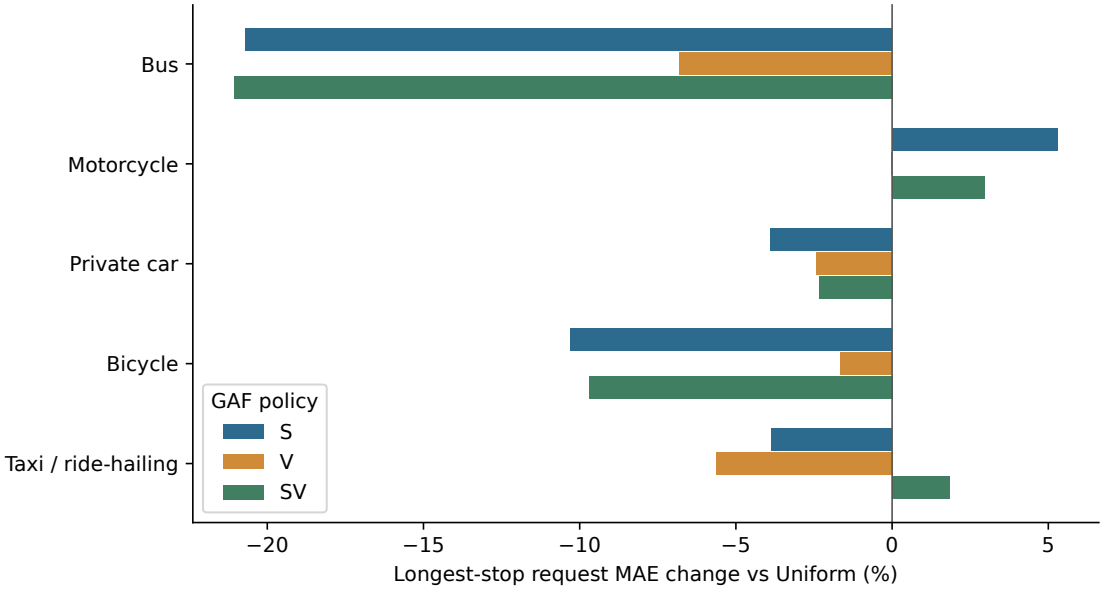


Figure A.10. Class-specific longest-stop request error after expanded GAF selection. The same five classes and all three policies are retained. The bus improvement coexists with a motorcycle cost; percentage reductions across classes are not a single common-unit effect size.

For buses, the spatial longest-stop request error falls by 20.7%, with all four cells improving. For motorcycles it rises by 5.3%, with all four cells worsening. The dual policy's corresponding bus reduction is 21.1%; that small difference from spatial-only feedback is not evidence of representation synergy. These observations motivate class-sensitive stopping analysis, but the metric does not determine whether the longest stop occurs at the correct bus station. It also does not distinguish a service stop from waiting in traffic.

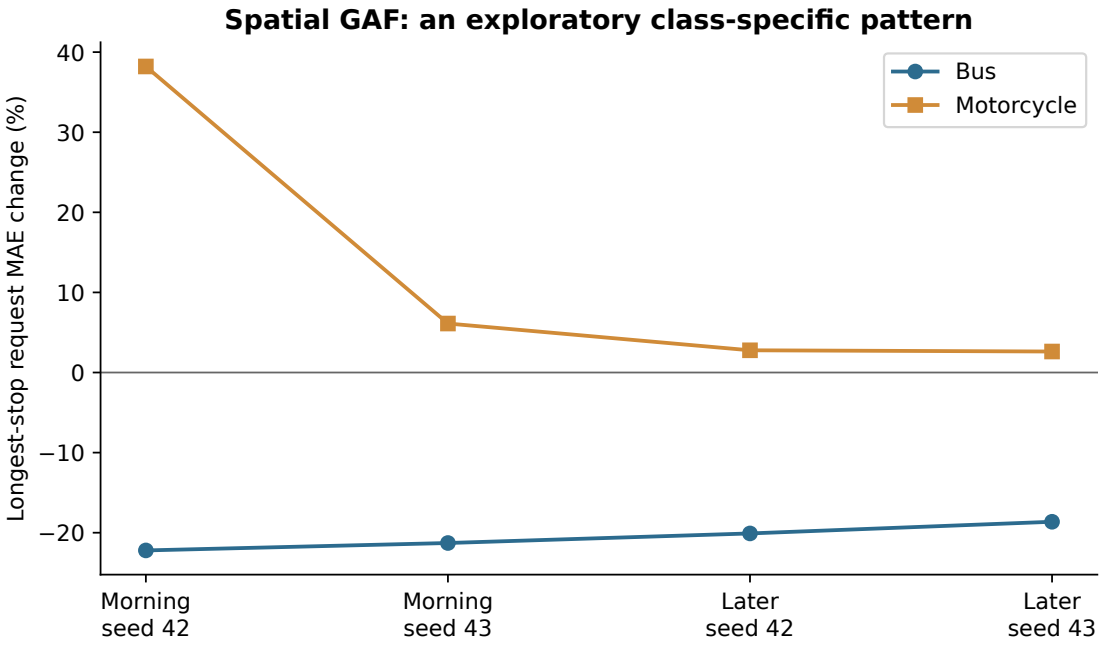


Figure A.11. Expanded spatial feedback versus Uniform for buses and motorcycles, shown separately in every scene/seed cell. The two classes illustrate opposing directions found in the complete class table. Connecting lines aid reading only; the four cells are not a time series or four independent traffic scenes.

A mechanism claim therefore requires a fresh, fixed comparison of stop location, service-stop coverage and non-service waiting, alongside the complete class matrix. The present finding was identified after inspecting results, and its consistency over two generator seeds cannot supply an independent confirmatory test. It is valuable as a specific link between the spatial trajectory representation and an output motion property, not as proof that the spatial encoding has become a universal realism objective.

Full-pool pair diversity describes the variation available before selection. With one retained sample per request, conditional pair diversity is undefined, not zero; across-request dispersion cannot replace it. The common 97-request map coverage is unchanged by selection and belongs to the geometric interface.

A.12 Frozen-reader supervision and fixed-marginal selection

The completed G39 frozen-feature alignment study includes seven arms, two fitted seeds and eight new reference episodes. Its 800 declared episode requests yield

782 route-supported requests and 754 physical pairs; 470 realised reference routes are present in their catalogs. The dual-view SV arm gives K1 speed/acceleration distances of 1.29337/0.18761 compared with 1.28383/0.18674 for B: increases of 0.743% and 0.466%. The speed-only arm slightly improves primary acceleration, but the same-class shuffled control changes it by a similar amount and the four-draw weighted comparison reverses the apparent speed-only advantage. Frozen-feature supervision does not establish a general improvement of motion.

Longer-timescale changes are also mixed. SV reduces longest-stop distance by 1.44%, while mean absolute acceleration on the one-second clock worsens by 4.86% and four-second speed-change distance worsens by 5.23%. Bicycle and motorcycle gains coexist with adverse bus and private-car results. The complete endpoints remain in the experiment records rather than being condensed into a favourable class score.

A separate same-pool diagnostic on the already exposed G38 scenes distinguishes global GAF weighting from weighting with fixed route and route–duration marginals. It uses four scene/seed cells and 10,784 candidates. Under the fixed joint marginals, speed-GAF weighting changes native speed/acceleration distances by only $-0.12\%/-0.13\%$, with improvements in one and two of the four cells respectively. Spatial and dual-view weighting worsen both macro endpoints. The preserved marginals are distributional constraints; an individual coupled random draw need not retain the same route and duration.

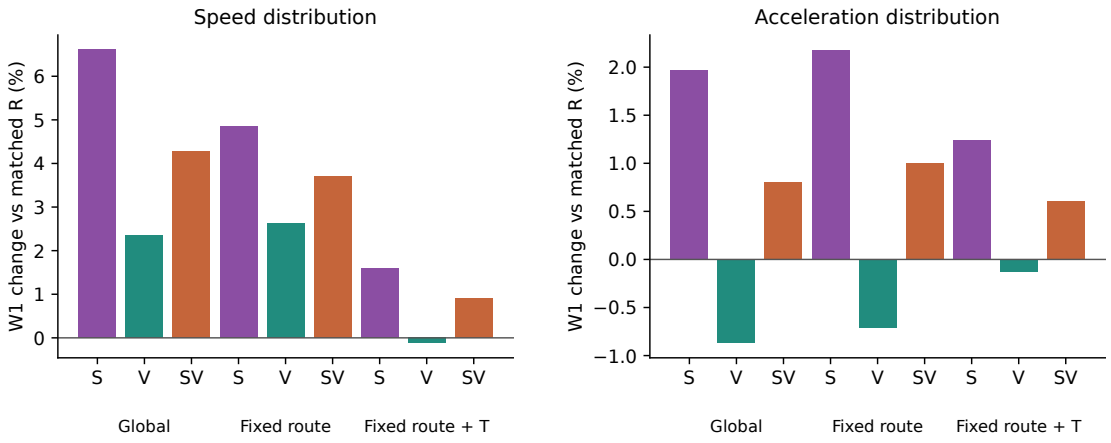


Figure A.12. Closed mechanism diagnostic: changes under global, fixed-route and fixed-route–duration weighting of identical candidate pools. The last comparison preserves the joint route/time marginal rather than asserting that each selected sample is identical. These reused scenes are development evidence; the small residual speed-view effects are not independent confirmation of guidance.

Independent augmentation nevertheless gives a useful, narrower finding. Across two raw-series learner seeds, real-only training reaches 91.25% macro F1, real plus

baseline diffusion data 93.17%, real plus speed-aligned data 94.10%, and real plus RAW-aligned data 94.06%. Real plus dual-view data reaches 93.57%; synthetic-only regimes remain below real-only. Thus diffusion samples can help this downstream task, while the almost equal speed-GAF and RAW results do not demonstrate an image-specific advantage. The generator has access to the full labelled corpus, so this is not a strict low-label learning claim.

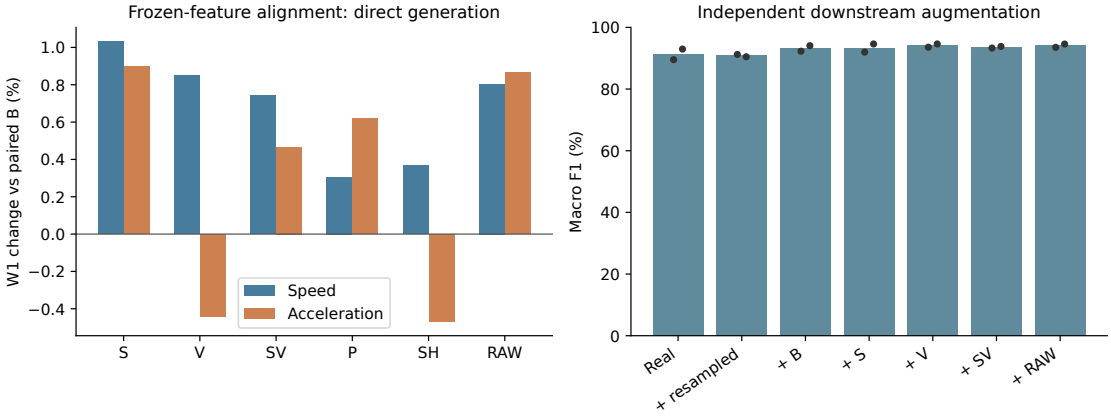


Figure A.13. Completed G39 frozen-feature alignment (left) and independent augmentation (right), with both downstream seeds shown. The evaluation model reads raw time series and is independent of the GAF readers. Better task performance does not override adverse physical endpoints or establish that GAF is necessary.

The main experiment extends these fixed-reader controls to jointly trained generative conditions.

A.13 Implementation and execution diagnostics

The prior pilot also matters. With 1,000 prior updates, sampled absolute codes reached approximately 119–178. The single declared-budget recovery used 5,000 updates and reduced the sampled pilot range substantially without refitting motion. The formal experiment uses that previously declared budget. Its largest recorded absolute code is approximately 6.50; some prospective samples still exceed their arm's observed training-code maximum. Counts are 31, 35, 14, 70, 18 and 86 out of 6,272 for F-SV, J-S, J-V, J-SV, J-RAW and J-SH. Exceeding an empirical maximum is not an invalidity threshold, and no samples are removed on that basis. Finite ranges alone do not establish an accurate code distribution.

A separate completed performance study reduces a repeated-archive evaluation unit from 104.168 to 10.017 seconds while reproducing its metrics and diagnostic JSON exactly. That I/O change is adopted. Proposed training caching gives no end-to-end speedup in the bounded benchmark and fails strict state equality,

including non-identical repeats of the original implementation; it is not adopted. The measured evaluation improvement must not be described as a tenfold training acceleration.

The final scientific queue completes without failure. Pilot verification errors, the unstable short-prior result and a presentation-only CUDA dependency failure remain recorded with their recoveries. Figure layout changes preserve the fixed-case coordinate/image hashes and numerical matrices.

A.14 Supplementary dataset and demand comparison

A.14.1 Archived trajectory exports and spatial alignment

The archived reference dataset (CNR export), `midimost_with_class2.csv`, contains 4,840,789 observations from 2,685 identifiers and six participant classes. A second export, `vehicle_positions_TS_New.csv`, contains 2,189,591 observations from 12,061 identifiers but only one recorded class. Their spatial support, durations and label composition differ. The secondary table is not pooled into the labelled complete-journey experiment.

The canonical record is `time,id,x,y,type`, with seconds and local planar metres. The principal export additionally contains simulator speed and the label `class2`; its class is mapped to the common type field. Speed for physical comparisons is derived from positions and timestamps on both observed and generated data, avoiding comparison between an internal simulator state and a numerical coordinate derivative.

No original network, route file or simulation configuration accompanied the tables. A check against the public Monaco SUMO Traffic network supports alignment of the principal source [3]: all 4,841 systematically sampled principal positions lie within five metres of a lane centreline, compared with 11.2% of the 2,190-point secondary sample. This identifies a useful projected spatial frame without recovering the original run settings. The positions are not treated as longitude and latitude, and map proximity alone is not a directed-road feasibility proof.

A.14.2 Demand structure and participant recognition

A controlled comparison evaluates class-permitted random routes against activity-based demand and the archived reference dataset. Road permissions determine feasible paths, while public services and activity plans introduce class-related spatial patterns.

The comparison matched six labels and class counts, with 1,983 training and 257 validation entities per source. Original spatial encoding, ResNet18 architecture,

fitting budget and final-checkpoint rule were fixed; two classifier seeds were used. Table A.21 separates an information readout from the image reader. The readouts evaluate recognition on observed trajectories.

Table A.21. Matched source diagnosis. Values are validation macro F1 in percent; the Hilbert–GAF row averages two fixed reader seeds.

Readout	Archived data	Random demand	Native MoST
Spatial geometry	79.12	34.85	76.95
Physical motion	88.08	80.57	90.35
Hilbert–GAF	59.88	30.15	58.69

Spatial information and GAF recognition both recover under native demand, while random demand still retains substantial motion-related class information. The source comparison identifies domain mismatch as a contributor to low spatial recognition in the initial expansion. Service routes, stops, density and activity chains change together in this comparison.

The native source is fixed to public revision `b29b2f65`, using the unchanged map, 24 bus services and 181 stops together with activities, parking and background traffic. These simulations use a separately specified configuration. In the matched source diagnosis, 204 of 257 validation entities share source identity with the archived dataset and native validation bus lines are represented in training. That study concerns recovery of domain structure, not unseen-line validation.

Participant labels and SUMO permission classes remain separate. Source vehicle types retain their actual physical and behavioural parameters [51]. A taxi/ride-hailing label does not change a passenger-permission vehicle into a taxi-permission vehicle. Such distinctions affect route construction and prevent manufactured label differences from being presented as learned transport behaviour.

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